The Short, the Long, and the Consequences of VAR Analysis For Structural DSGE Estimation

Master's Degree in Applied Economics; Economic Data Analytics

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Abstract

Truncation bias could be a severe problem regarding structural VAR analysis (De Greave and Westermark) (DGW2025) claims that the role of lag choice in SVAR is crucial to correctly identify IRFs, and they claim that choosing long lags solve the bias and, surprisingly, this reduce the variance of the point estimate. CEE2005 is a classic paper that studies the effects of monetary policy shock in the economy. Particularly their DSGE is calibrated ex-ante using matching the model's IRFs with the ones estimated from the VAR analysis. The aim of this thesis is to perform the (DGW2025) exercise, see if and how the IRFs changes, analys how the DSGE model calibration vary and if the model is consistent to this new empirical way to estimate the VARs. If so, we want to understand more about the claimed variance reduction caused by the solution of the truncation bias how works in the Frequentist and Bayesian world.

Contents

1 Introduction **4**

1.1 Research Question 4

2 Theoretical Framework **4**

3 Methodology **4**

4 Empirical Results **4**

4.1 From the state-space solution to a VAR(∞) 4

5 Conclusion **6**

A Replication Files and Code **8**

B Additional Tables **8**

B.1 DSGE Parameters Table: all lags 8

B.2 Implied Duration from Calvo Price Table: all lags 9

B.3 Implied Duration from Calvo Price Table: lags 4 8 12 9

B.4 VAR ALIGNMENT DSGE Parameters Table: all lags 9

B.5 Decomposition of Variance across different lags 9

C Additional Figures **9**

C.1 ACEL 2005 VAR Variables 10

C.2 ACEL 2005 VAR Across different lag structures: Monetary Policy Shock . . . 11

C.3 ACEL 2005 VAR Across different lag structures: Embodied Technology Shock 14

C.4 ACEL 2005 VAR Across different lag structures: Neutral Technology Shock . . 17

C.5 ACEL 2005 DSGE Matched with VAR 4, compared with VAR estimated across different lag structures: Monetary Policy Shock 20

C.6 ACEL 2005 DSGE Matched with VAR 4, compared with VAR estimated across different lag structures: Embodied Technology Shock 23

C.7 ACEL 2005 DSGE Matched with VAR 4, compared with VAR estimated across different lag structures: Neutral Technology Shock 26

C.8 ACEL 2005 DSGE Matched and compared with VAR estimated across different lag structures: Monetary Policy Shock 29

C.9 ACEL 2005 DSGE Matched and compared with VAR estimated across different lag structures: Embodied Technology Shock 32

C.10 ACEL 2005 DSGE Matched and compared with VAR estimated across different lag structures: Neutral Technology Shock 35

C.11 VAR Residual Analysis 38

C.12 ACEL 2005 VAR Across different lag structures Progressively Comparison: Monetary Policy Shock 40
C.13 ACEL 2005 VAR Across different lag structures Progressively Comparison: Embodied Technology Shock 45
C.14 ACEL 2005 VAR Across different lag structures Progressively Comparison: Neutral Technology Shock 50
C.15 ACEL 2005 DSGE Across different lag structures Progressively Comparison: Monetary Policy Shock 55
C.16 ACEL 2005 DSGE Across different lag structures Progressively Comparison: Embodied Technology Shock 60
C.17 ACEL 2005 DSGE Across different lag structures Progressively Comparison: Neutral Technology Shock 65

1 Introduction

Provide the background of your topic, state the research problem, and outline the structure of your thesis here. Maybe I have switched the latex writing in VS Code

1.1 Research Question

Clearly define what your thesis aims to solve or investigate.

2 Theoretical Framework

Discuss the existing literature and the theoretical models that underpin your research.

3 Methodology

Explain the empirical strategy, models, and data you are using.

4 Empirical Results

4.1 From the state-space solution to a VAR(∞)

The state-space solution. Following Ravenna (2007), the recursive equilibrium law of motion of a linearised DSGE model is

$$y_t = Px_{t-1} + Qz_t, \tag{1}$$

$$x_t = Rx_{t-1} + Sz_t, \tag{2}$$

$$Z(L)\, z_t = \varepsilon_t, \tag{3}$$

where x_t is the $n \times 1$ vector of endogenous states, z_t the $m \times 1$ vector of exogenous states (structural shocks), y_t the $r \times 1$ vector of controls, and ε_t is $m \times 1$ white noise with $\mathbb{E}(\varepsilon_t \varepsilon_t') = \Sigma$ diagonal. The polynomial $Z(L) = [I - Z_1 L - \cdots - Z_q L^q]$ governs the persistence of the shocks.

Step 1: eliminate the endogenous states. Define $G(L) \equiv [I - RL]$ and write (2) as $G(L)x_t = Sz_t$. Assuming $G(L)$ invertible,

$$x_{t-1} = G(L)^{-1} SL z_t. \tag{4}$$

Step 2: the moving-average form. Substituting (4) into (1), and taking the observable vector to be $\widehat{Y}_t = y_t$ (the econometrician observes the controls but *not* the states),

$$y_t = Qz_t + PG(L)^{-1} SL z_t. \tag{5}$$

This is a VMA(∞). The moving-average term is the direct consequence of x_t being unobserved.

Step 3: the VARMA (i.i.d. shocks). Let $Z(L) = I$, so $z_t = \varepsilon_t$. Assume Q is invertible, which requires $r = m$, and define the reduced-form innovations

$$\eta_t \equiv Q\varepsilon_t = Qz_t, \tag{6}$$

so that Q is the theoretical matrix mapping structural shocks into reduced-form innovations. Then (5) becomes $y_t = \eta_t + PG(L)^{-1}SQ^{-1}L\,\eta_t$. Using the determinant–adjoint identity $G(L)^{-1} = D_G(L)\,|G(L)|^{-1}$, where $|G(L)|$ is a scalar polynomial of order n and the adjoint $D_G(L)$ is of order $n-1$, and multiplying through by $|G(L)|$:

$$\underbrace{|G(L)|\,y_t}_{\text{AR part}} = \underbrace{\Big[\,|G(L)|\,I + PD_G(L)SQ^{-1}L\,\Big]}_{\equiv\, G^*(L)}\eta_t \;=\; G^*(L)\,\eta_t. \tag{7}$$

Equation (7) is a VARMA(n,n) in *final equations form*: the autoregressive operator is scalar times the identity, so each component of y_t depends only on its own lags. Since $|G(0)| = 1$ and $G^*(0) = I$, the representation is unique.

Step 4: inverting the MA part. If $G^*(L)$ is invertible, premultiplying (7) by $G^*(L)^{-1}$ isolates the innovations:

$$|G(L)|\,G^*(L)^{-1}y_t = \eta_t. \tag{8}$$

Writing $G^*(L)^{-1} = D_{G^*}(L)\,|G^*(L)|^{-1}$, the adjoint $D_{G^*}(L)$ is of finite order, but the reciprocal of the scalar operator $|G^*(L)|$ expands as

$$\frac{1}{1-aL} = \sum_{j=0}^{\infty} a^j L^j, \tag{9}$$

which is of infinite order unless $|G^*(L)|$ is constant. Hence:

$$\boxed{\text{the DSGE solution generically implies a VAR}(\infty);} \tag{10}$$

a finite-order VAR(p) exists *if and only if* $|G^*(L)|$ is of degree zero in L, i.e. $G^*(L)$ is unimodular (Lütkepohl, 1993, p. 245).

Step 5: persistent shocks. With $Z(L) \neq I$, define $\widetilde{Z}(L) \equiv QZ(L)Q^{-1}$, so that $QZ(L)^{-1}\varepsilon_t = \widetilde{Z}(L)^{-1}\eta_t$. Then

$$|\widetilde{Z}(L)|\,|G(L)|\,y_t = G^*(L)\,D_{\widetilde{Z}}(L)\,\eta_t, \tag{11}$$

which, since $|\widetilde{Z}(L)|$ is of order qm, is a

$$\text{VARMA}\big(n+qm,\; n+q(m-1)\big) \tag{12}$$

process. Shock persistence therefore *raises* the autoregressive order required to represent the data.

Step 6: the VAR(∞) in explicit form. An equivalent and more transparent route (Corollary 2.2 in Ravenna, 2007) solves (1) for $z_{t-1} = Q^{-1}(y_{t-1} - Px_{t-1})$ and substitutes into (2), giving $H(L)x_t = SQ^{-1}y_{t-1}$ with

$$H(L) \equiv \big[I - (R - SQ^{-1}P)L\big]. \tag{13}$$

If $H(L)$ is invertible, the observables satisfy

$$y_t = P\sum_{j=0}^{\infty}\big[R - SQ^{-1}P\big]^{j} SQ^{-1}\, y_{t-j-1} \;+\; \eta_t. \tag{14}$$

Equation (14) *is* the VAR(∞), written out: the coefficient matrix attached to y_{t-j-1} is $P\,[R - SQ^{-1}P]^{j}\,SQ^{-1}$.

Interpretation. The matrix $R - SQ^{-1}P$ is the counterpart of $A - BD^{-1}C$ in Fernández-Villaverde et al. (2005); Ravenna notes that (14) "is the derivation obtained in Fernandez-Villaverde et al. (2005)". Two distinct questions follow. *Existence*: the VAR(∞) is well defined if all eigenvalues of $R - SQ^{-1}P$ lie inside the unit circle — the "poor man's invertibility condition". *Accuracy*: the eigenvalue *moduli* govern the geometric rate at which the coefficients in (14) decay, and hence the size of the bias incurred by truncating the lag structure at finite p. An eigenvalue close to unity implies coefficients that remain non-negligible at long lags, so that a short-lag VAR is a poor approximation to the model-consistent representation of the data.

5 Conclusion

Summarize the main takeaways, discuss the policy implications, and mention any limitations of your work.

References

Fernández-Villaverde, J., Rubio-Ramírez, J., and Sargent, T. J. (2005). A, b, c's (and d)'s for understanding vars. NBER Technical Working Paper 308, National Bureau of Economic Research.

Lütkepohl, H. (1993). *Introduction to Multiple Time Series Analysis*. Springer-Verlag, Berlin, 2nd edition.

Ravenna, F. (2007). Vector autoregressions and reduced form representations of dsge models. *Journal of Monetary Economics*, 54(7):2048–2064.

A Replication Files and Code

Detail where the reader can find your code (e.g., a GitHub link).

B Additional Tables

B.1 DSGE Parameters Table: all lags

Table 1: DSGE Structural Parameters Comparison Across Lag Lengths

Parameter	**VAR** $p = 4$	**VAR** $p = 6$	**VAR** $p = 8$	**VAR** $p = 10$	**VAR** $p = 12$
			Structural Estimated Frictions		
Price markup (λ_f)	1.0100	1.0100	1.0100	1.0100	1.0100
	(-)	(-)	(-)	(-)	(-)
Wage stickiness (ξ_w)	0.7221***	0.9269***	0.5617***	0.6918***	0.4838***
	(0.1506)	(0.0705)	(0.1142)	(0.1276)	(0.0757)
Phillips curve slope (γ)	0.0402	0.1519	0.0402	0.2827	0.0129***
	(0.0246)	(0.2283)	(0.0598)	(0.3277)	(0.0042)
Capacity util. cost (σ_a)	1.9938	10.5446	0.1175	0.0853	0.0042
	(2.6066)	(27.8459)	(0.6299)	(0.2326)	(0.0230)
Habit formation (b)	0.7066***	0.7183***	0.6935***	0.7802***	0.8596***
	(0.0463)	(0.0573)	(0.0570)	(0.0407)	(0.0203)
Inv. adj. cost (S'')	3.2807*	2413.4944	3.0372*	10.0700	6.8431***
	(1.8874)	(7400.6891)	(1.5506)	(7.2363)	(2.3306)
Money demand elast. (ε)	0.8077***	0.8089***	0.7531***	0.9409***	0.8346***
	(0.2111)	(0.1996)	(0.1473)	(0.2900)	(0.1944)
			Implied Price Stickiness		
Homogeneous capital ($\xi_{p,hom}$)	0.8213	0.6809	0.8213	0.5928	0.8958
	(-)	(-)	(-)	(-)	(-)
Firm-specific capital ($\xi_{p,fs}$)	0.3372	0.1055	0.6448	0.3655	0.8884
	(-)	(-)	(-)	(-)	(-)

Investment adjustment cost in VAR with 6 lags has a point estimate of 2413 with a standard deviation of 7400. This is a good result in the sense that matches perfectly the ACEL2005 working paper, they estimate the baseline model with a VAR with 6 lags as robustness. The second thing that we can learn from this is that probably the likelihood function of this parameter int he underlying DSGE model, when matched with a VAR 6 is complitely flat. The maximum likelihood estimator is not able to find a maximum point

B.2 Implied Duration from Calvo Price Table: all lags

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B.3 Implied Duration from Calvo Price Table: lags 4 8 12

Table 2: Implied Average Time between Re-Optimization

Model Specification	**VAR** $p = 4$	**VAR** $p = 8$	**VAR** $p = 12$
Homogeneous Capital Model			
Calvo parameter ($\xi_{p,hom}$)	0.82	0.82	0.90
Implied duration (quarters)	**5.6**	**5.6**	**9.6**
Firm-Specific Capital Model			
Calvo parameter ($\xi_{p,fs}$)	0.34	0.64	0.89
Implied duration (quarters)	**1.5**	**2.8**	**9.0**

- **Baseline VAR** ($p = 4$): Homogeneous capital implies price changes every **5.6 quarters**.
- **Firm-specific capital**: Drastically reduces this to **1.5 quarters**, matching microeconomic evidence.

Note: Implied duration is calculated as $D = 1/(1 - \xi_p)$.

B.4 VAR ALIGNMENT DSGE Parameters Table: all lags

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Phillips Curve Slope in VAR 10 is a monster, is 0.6 omplitely messed up.

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Results are robust to this exercise, no major changes, but this exercise exacerbates the Phillips Curve Slope of the DSGE when estimated matching with the VAR 6 and expecially 10. When using the VAR 10 the DSGE Phillips curve is tremendously hig 0.6 –¿ this is the driver of the low implied time from the calvo parameter, hence the ACEL result seems to to be robust by increasing the lag choice, an expecially there are some problems with the Phillips curve when chooising 6 and 10 lags

B.5 Decomposition of Variance across different lags

C Additional Figures

C.1 ACEL 2005 VAR Variables

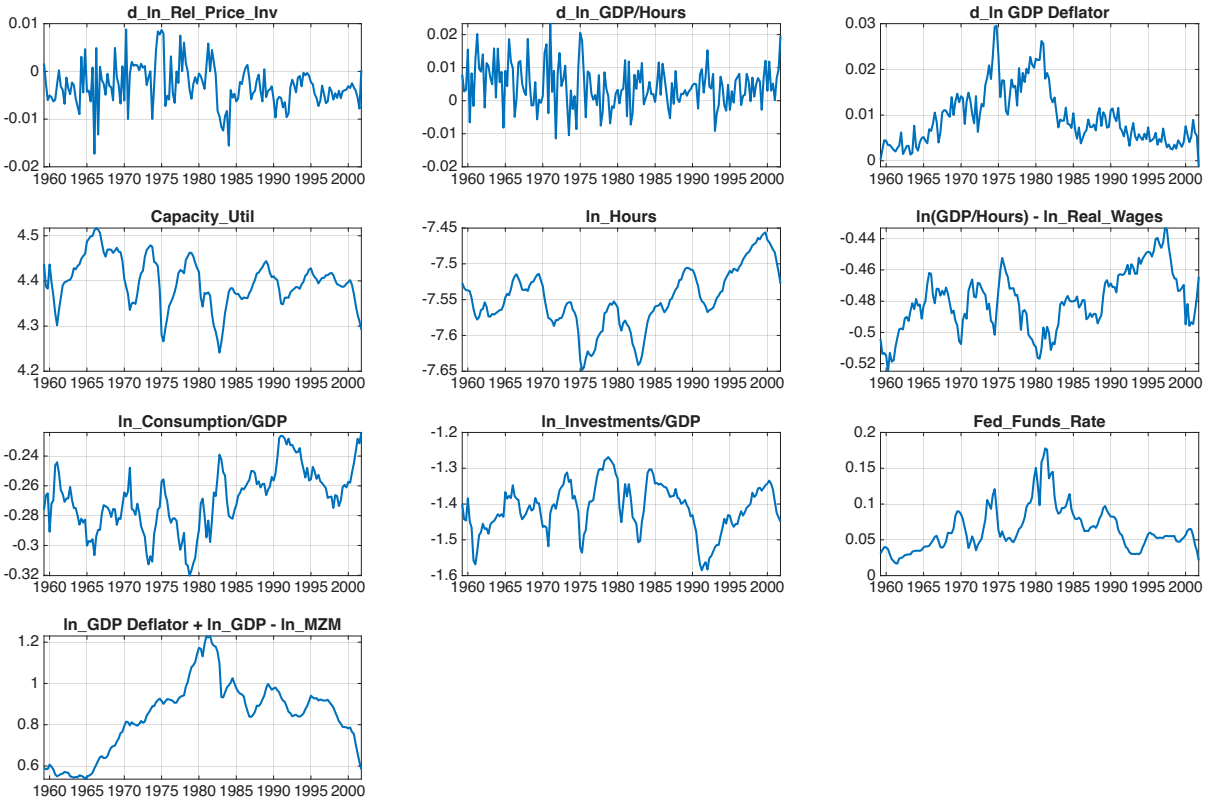


Figure 1: Raw Historical Series

C.2 ACEL 2005 VAR Across different lag structures: Monetary Policy Shock

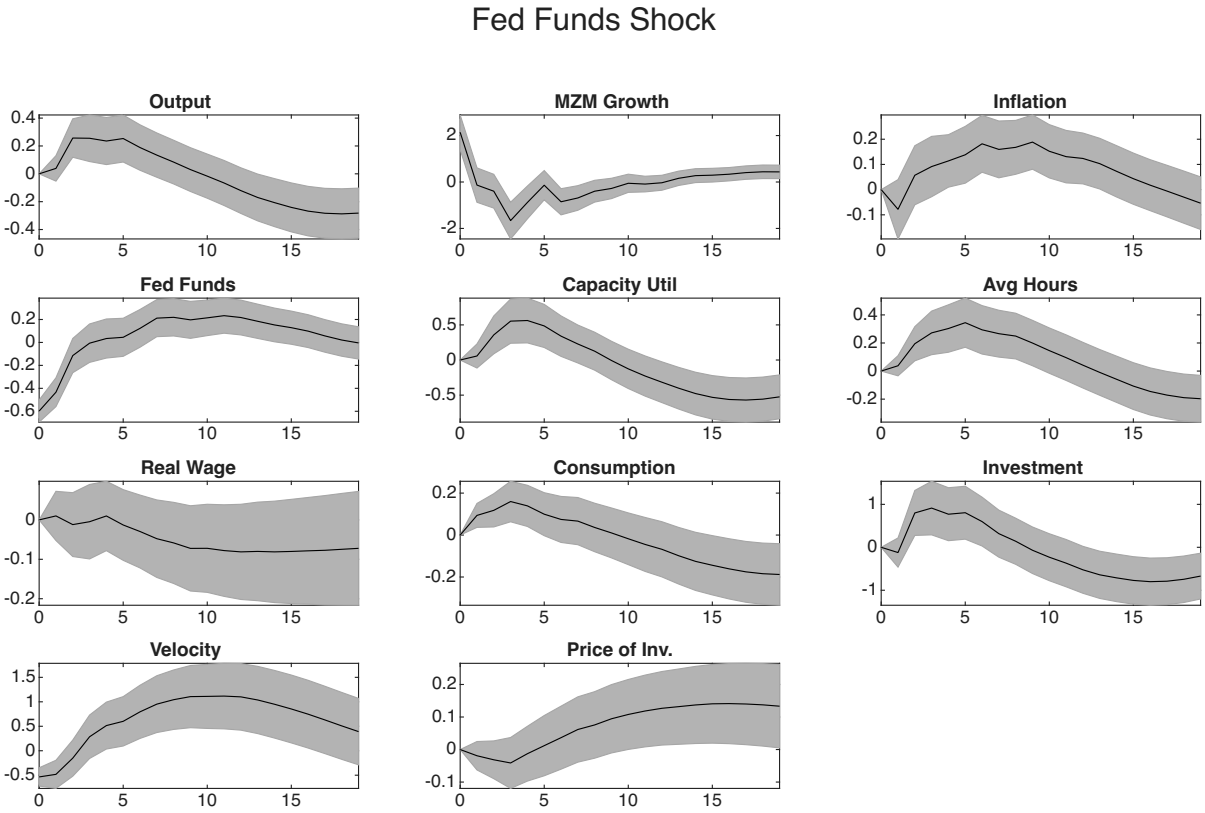


Figure 2: Monetary Policy Shock lag 4

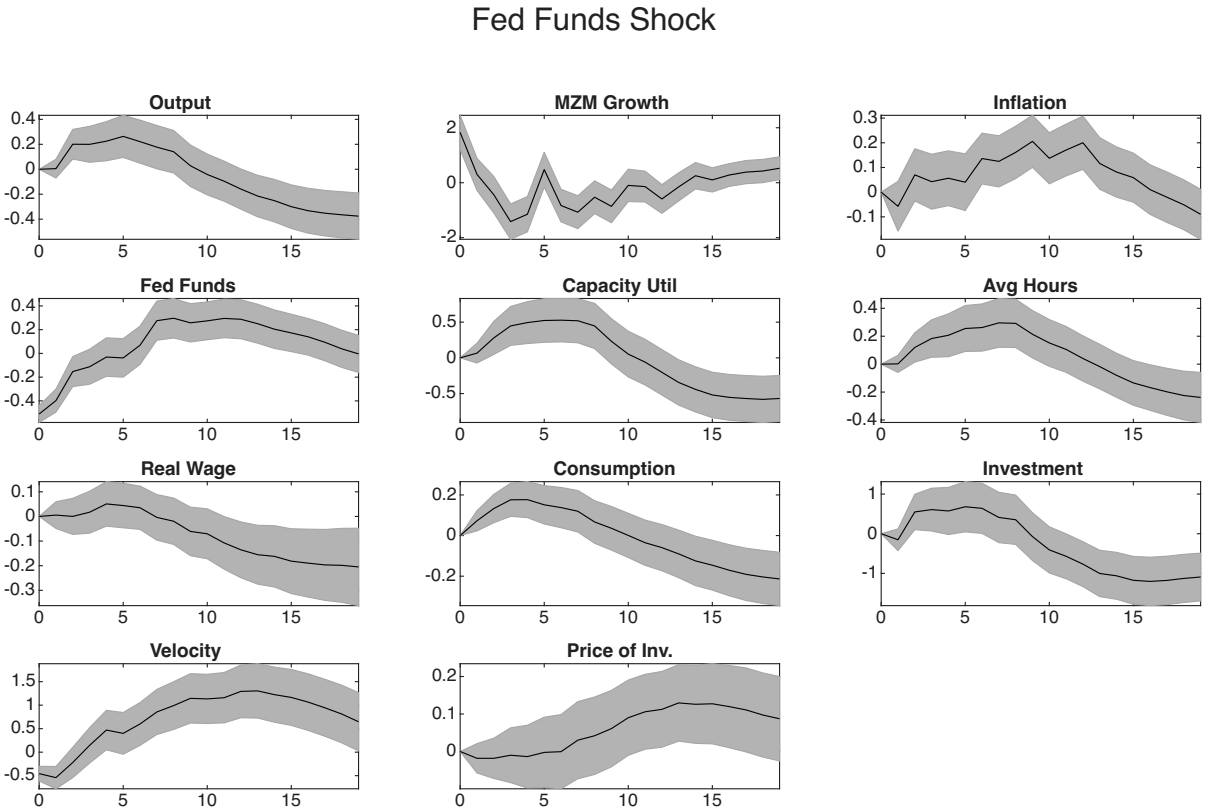


Figure 3: Monetary Policy Shock lag 6

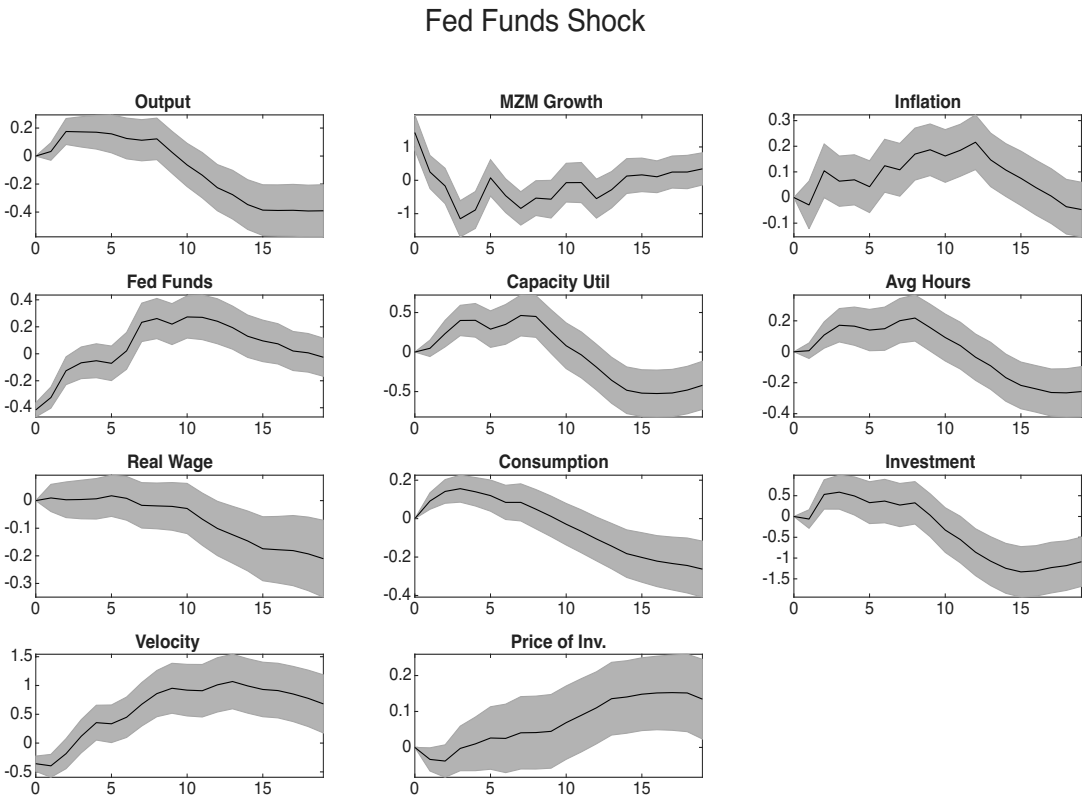


Figure 4: Monetary Policy Shock lag 8

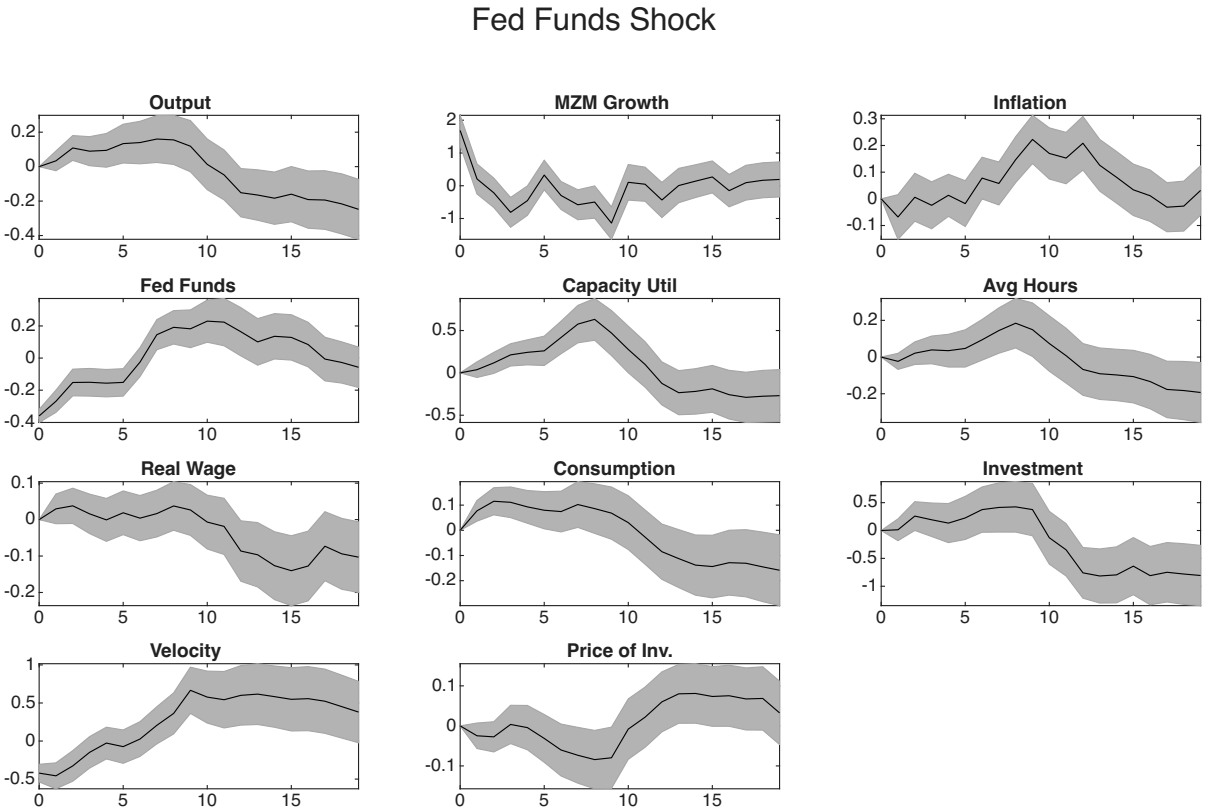


Figure 5: Monetary Policy Shock lag 10

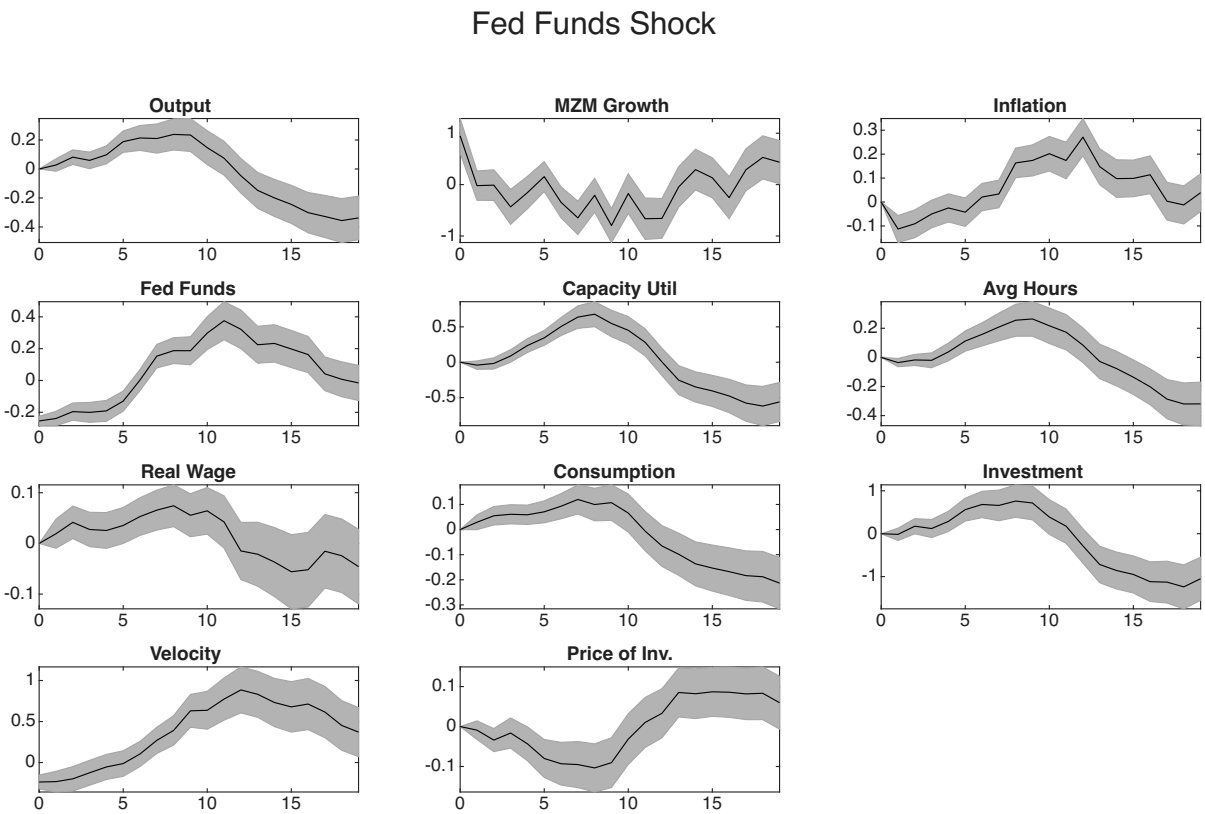


Figure 6: Monetary Policy Shock lag 12

C.3 ACEL 2005 VAR Across different lag structures: Embodied Technology Shock

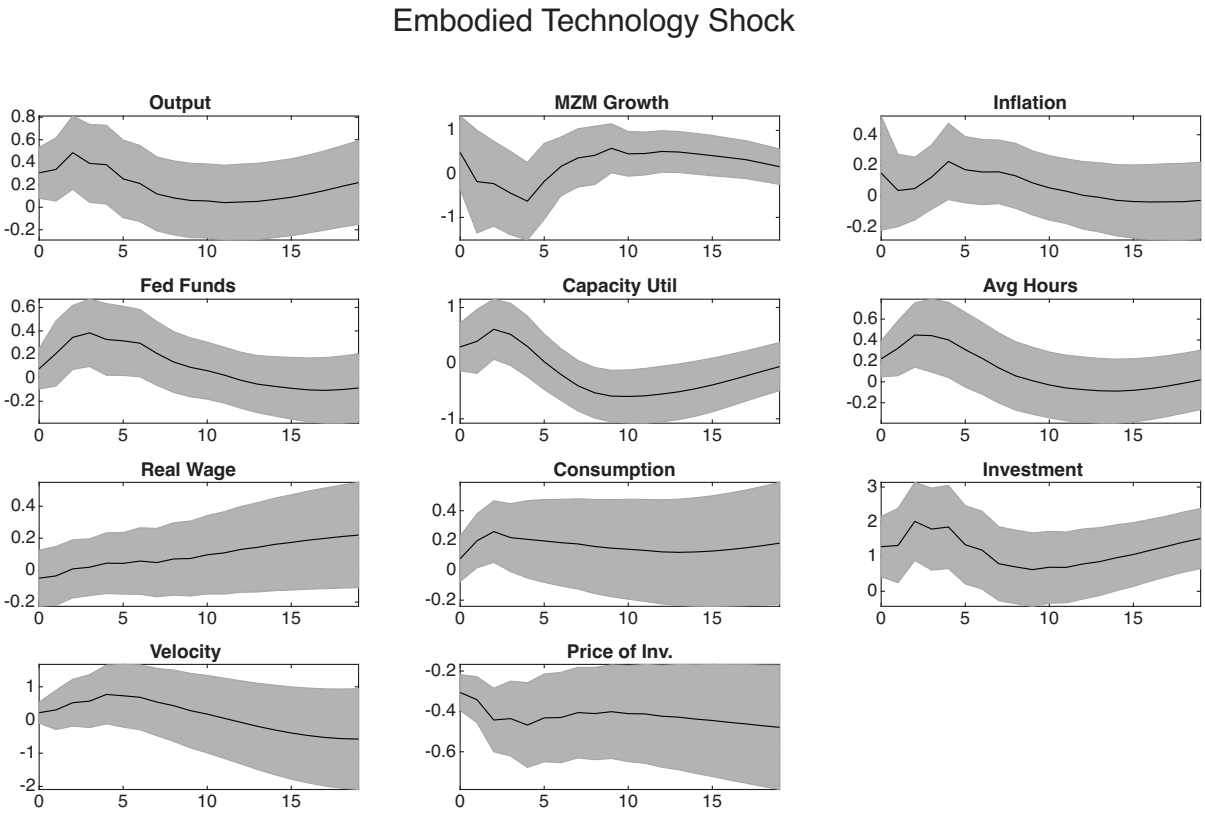


Figure 7: Embodied Technology Shock lag 4

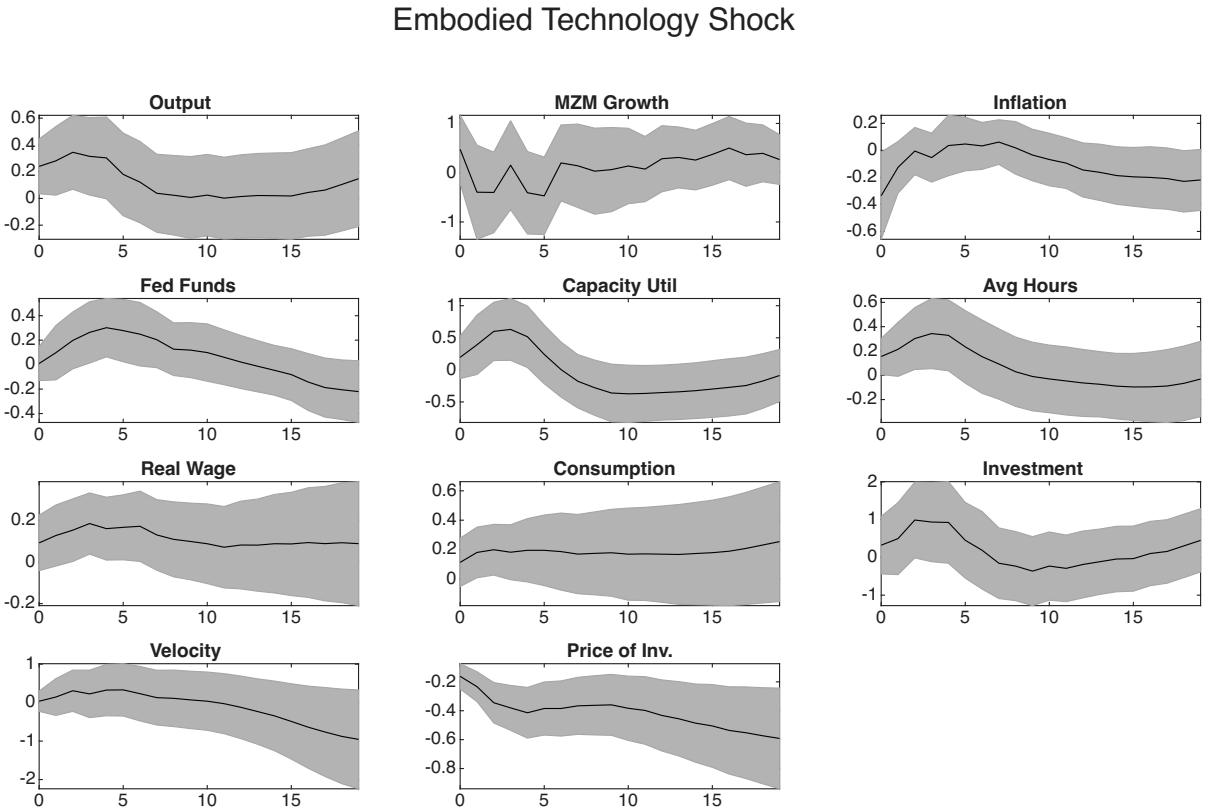


Figure 8: Embodied Technology Shock lag 6

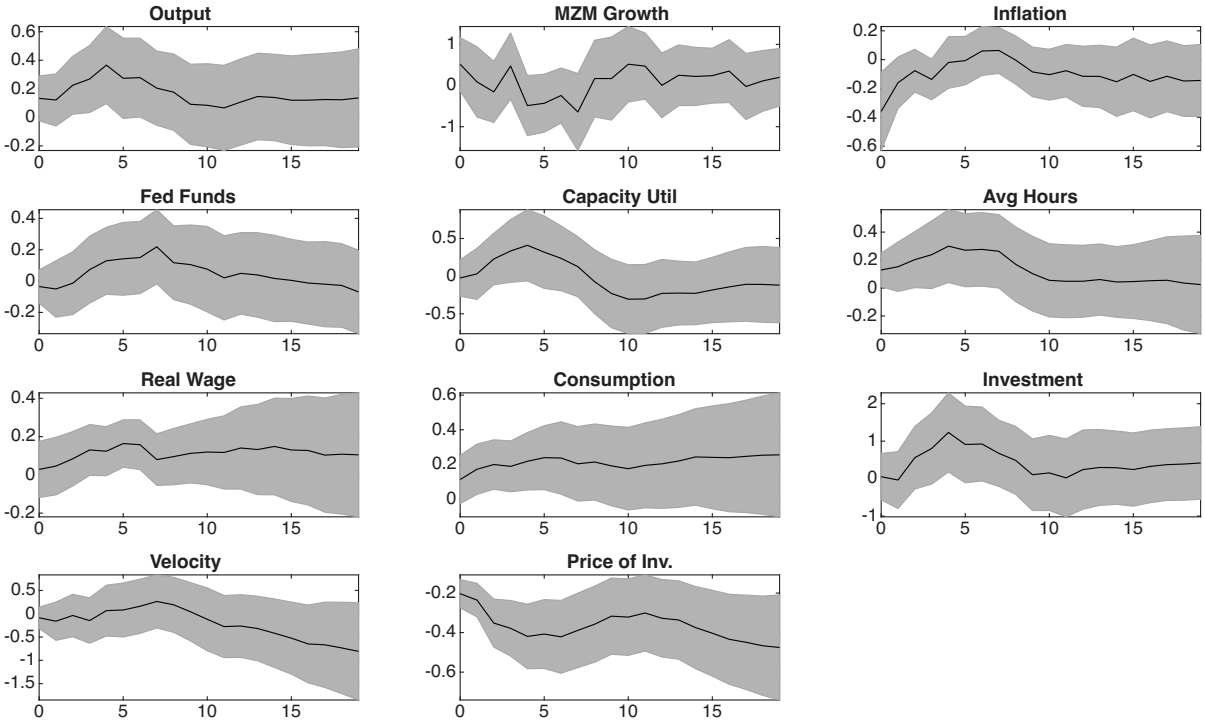


Figure 9: Embodied Technology Shock lag 8

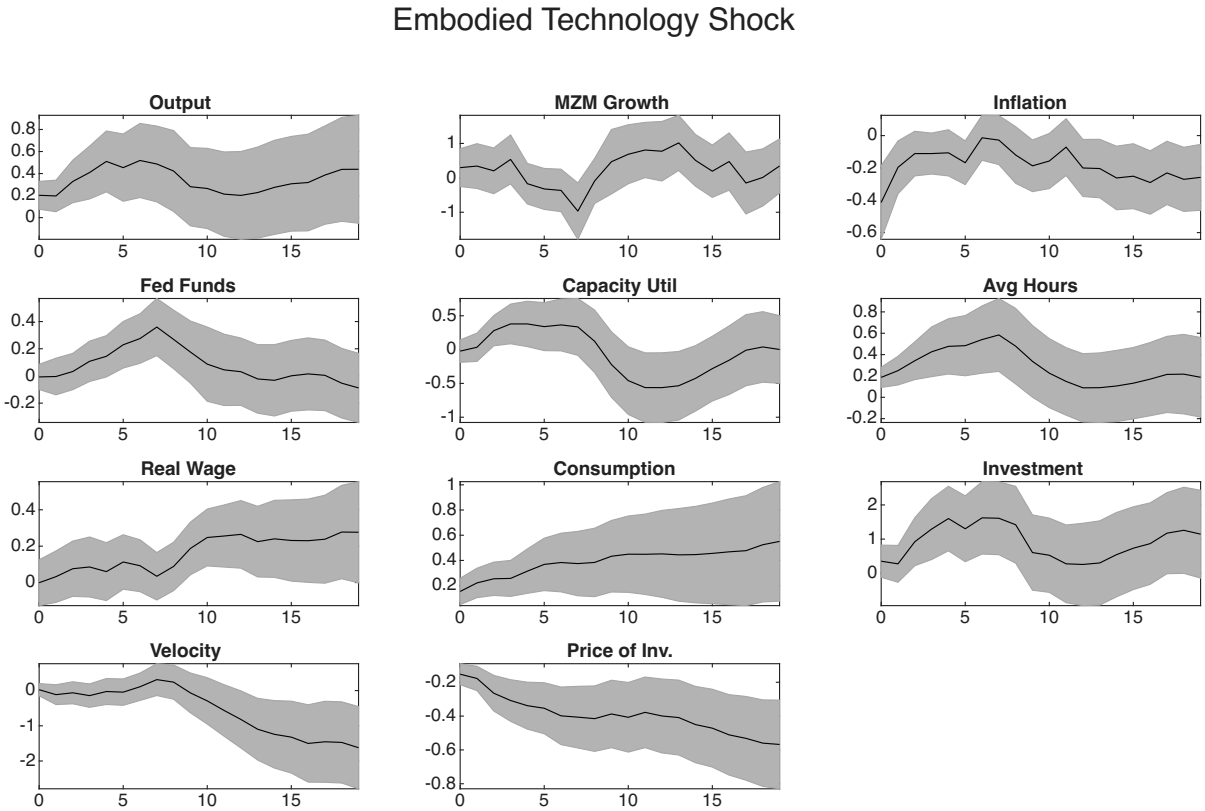


Figure 10: Embodied Technology Shock lag 10

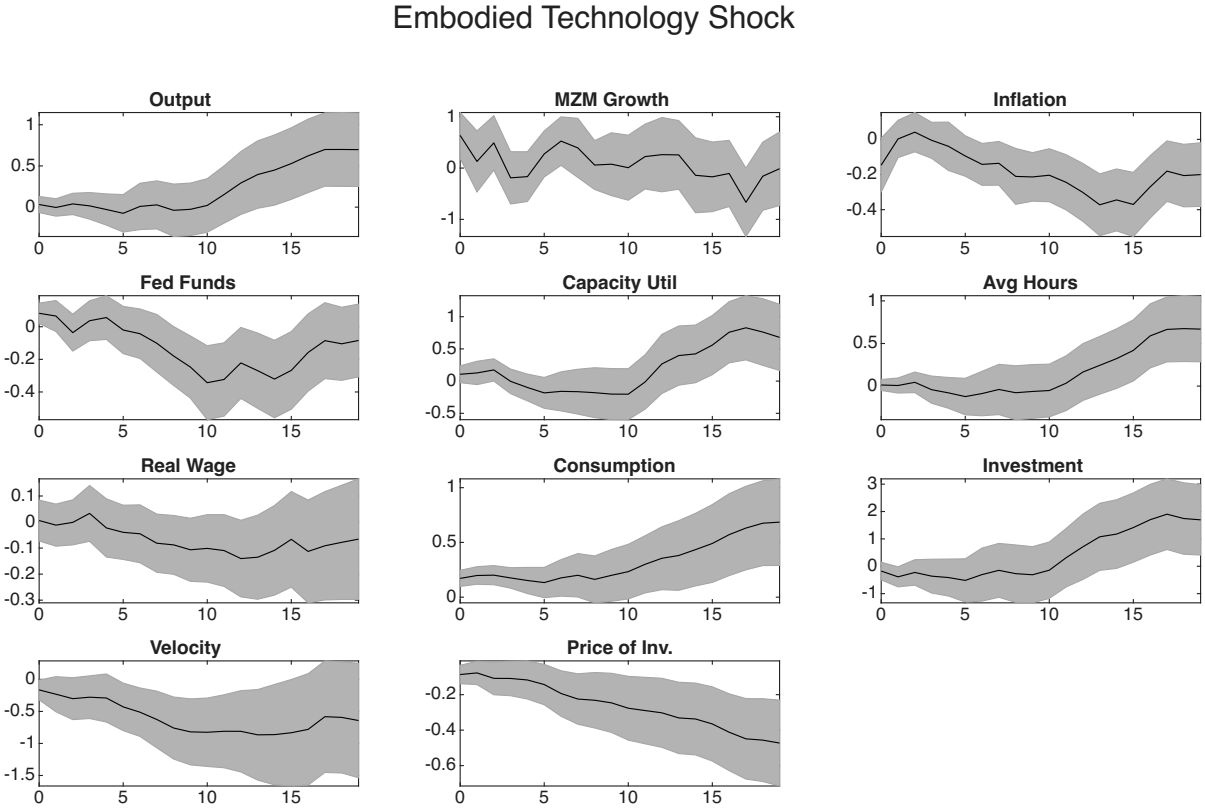


Figure 11: Embodied Technology Shock lag 12

C.4 ACEL 2005 VAR Across different lag structures: Neutral Technology Shock

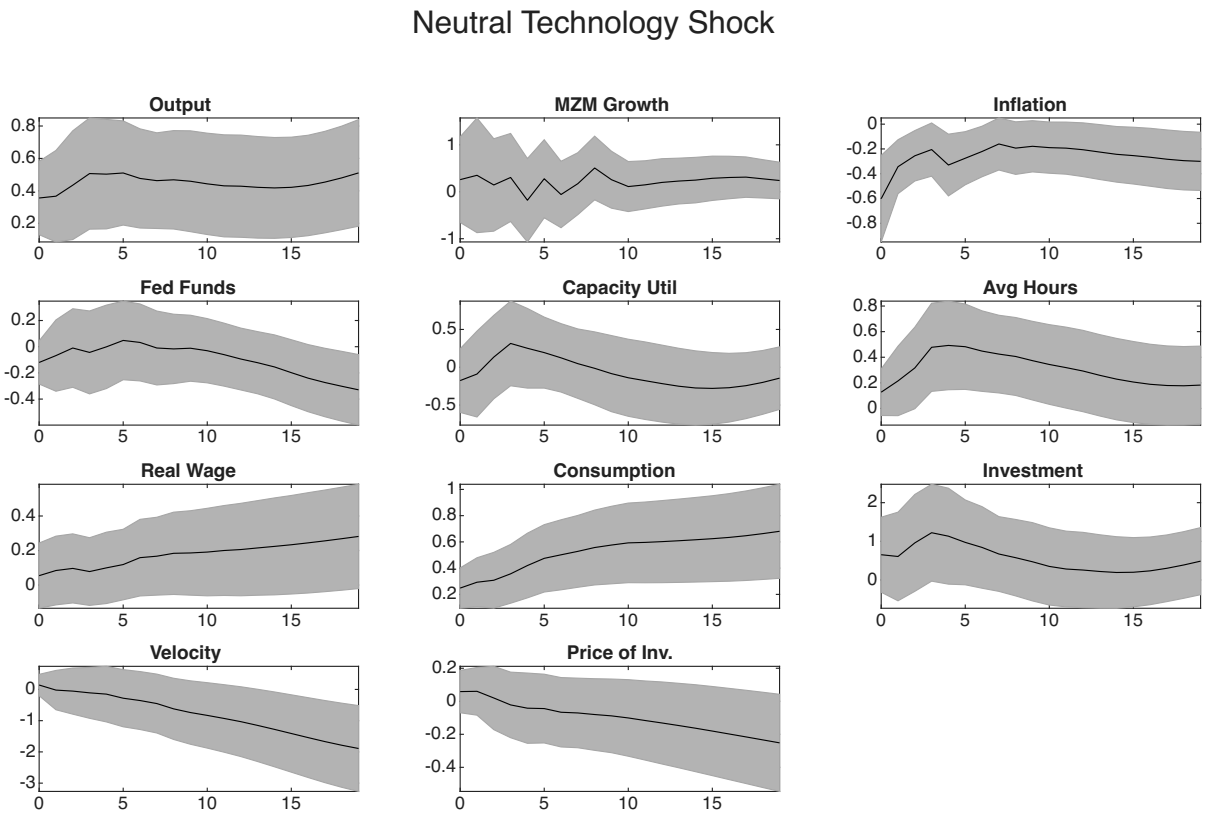


Figure 12: Neutral Technology Shock lag 4

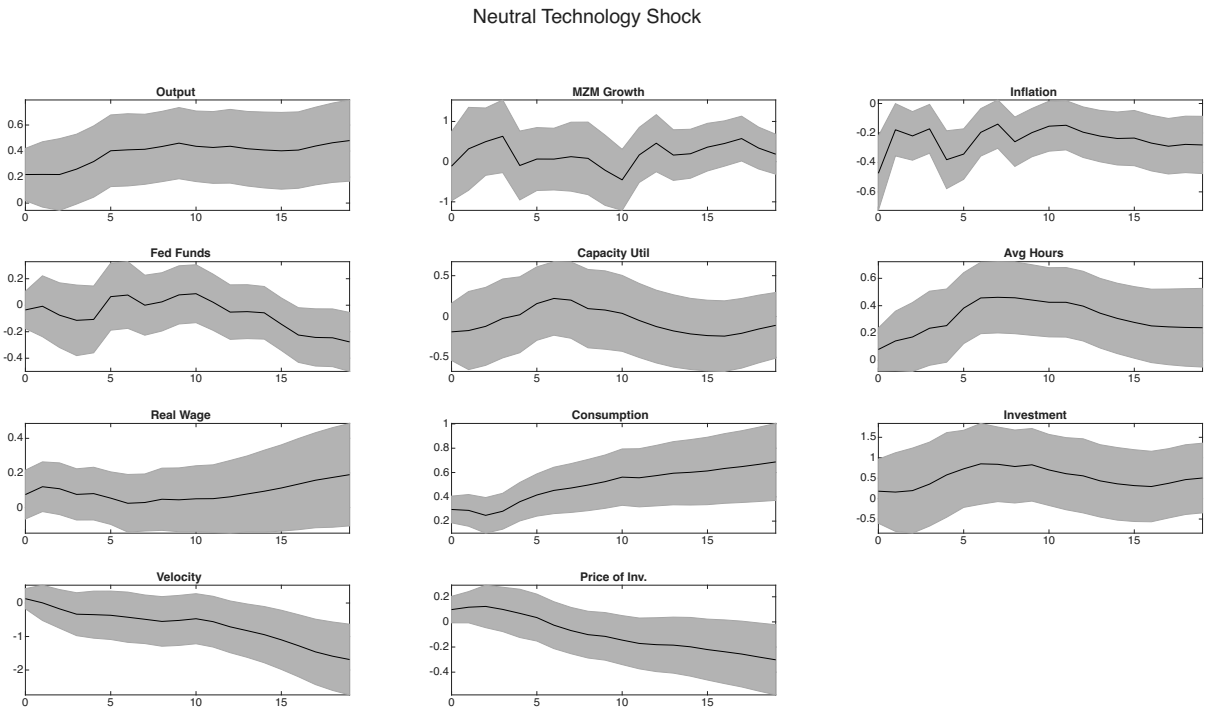


Figure 13: Neutral Technology Shock lag 6

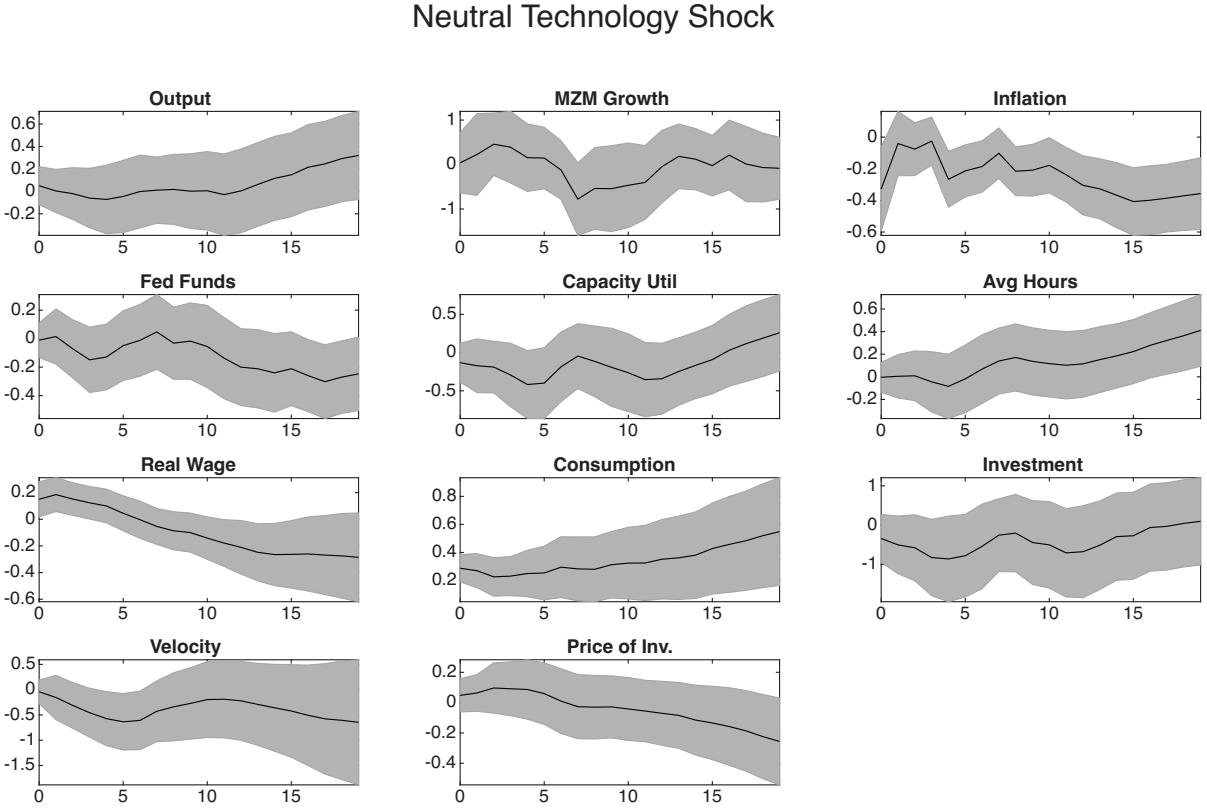


Figure 14: Neutral Technology Shock lag 8

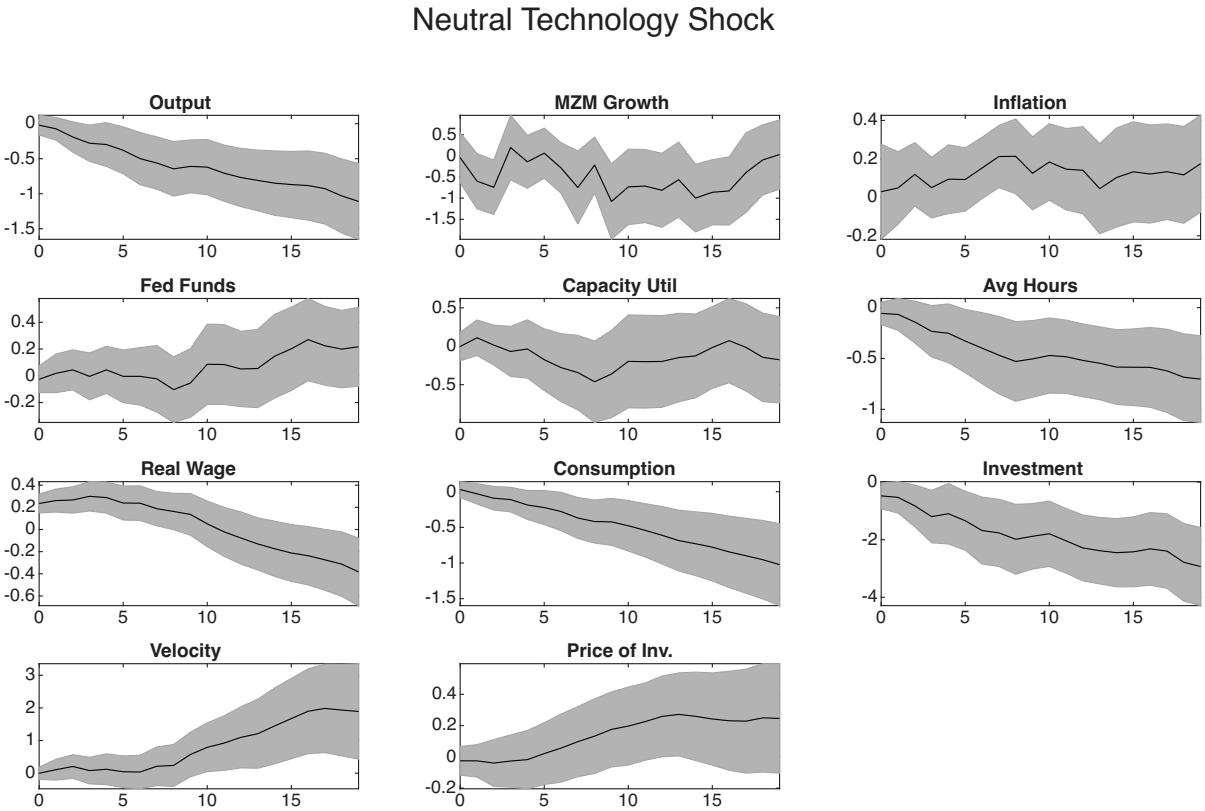


Figure 15: Neutral Technology Shock lag 10

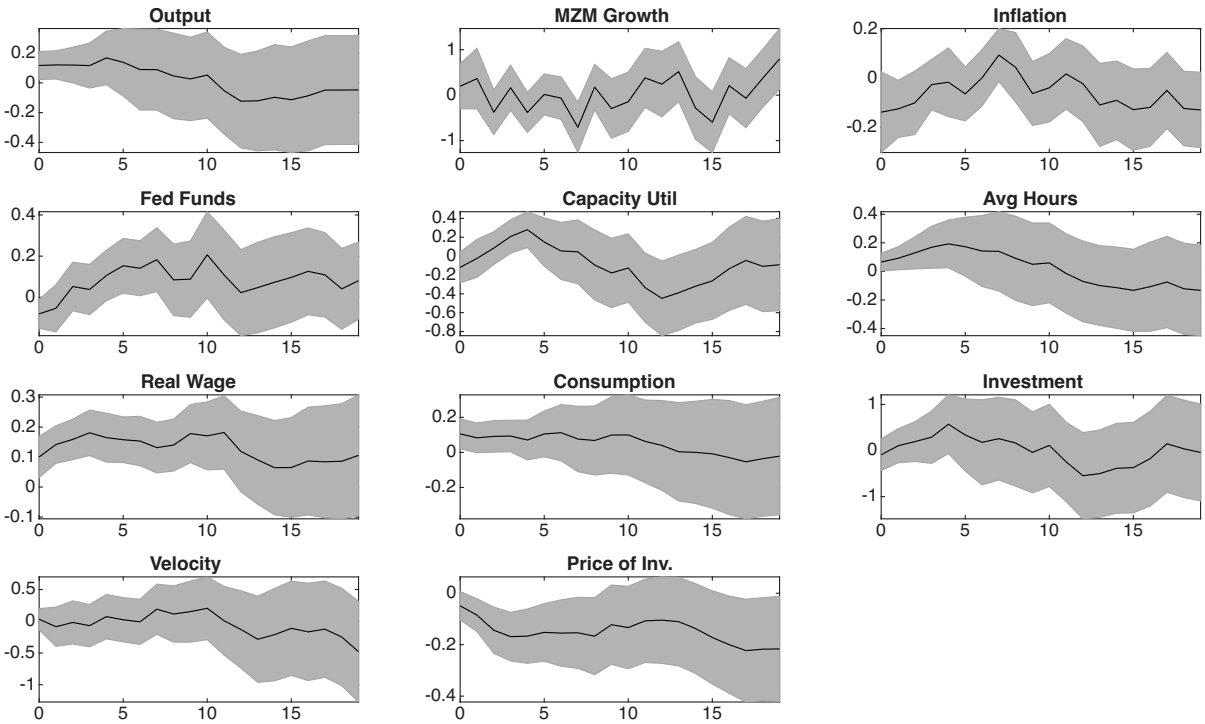


Figure 16: Neutral Technology Shock lag 12

C.5 ACEL 2005 DSGE Matched with VAR 4, compared with VAR estimated across different lag structures: Monetary Policy Shock

Figure 1: Response to a monetary policy shock (o - Model, - VAR, grey area - 95 % Confidence Interval)

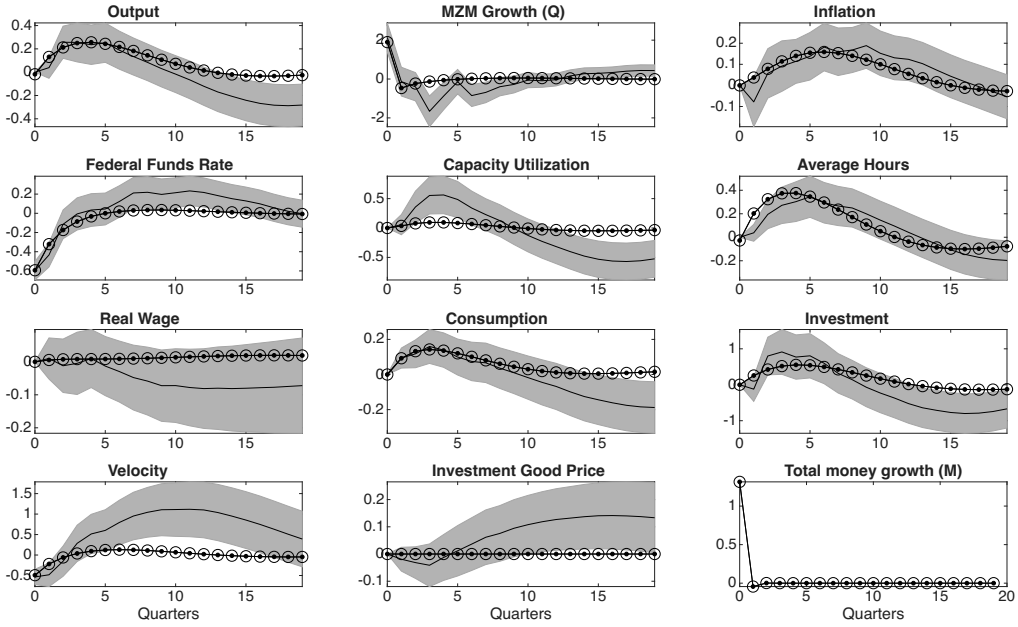


Figure 17: Monetary Policy Shock DSGE matched with VAR lag 4 compared to VAR lag 4

Figure 1: Response to a monetary policy shock (o - Model, - VAR, grey area - 95 % Confidence Interval)

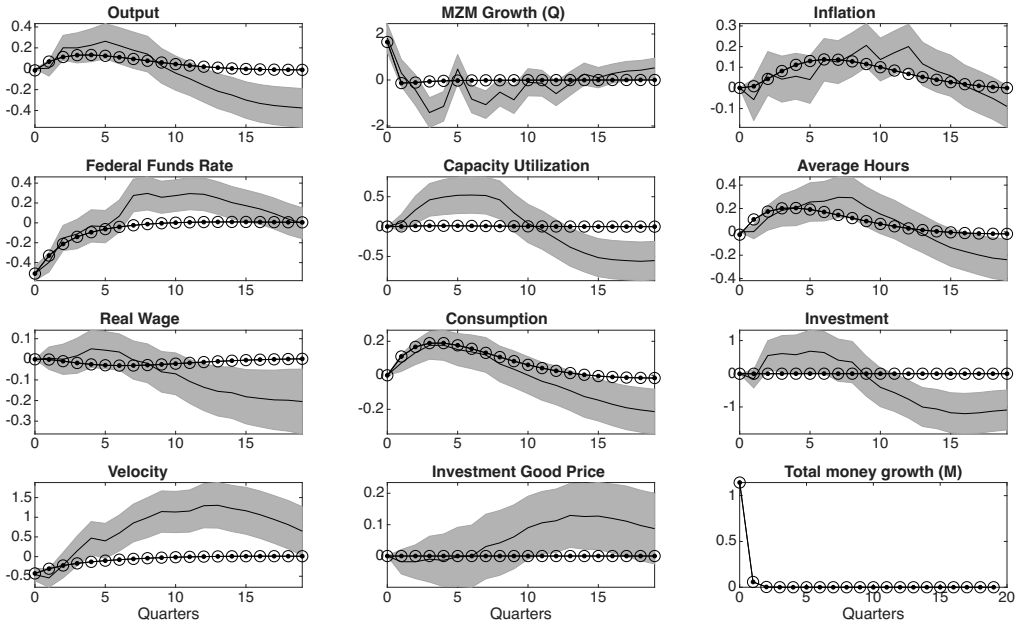


Figure 18: Monetary Policy Shock DSGE matched with VAR lag 4 compared to VAR lag 6

Figure 1: Response to a monetary policy shock (o - Model, - VAR, grey area - 95 % Confidence Interval)

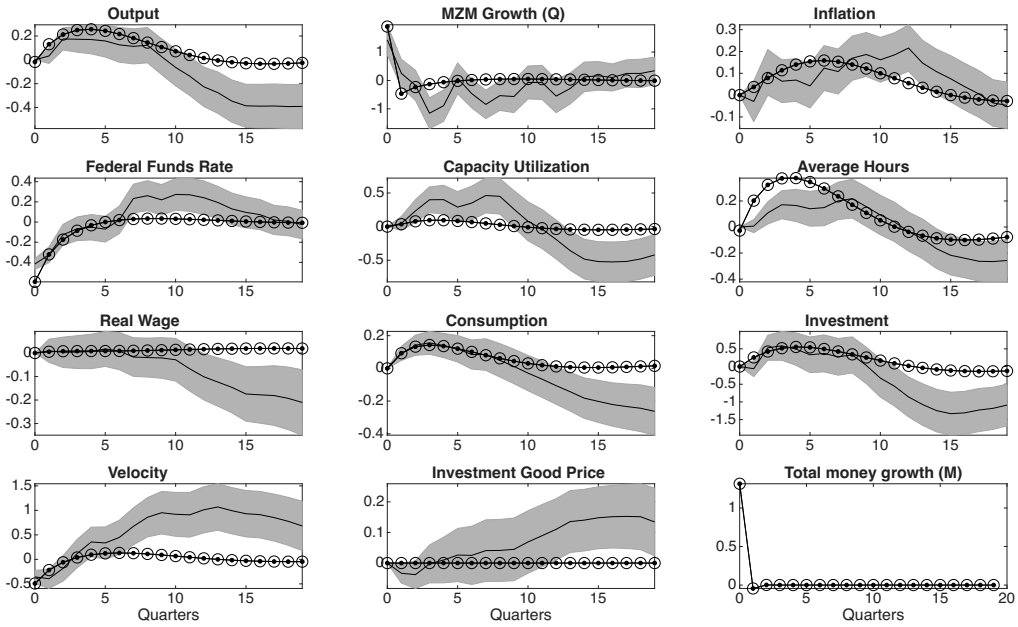


Figure 19: Monetary Policy Shock DSGE matched with VAR lag 4 compared to VAR lag 8

Figure 1: Response to a monetary policy shock (o - Model, - VAR, grey area - 95 % Confidence Interval)

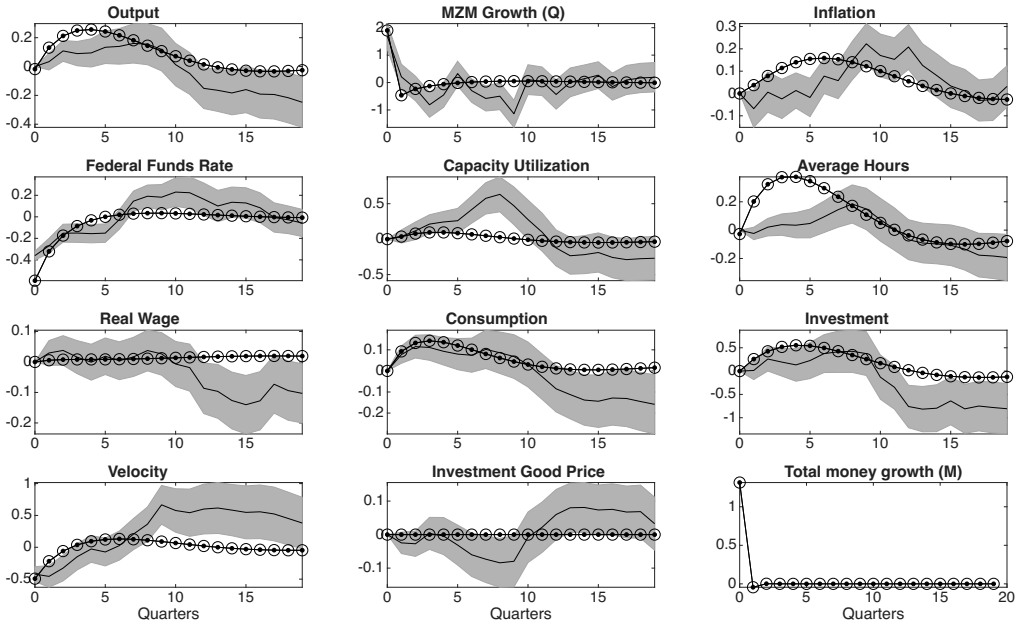


Figure 20: Monetary Policy Shock DSGE matched with VAR lag 4 compared to VAR lag 10

Figure 1: Response to a monetary policy shock (o - Model, - VAR, grey area - 95 % Confidence Interval)

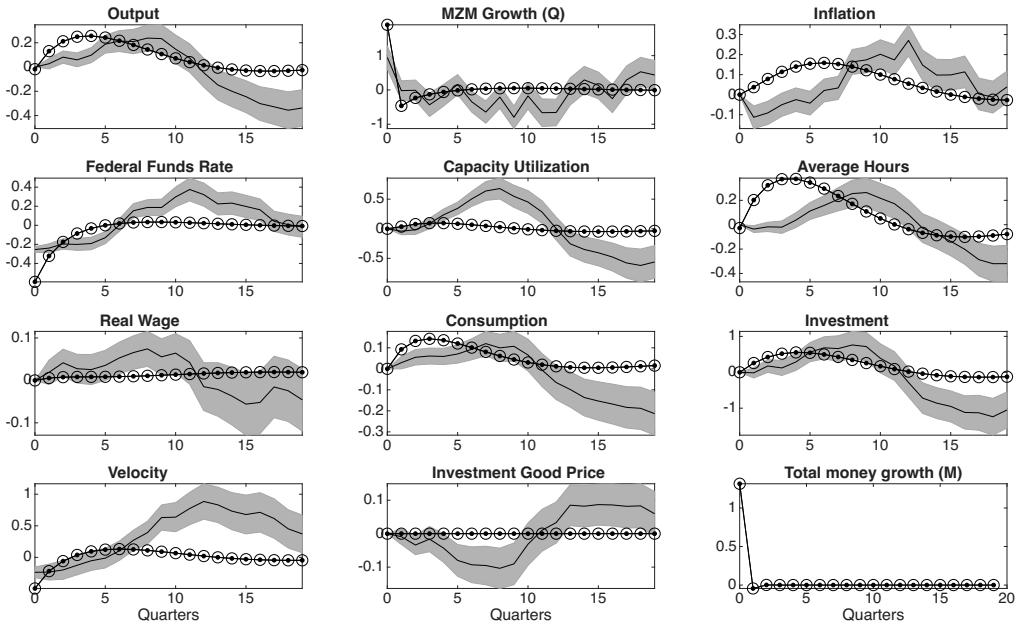


Figure 21: Monetary Policy Shock DSGE matched with VAR lag 4 compared to VAR lag 12

C.6 ACEL 2005 DSGE Matched with VAR 4, compared with VAR estimated across different lag structures: Embodied Technology Shock

Figure 3: Response to an embodied technology shock (o - Model, - VAR, grey area - 95 % Confidence Interval)

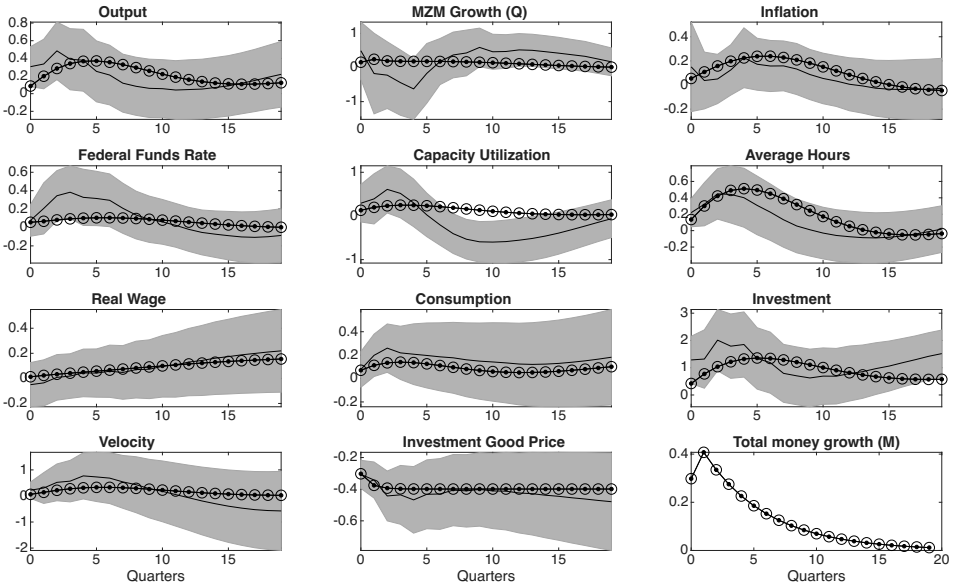


Figure 22: Embodied Technology Shock DSGE matched with VAR lag 4 compared to VAR lag 4

Figure 3: Response to an embodied technology shock (o - Model, - VAR, grey area - 95 % Confidence Interval)

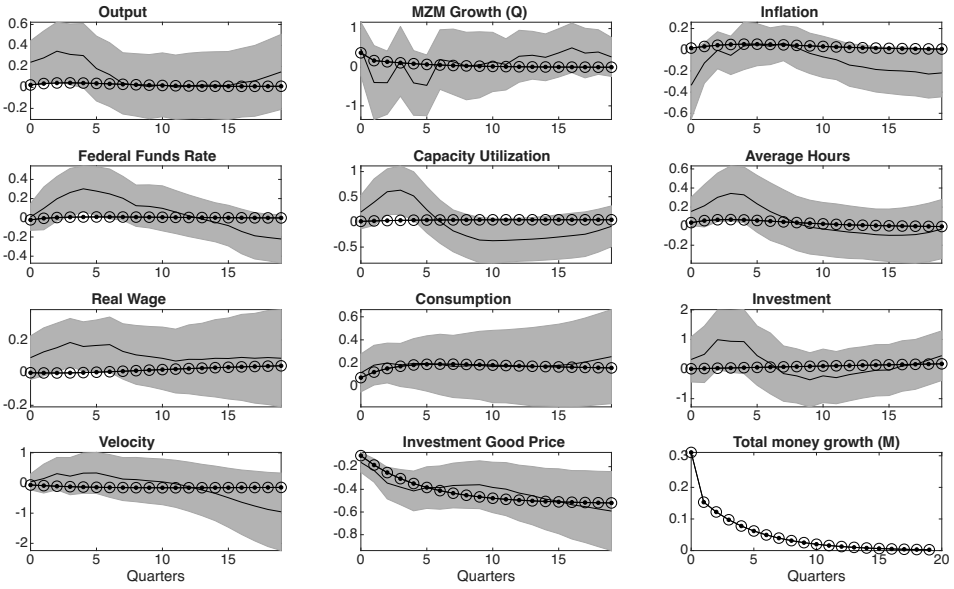


Figure 23: Embodied Technology Shock DSGE matched with VAR lag 4 compared to VAR lag 6

Figure 3: Response to an embodied technology shock (o - Model, - VAR, grey area - 95 % Confidence Interval)

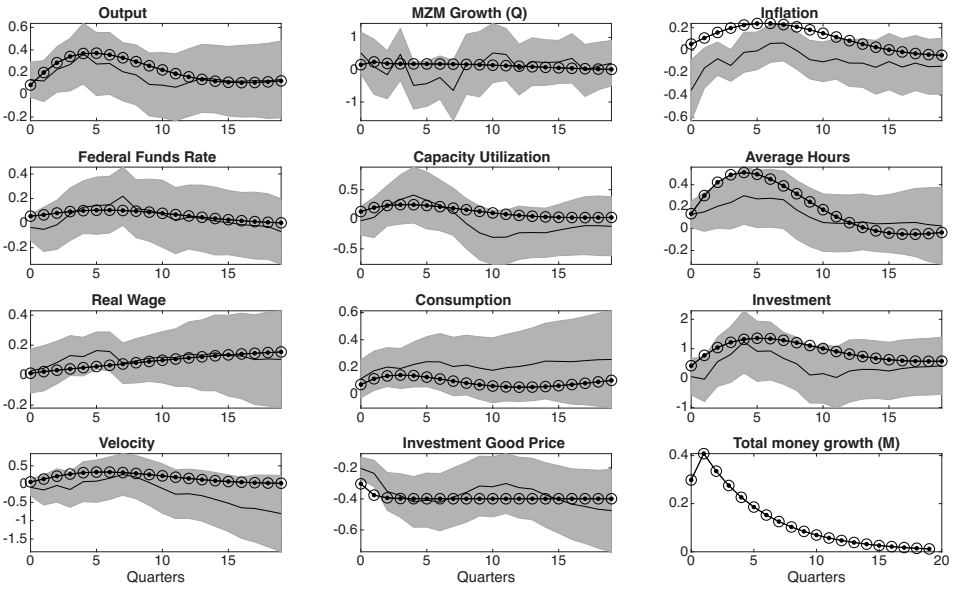


Figure 24: Embodied Technology Shock DSGE matched with VAR lag 4 compared to VAR lag 8

Figure 3: Response to an embodied technology shock (o - Model, - VAR, grey area - 95 % Confidence Interval)

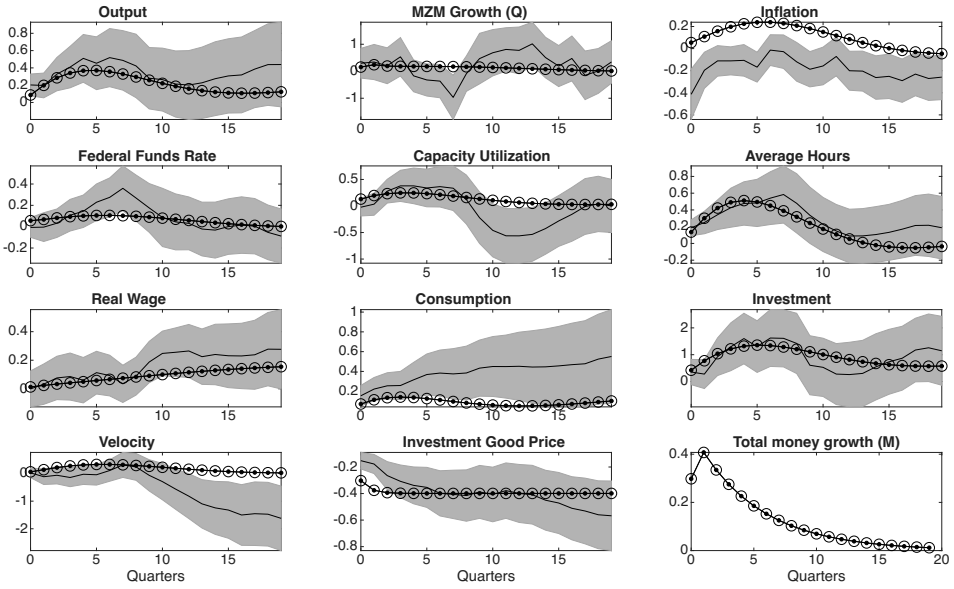


Figure 25: Embodied Technology Shock DSGE matched with VAR lag 4 compared to VAR lag 10

Figure 3: Response to an embodied technology shock (o - Model, - VAR, grey area - 95 % Confidence Interval)

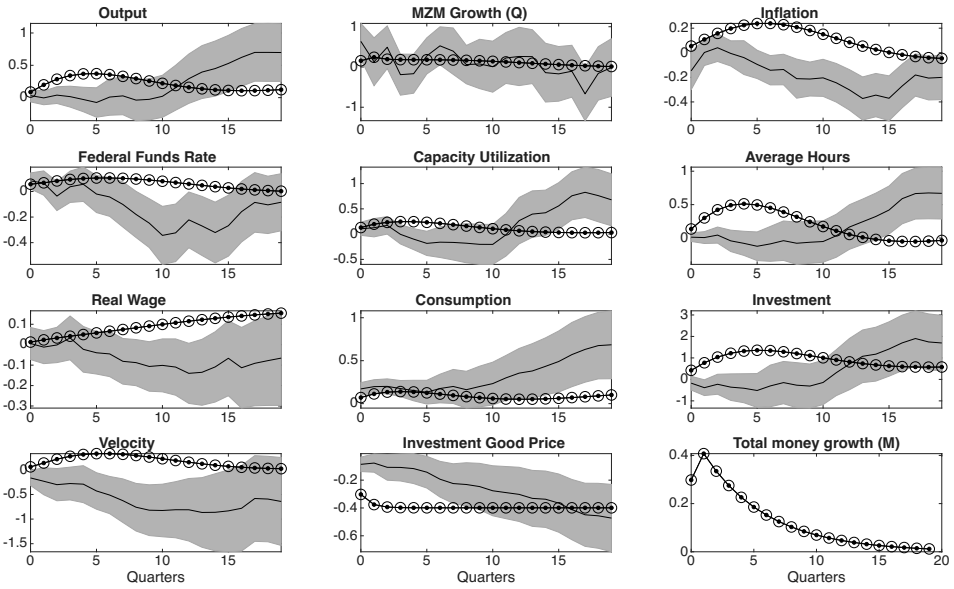


Figure 26: Embodied Technology Shock DSGE matched with VAR lag 4 compared to VAR lag 12

C.7 ACEL 2005 DSGE Matched with VAR 4, compared with VAR estimated across different lag structures: Neutral Technology Shock

Figure 2: Response to a neutral technology shock (o - Model, - VAR, grey area - 95 % Confidence Interval)

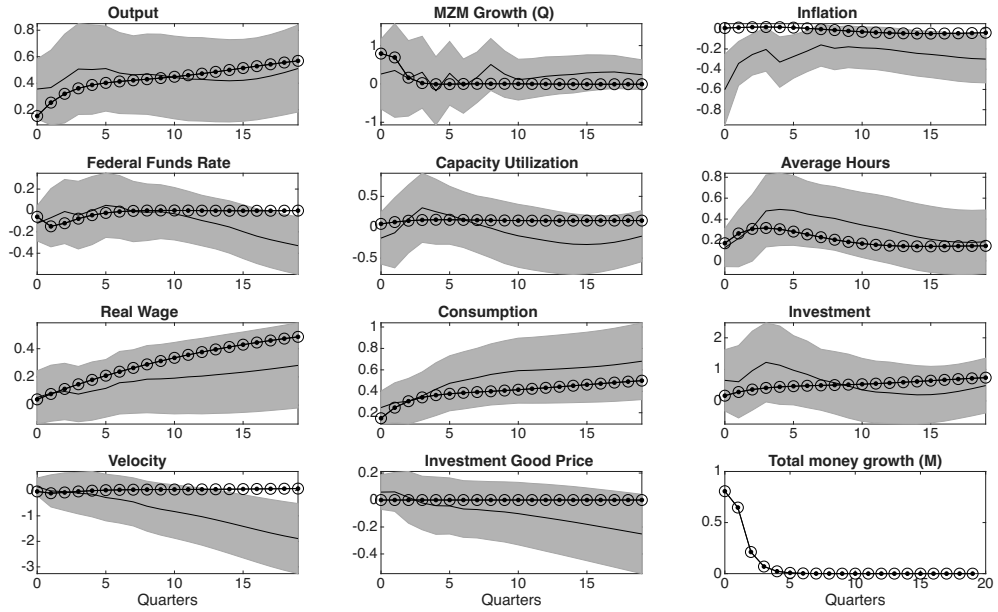


Figure 27: Neutral Technology Shock DSGE matched with VAR lag 4 compared to VAR lag 4

Figure 2: Response to a neutral technology shock (o - Model, - VAR, grey area - 95 % Confidence Interval)

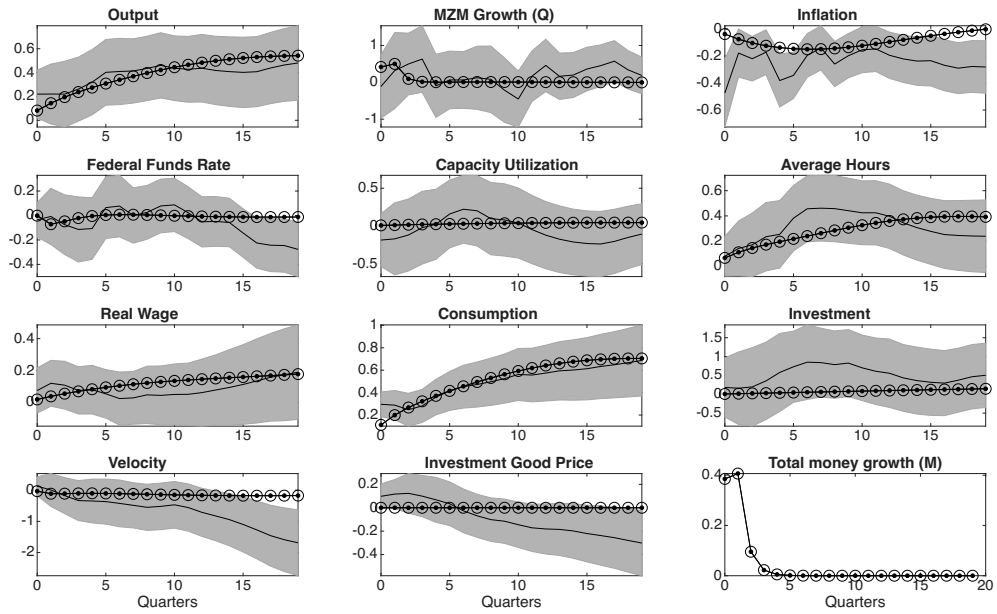


Figure 28: Neutral Technology Shock DSGE matched with VAR lag 4 compared to VAR lag 6

Figure 2: Response to a neutral technology shock (o - Model, - VAR, grey area - 95 % Confidence Interval)

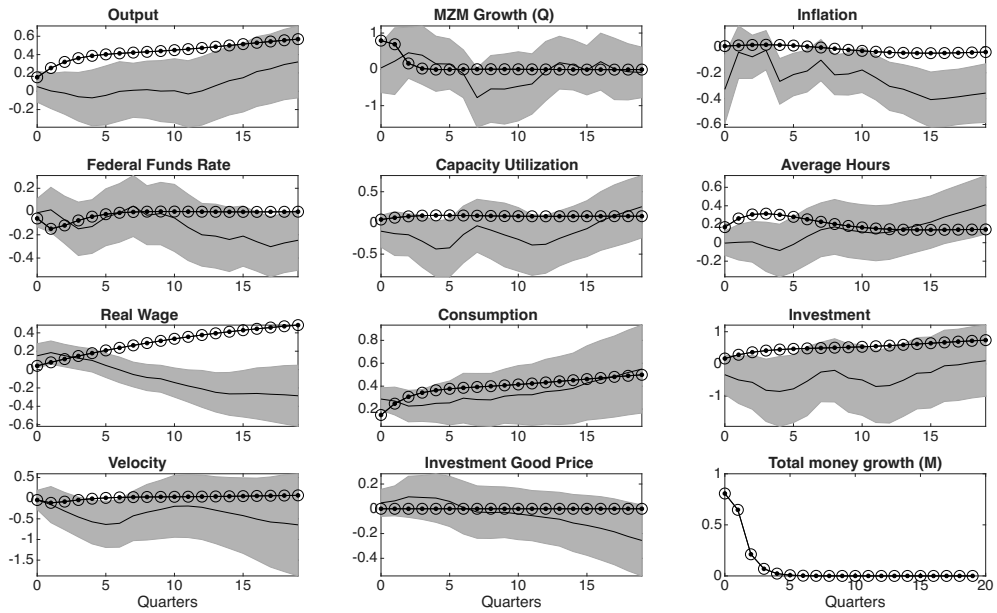


Figure 29: Neutral Technology Shock DSGE matched with VAR lag 4 compared to VAR lag 8

Figure 2: Response to a neutral technology shock (o - Model, - VAR, grey area - 95 % Confidence Interval)

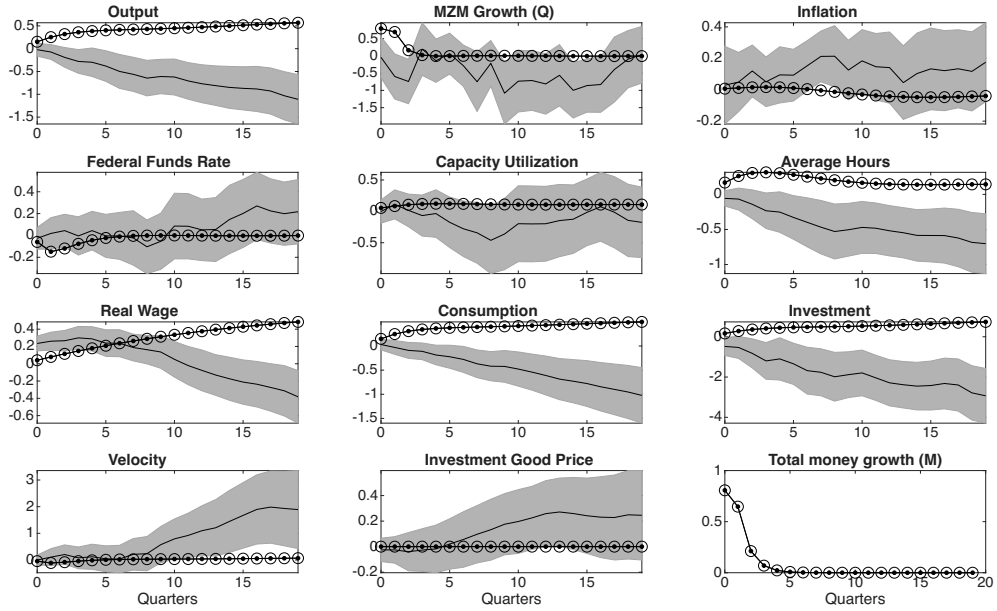


Figure 30: Neutral Technology Shock DSGE matched with VAR lag 4 compared to VAR lag 10

Figure 2: Response to a neutral technology shock (o - Model, - VAR, grey area - 95 % Confidence Interval)

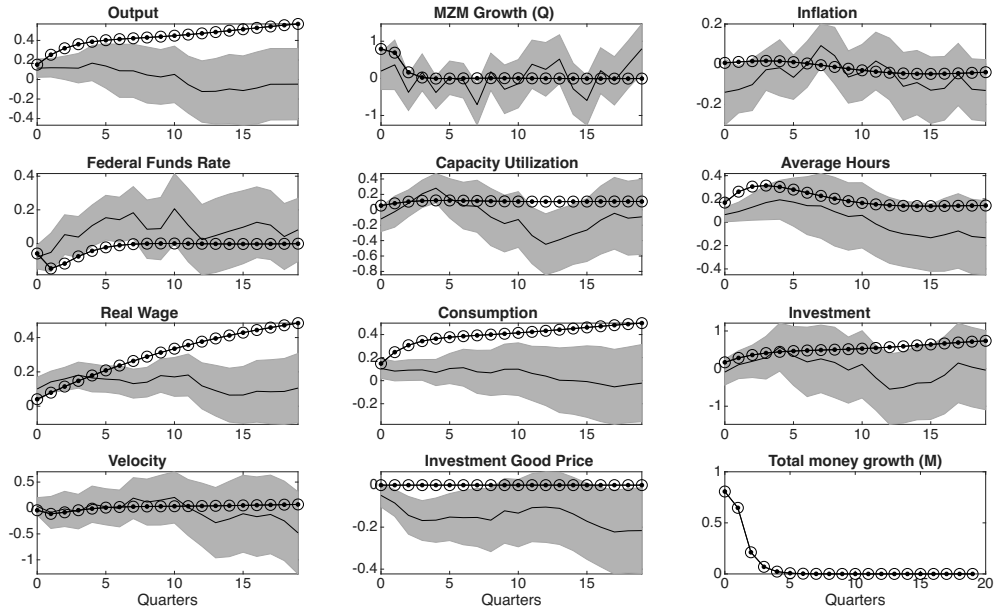


Figure 31: Neutral Technology Shock DSGE matched with VAR lag 4 compared to VAR lag 12

C.8 ACEL 2005 DSGE Matched and compared with VAR estimated across different lag structures: Monetary Policy Shock

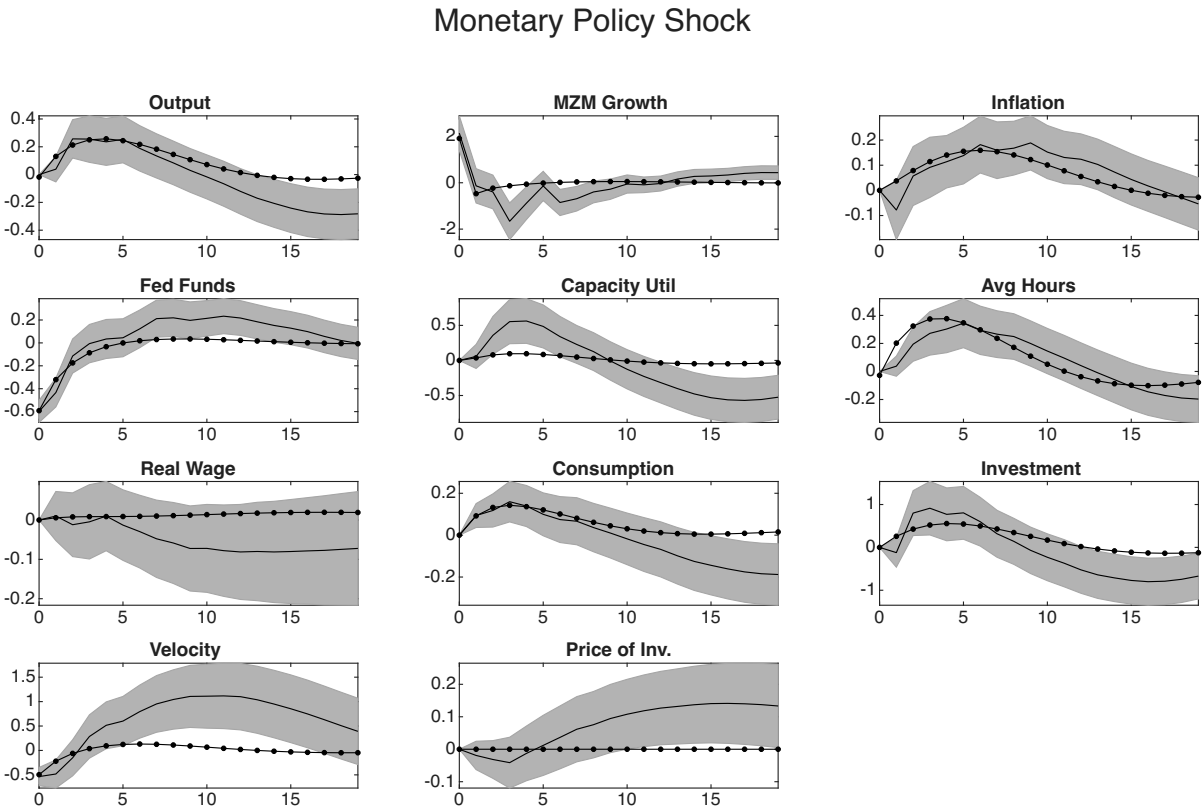


Figure 32: Monetary Policy Shock DSGE matched with VAR lag 4 compared to VAR lag 4

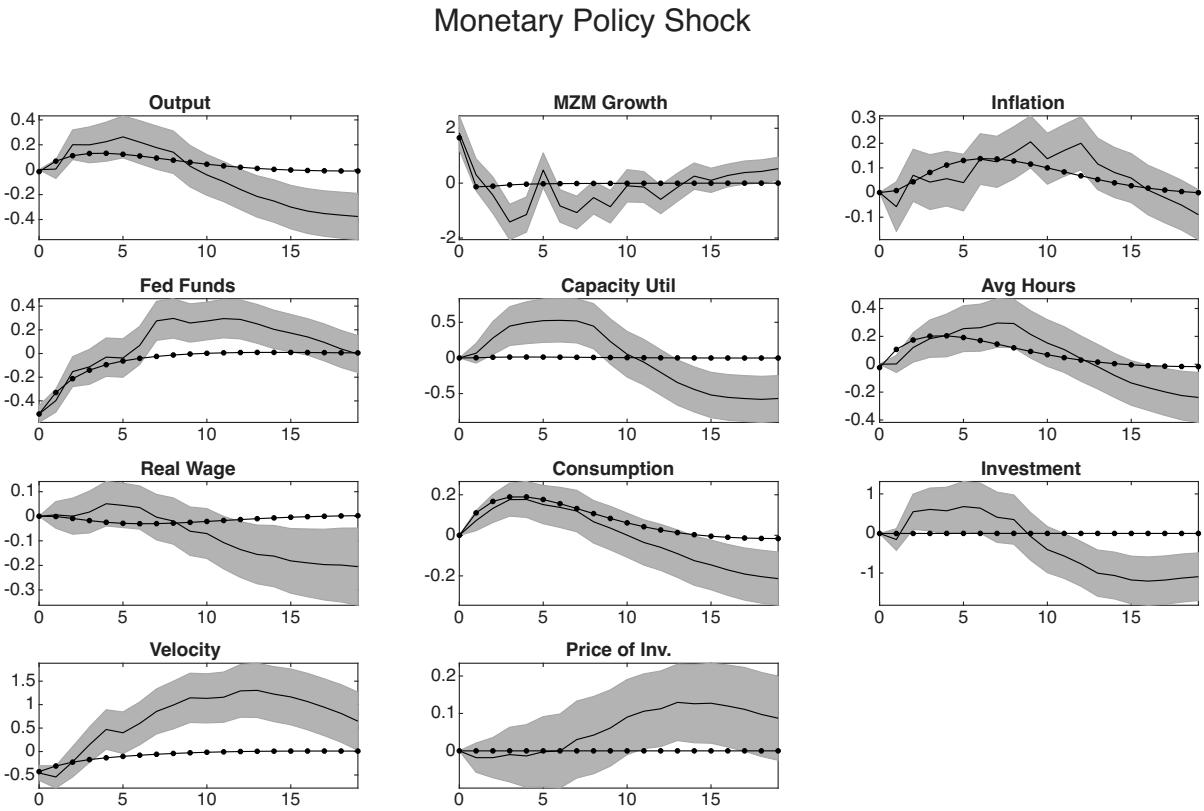


Figure 33: Monetary Policy Shock DSGE matched with VAR lag 6 compared to VAR lag 6

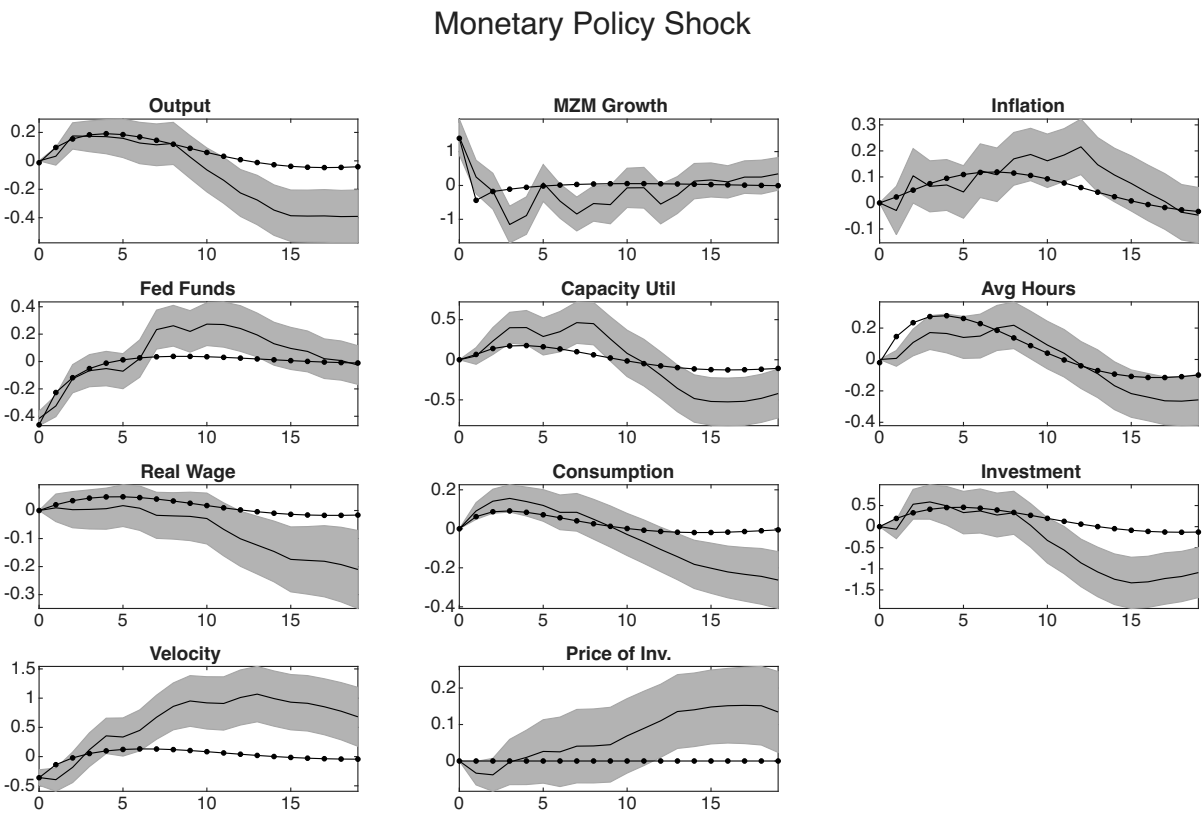


Figure 34: Monetary Policy Shock DSGE matched with VAR lag 8 compared to VAR lag 8

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Figure 35: Monetary Policy Shock DSGE matched with VAR lag 10 compared to VAR lag 10

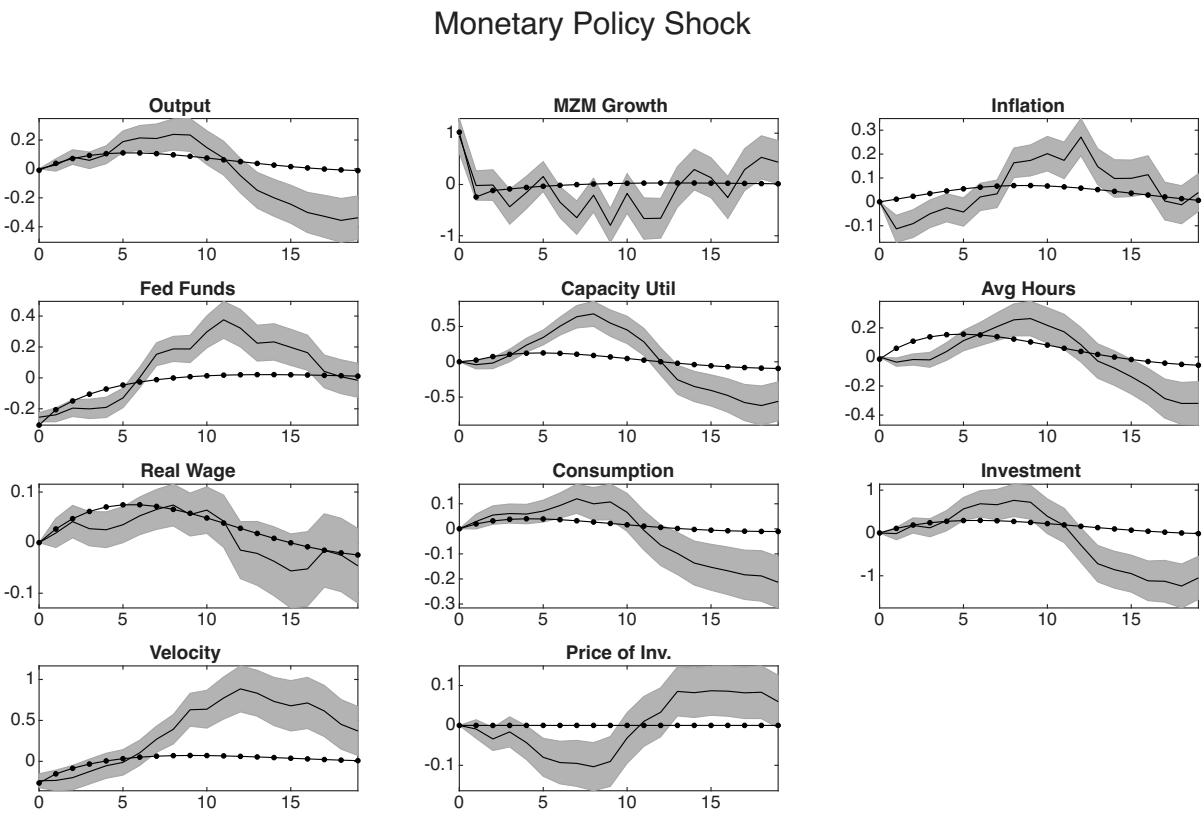


Figure 36: Monetary Policy Shock DSGE matched with VAR lag 12 compared to VAR lag 12

C.9 ACEL 2005 DSGE Matched and compared with VAR estimated across different lag structures: Embodied Technology Shock

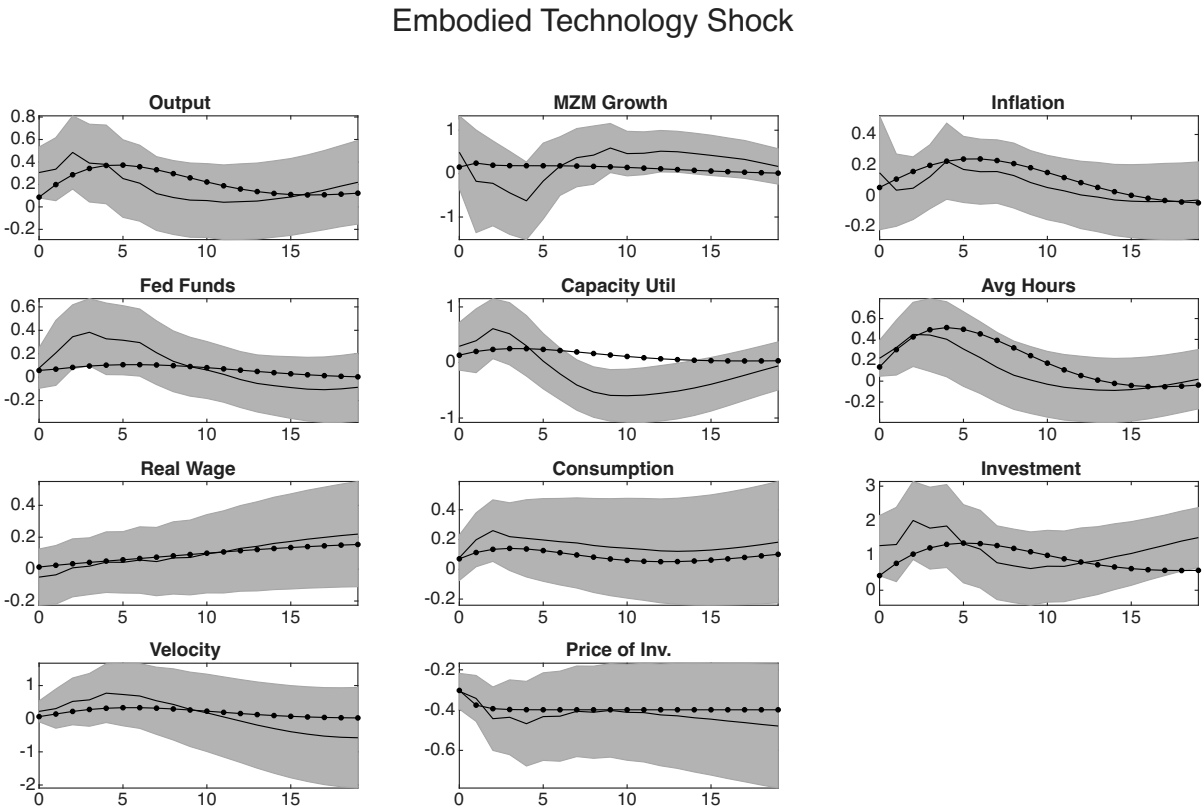


Figure 37: Embodied Technology Shock DSGE matched with VAR lag 4 compared to VAR lag 4

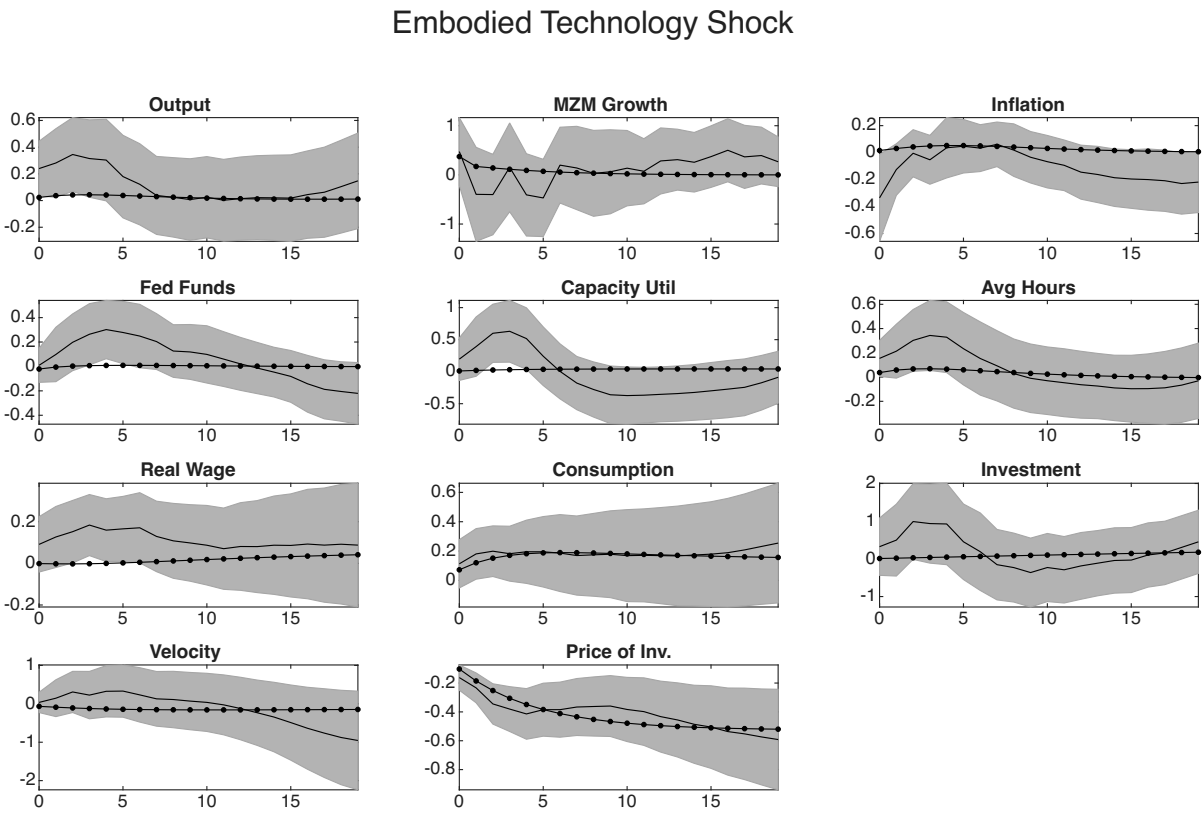


Figure 38: Embodied Technology Shock DSGE matched with VAR lag 6 compared to VAR lag 6

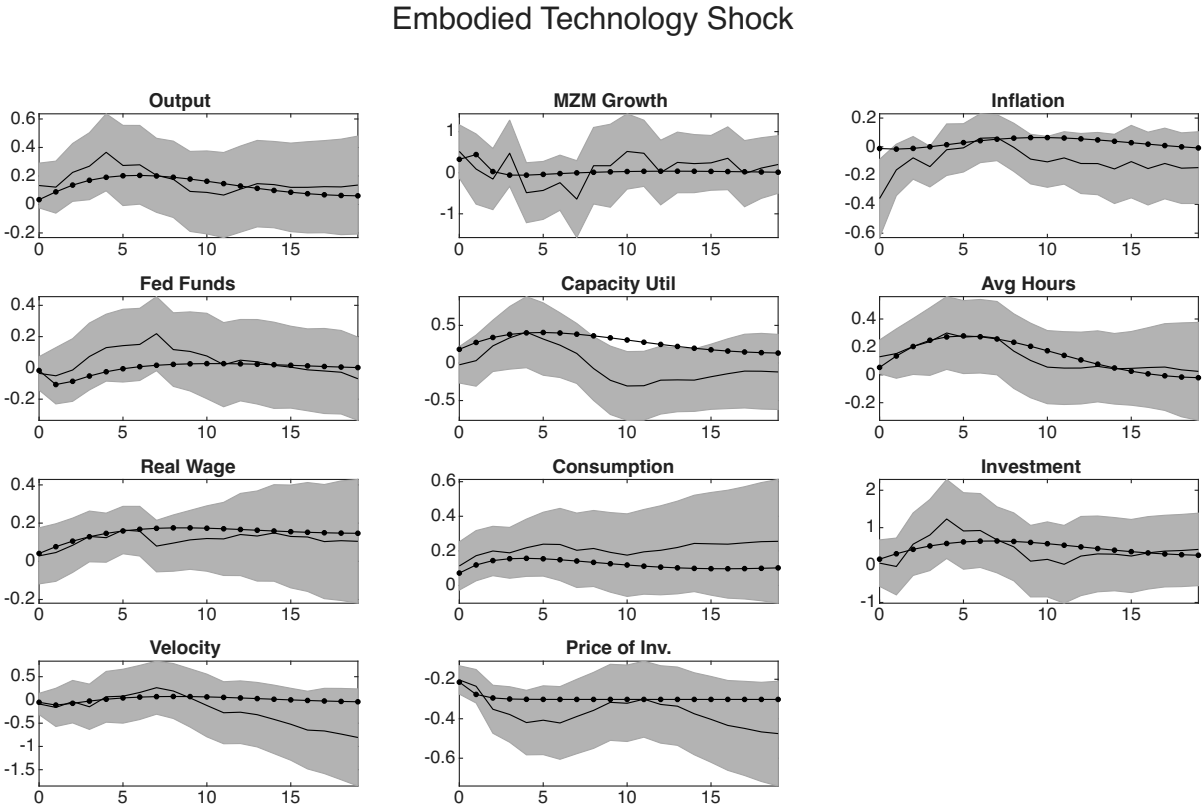


Figure 39: Embodied Technology Shock DSGE matched with VAR lag 8 compared to VAR lag 8

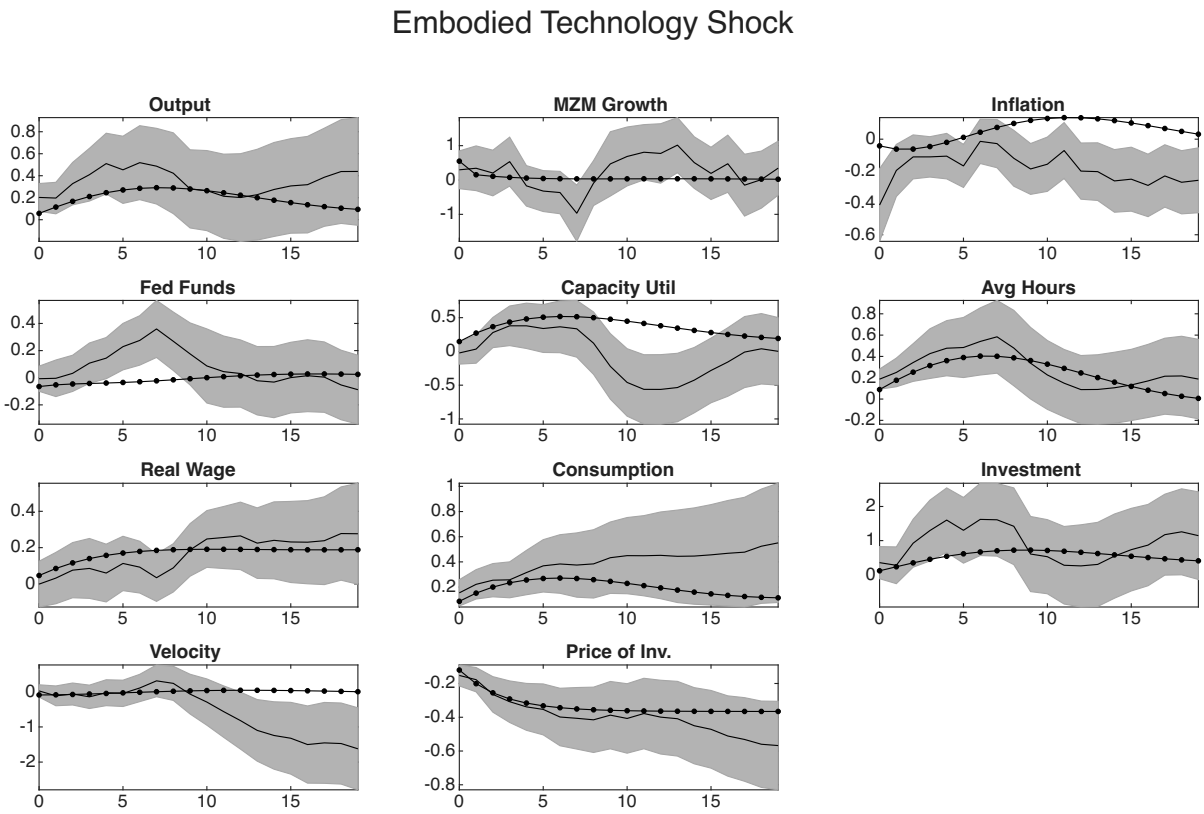


Figure 40: Embodied Technology Shock DSGE matched with VAR lag 10 compared to VAR lag 10

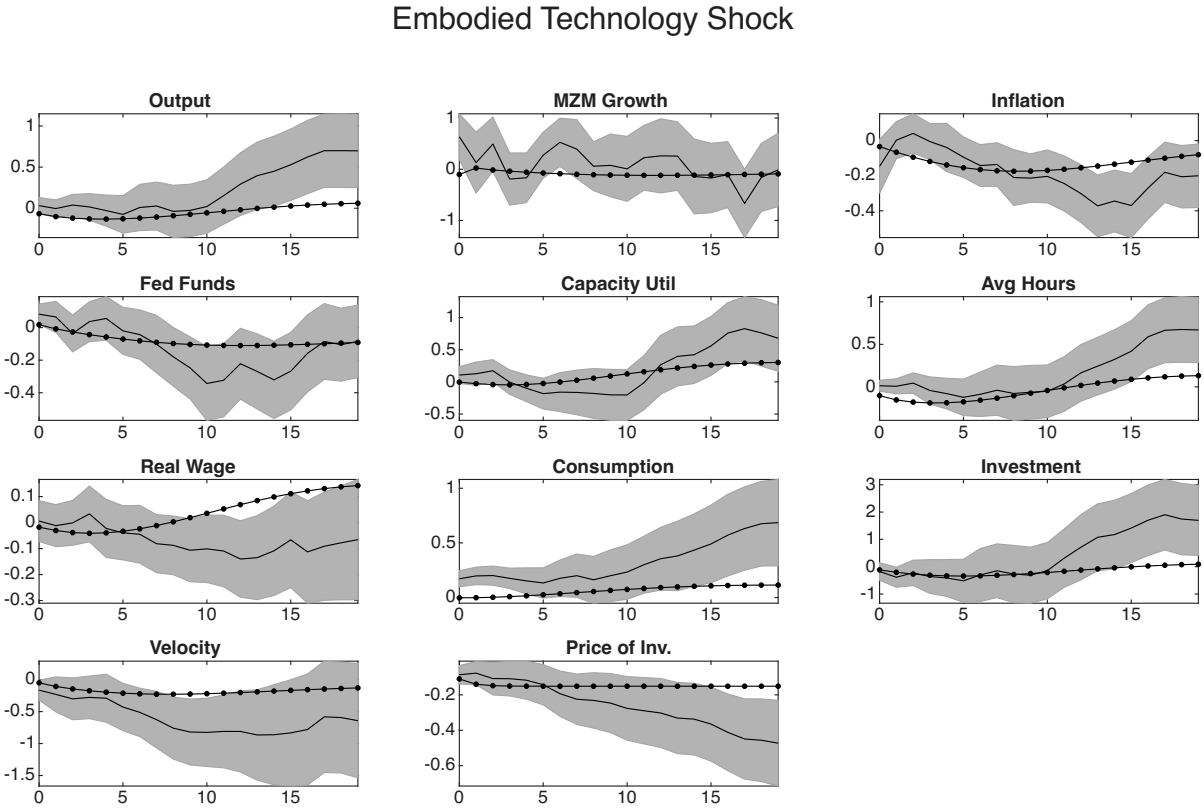


Figure 41: Embodied Technology Shock DSGE matched with VAR lag 12 compared to VAR lag 12

C.10 ACEL 2005 DSGE Matched and compared with VAR estimated across different lag structures: Neutral Technology Shock

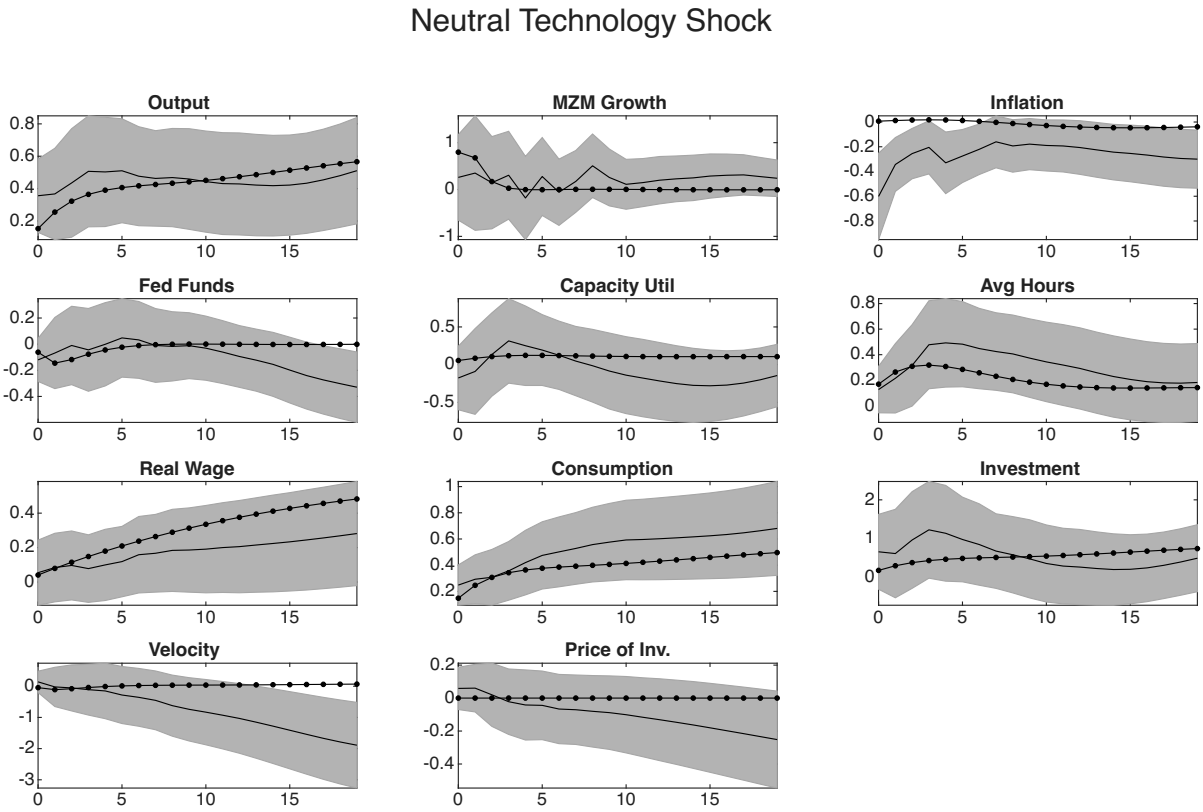


Figure 42: Neutral Technology Shock DSGE matched with VAR lag 4 compared to VAR lag 4

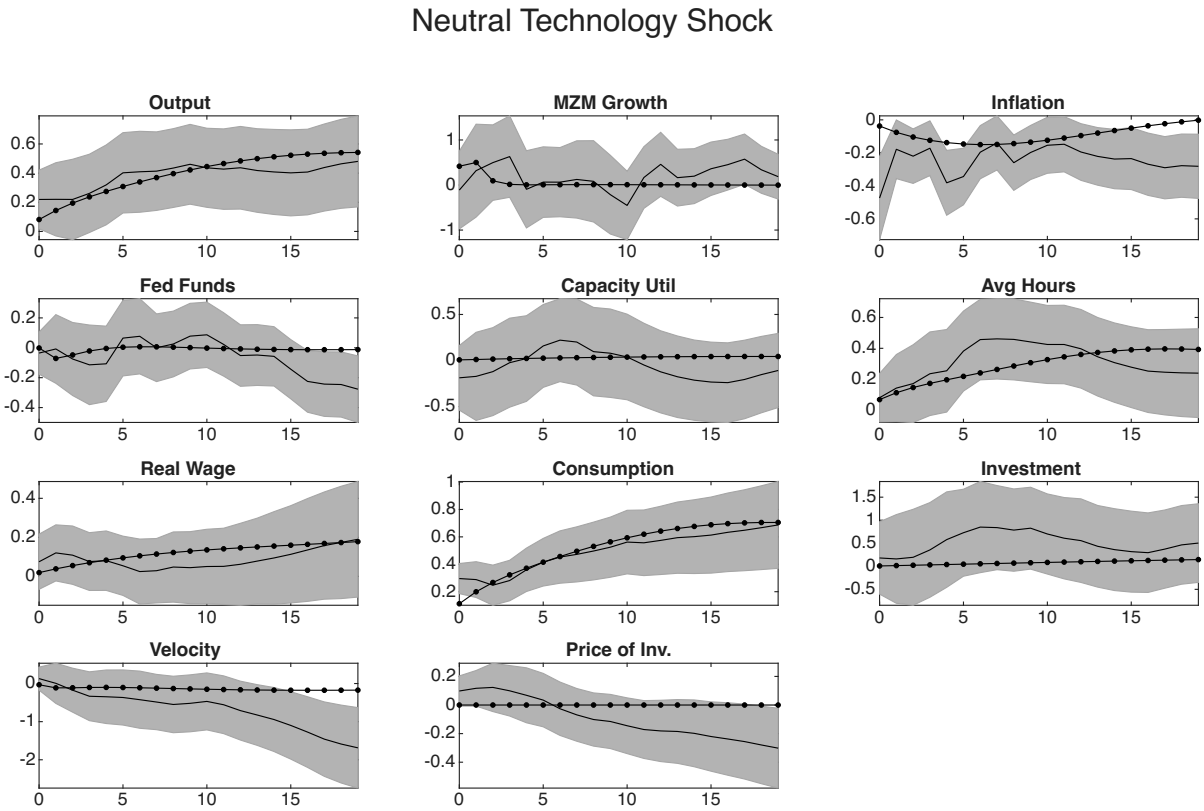


Figure 43: Neutral Technology Shock DSGE matched with VAR lag 6 compared to VAR lag 6

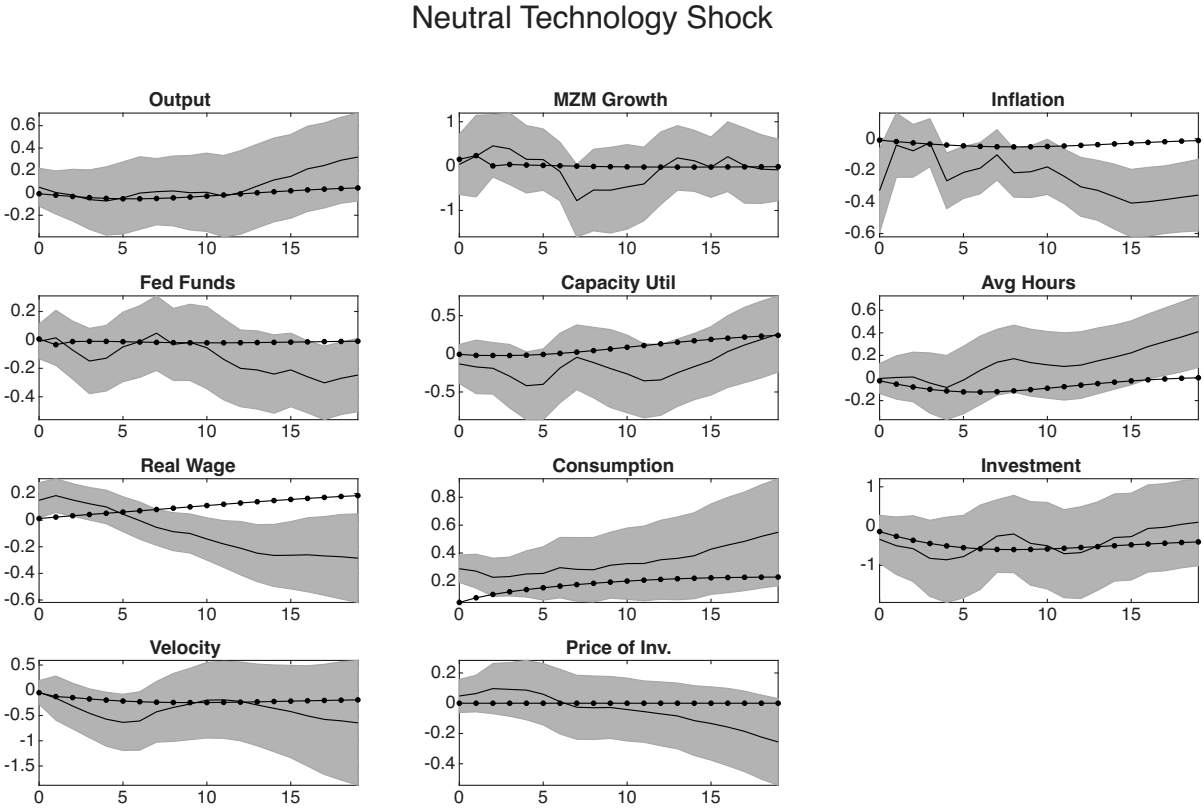


Figure 44: Neutral Technology Shock DSGE matched with VAR lag 8 compared to VAR lag 8

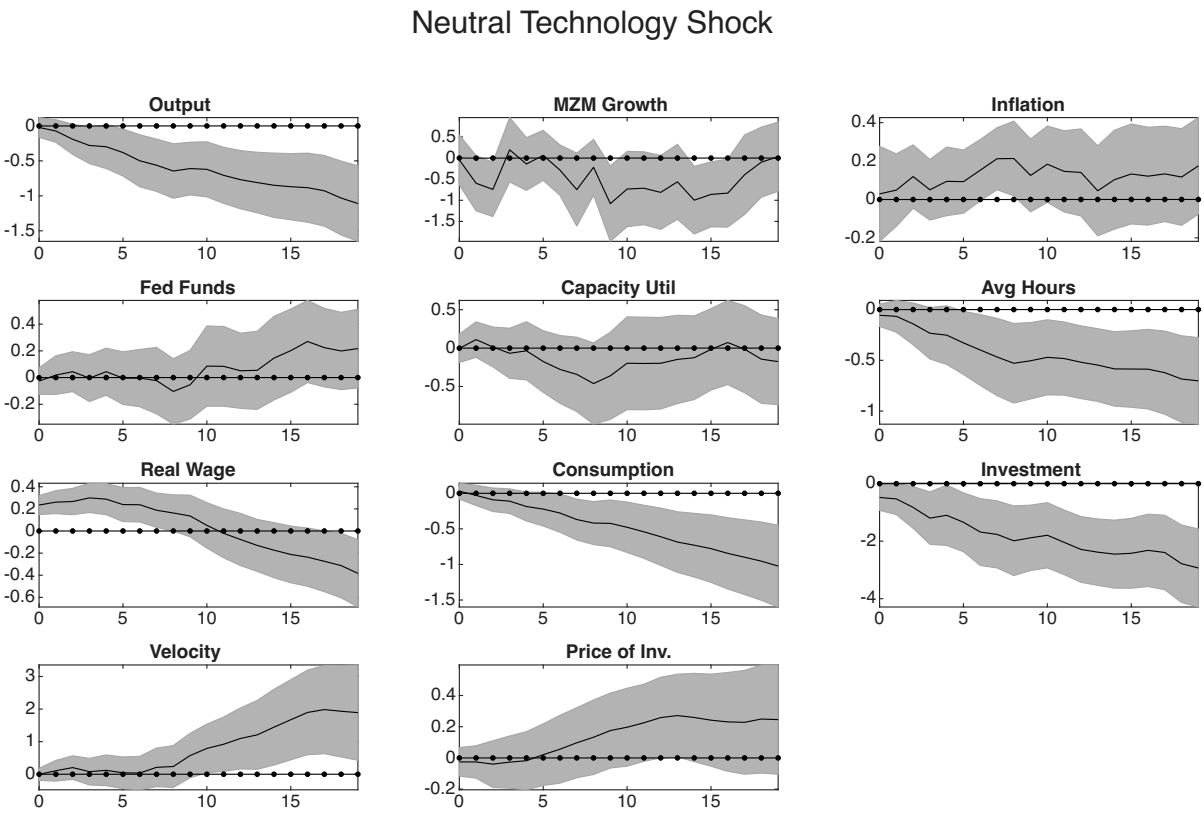


Figure 45: Neutral Technology Shock DSGE matched with VAR lag 10 compared to VAR lag 10

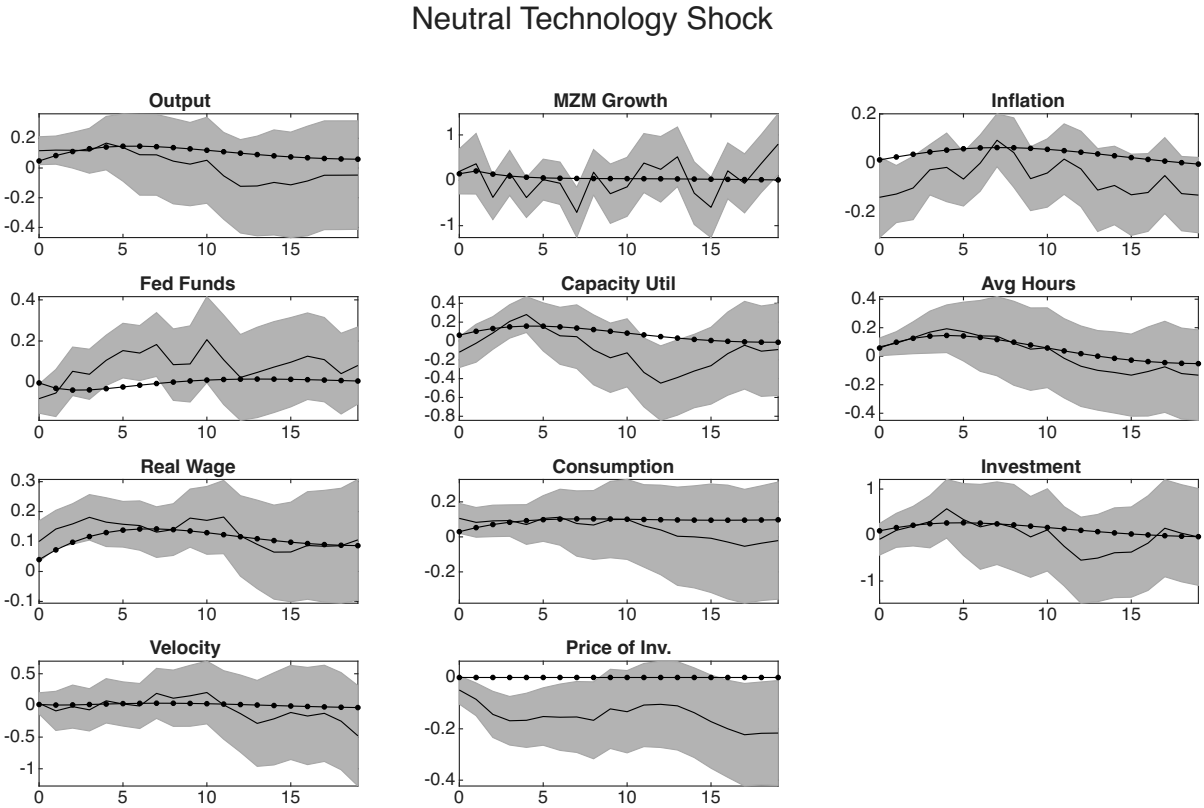


Figure 46: Neutral Technology Shock DSGE matched with VAR lag 12 compared to VAR lag 12

C.11 VAR Residual Analysis

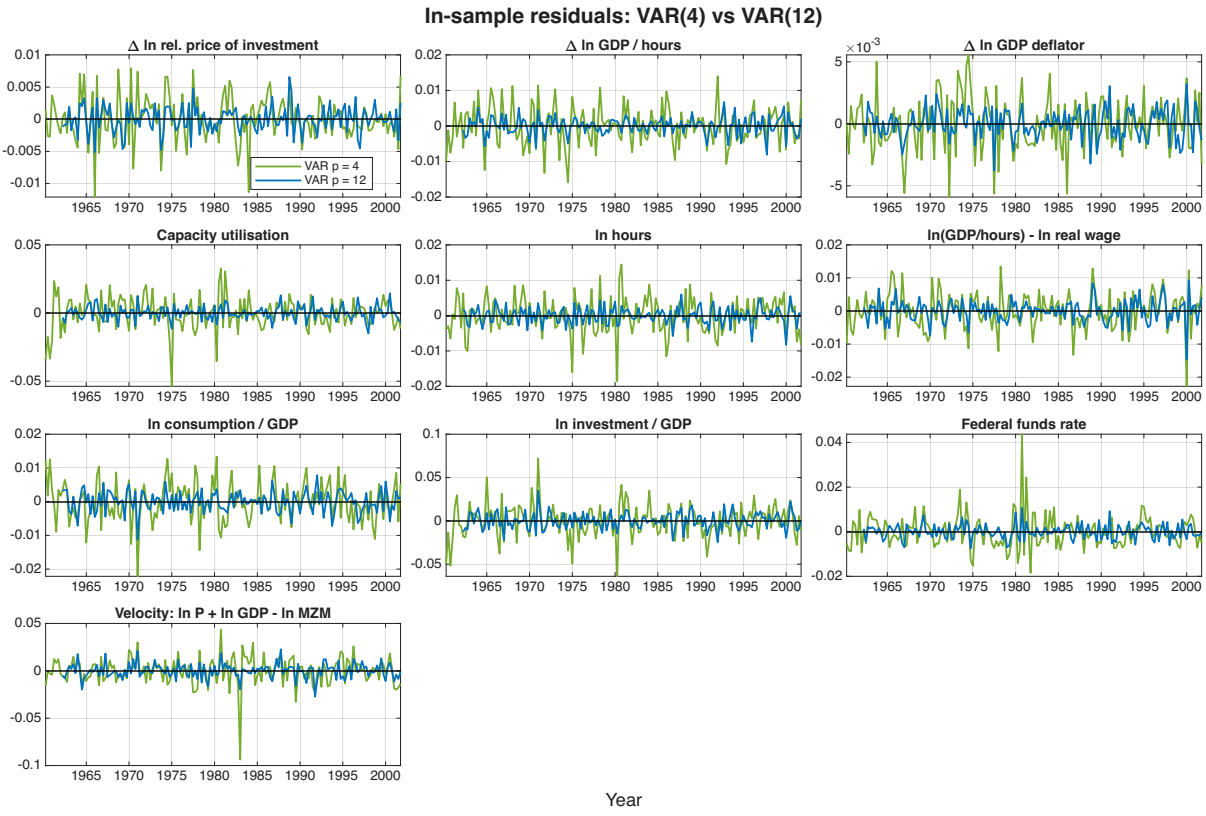


Figure 47: VAR Residuals Comparison: VAR(4) vs VAR(12)

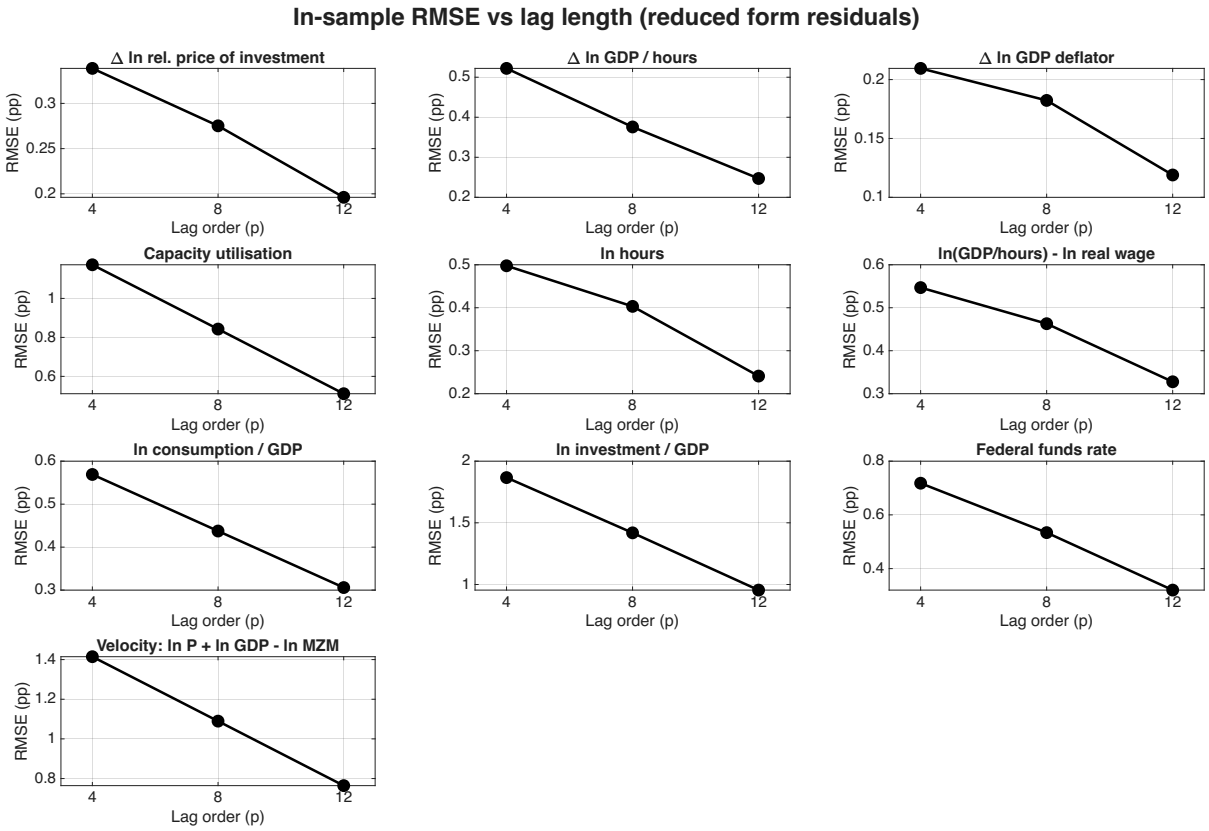


Figure 48: VAR RMSE Comparison across different lag structures

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Figure 49: Ljung-Box P-Values Across VAR Lags

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Figure 50: Standard Deviation Ratios

C.12 ACEL 2005 VAR Across different lag structures Progressively Comparison: Monetary Policy Shock

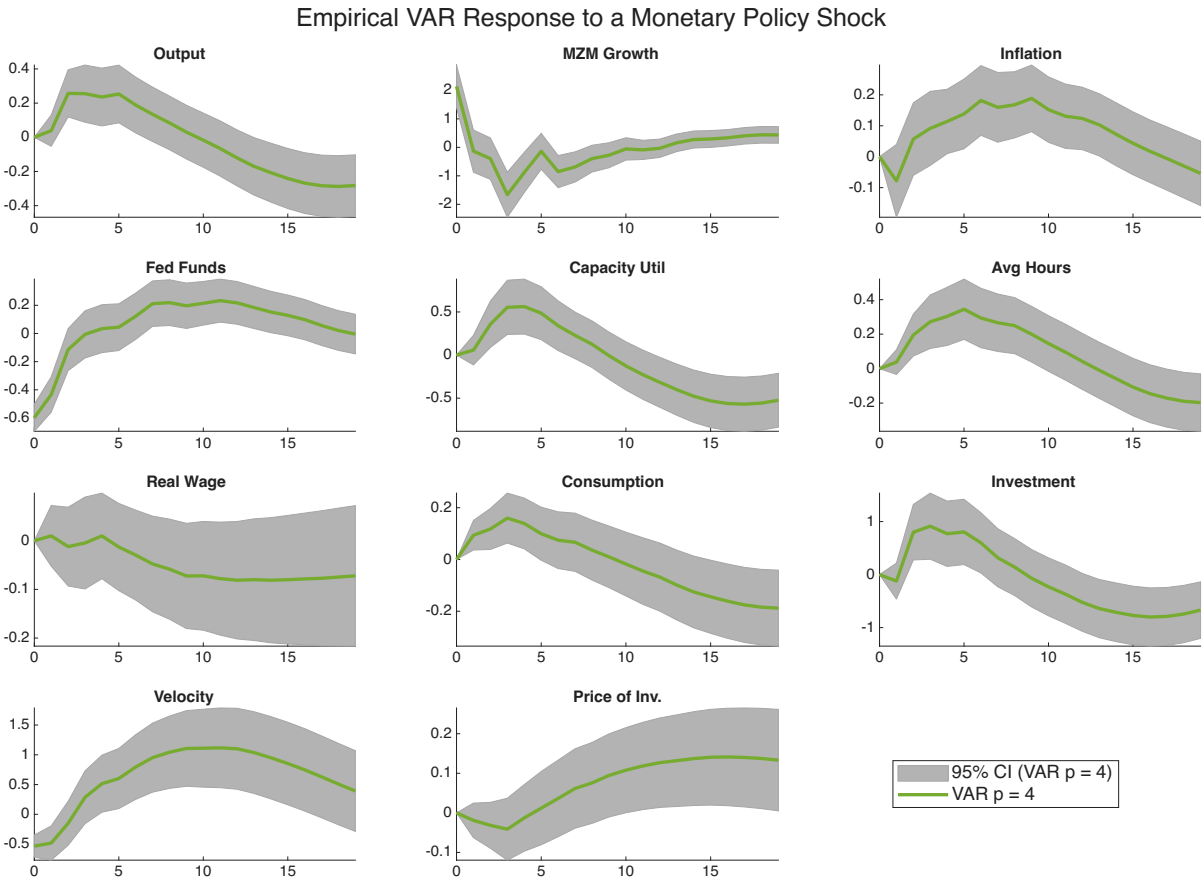


Figure 51: Monetary Policy Shock lag 4

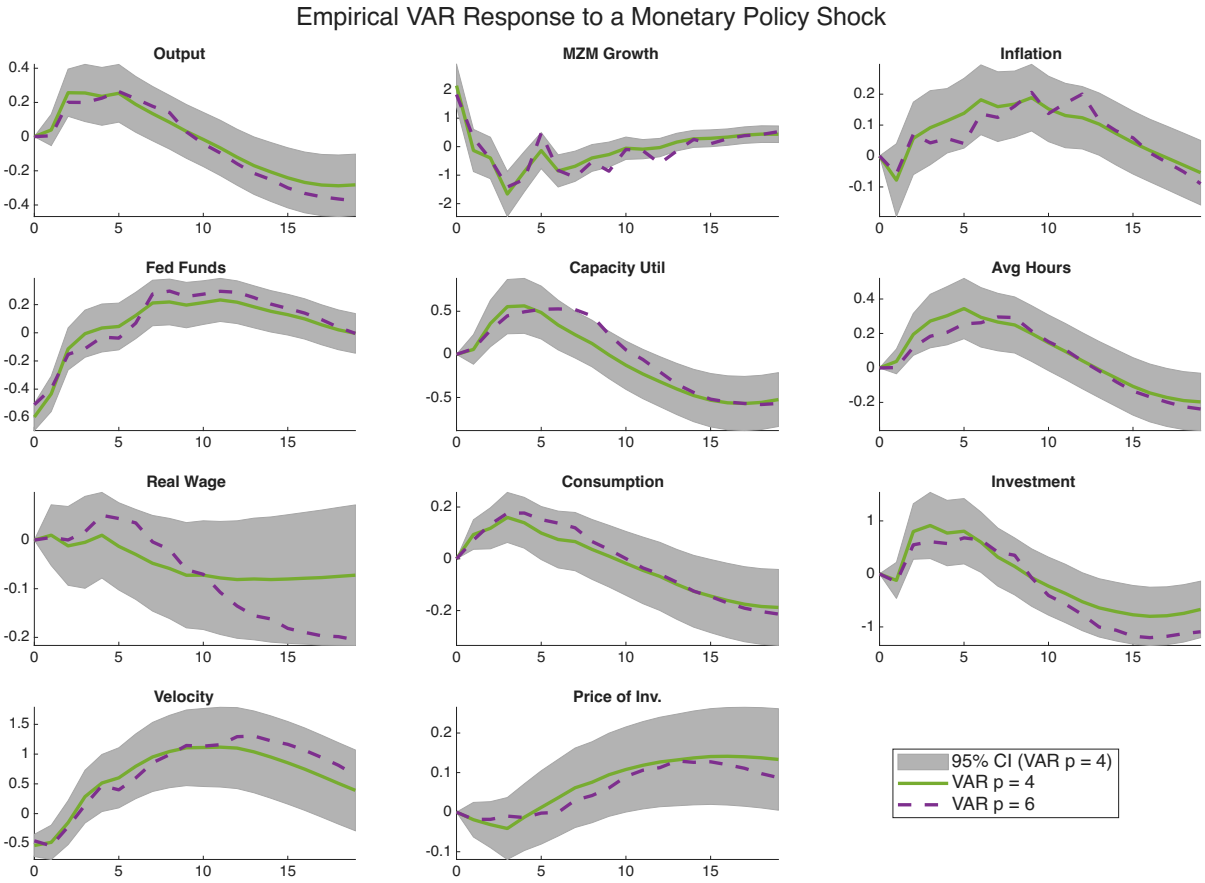


Figure 52: Monetary Policy Shock lag 6

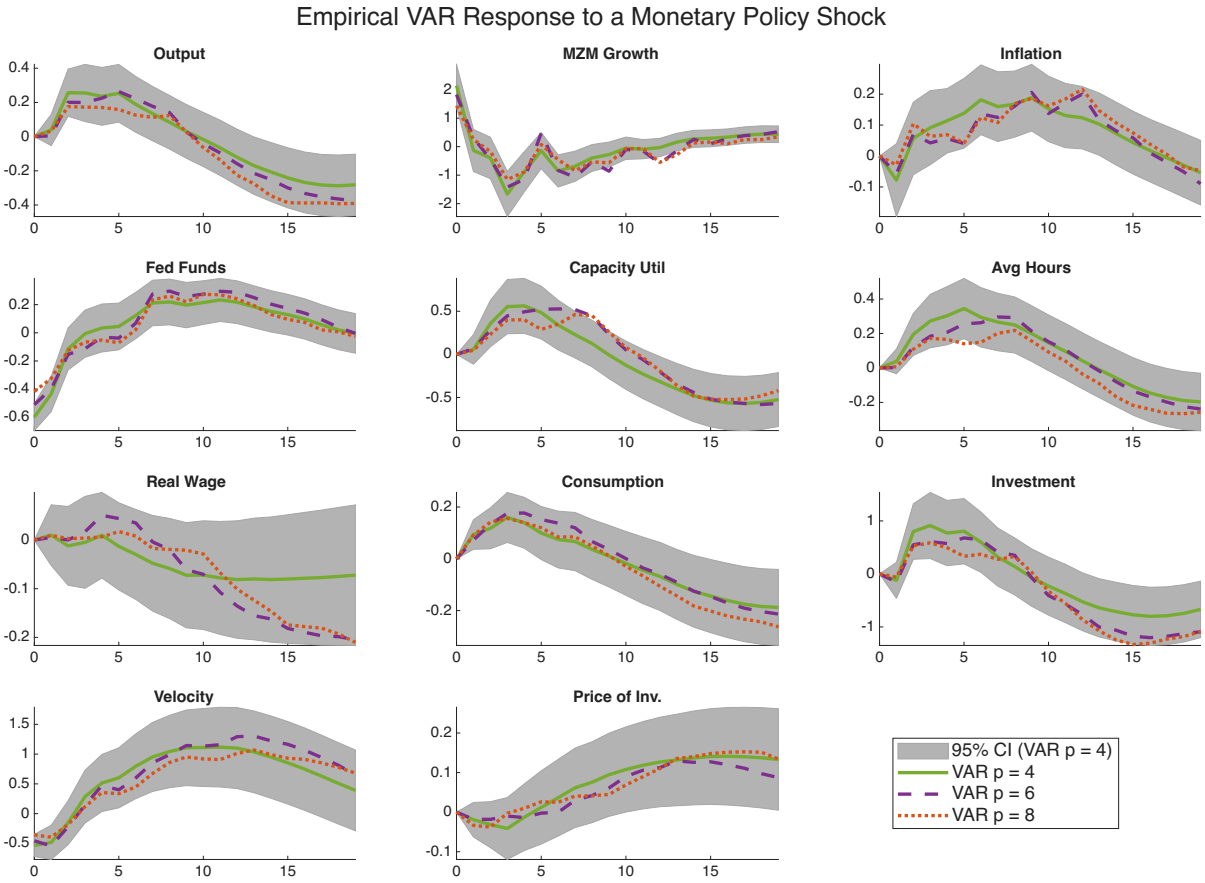


Figure 53: Monetary Policy Shock lag 8

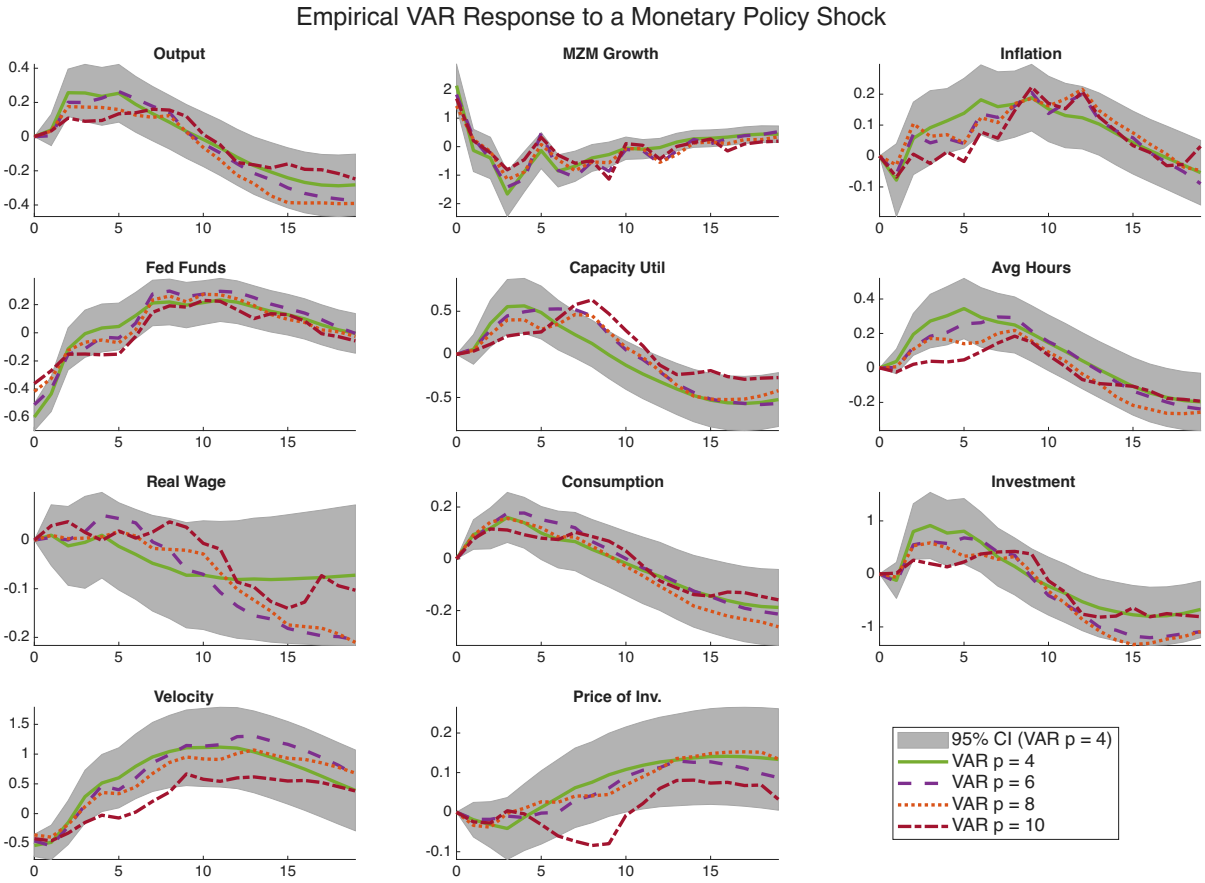


Figure 54: Monetary Policy Shock lag 10

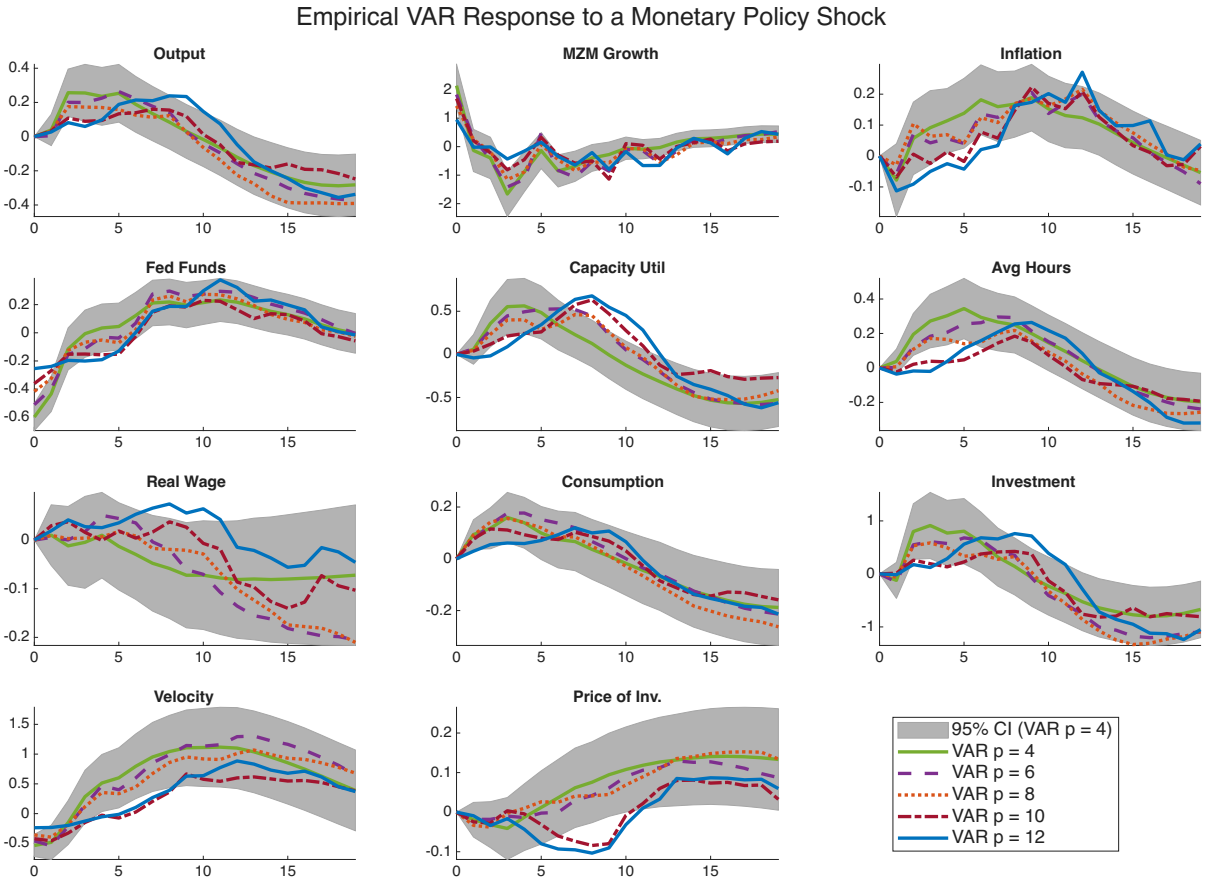


Figure 55: Monetary Policy Shock lag 12

C.13 ACEL 2005 VAR Across different lag structures Progressively Comparison: Embodied Technology Shock

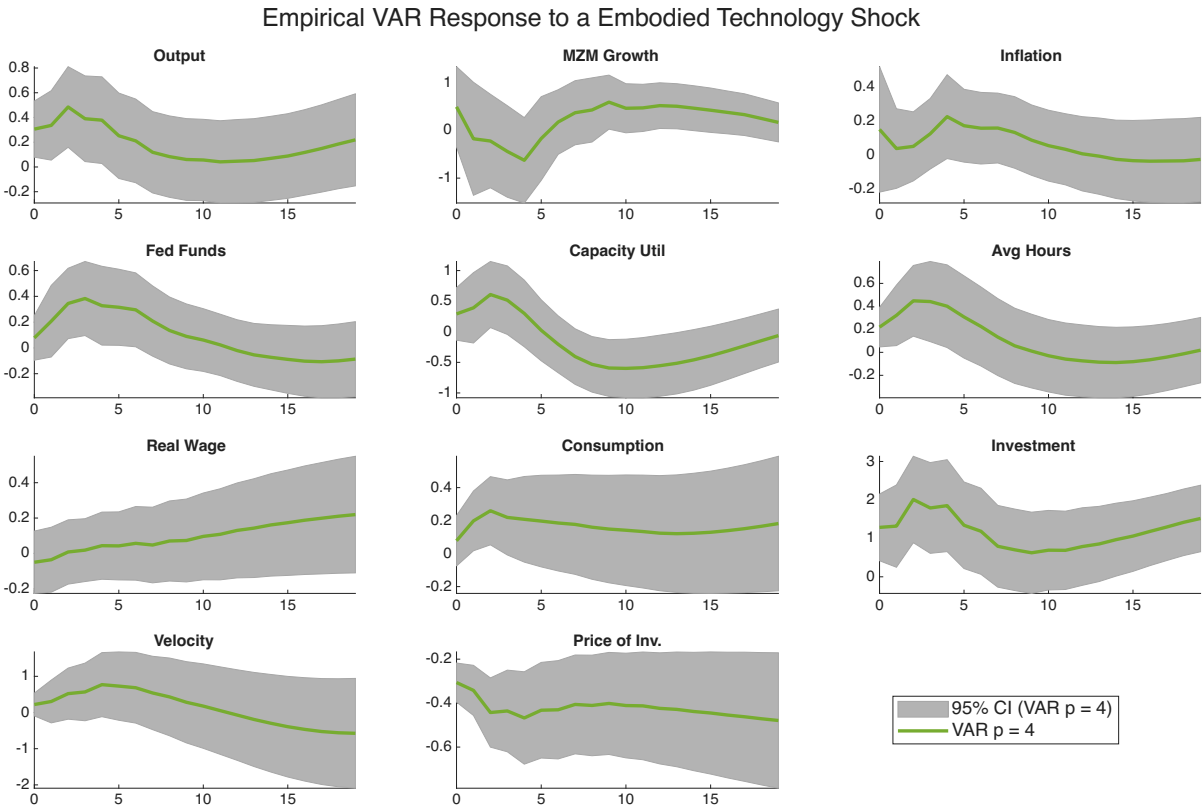


Figure 56: Embodied Technology Shock lag 4

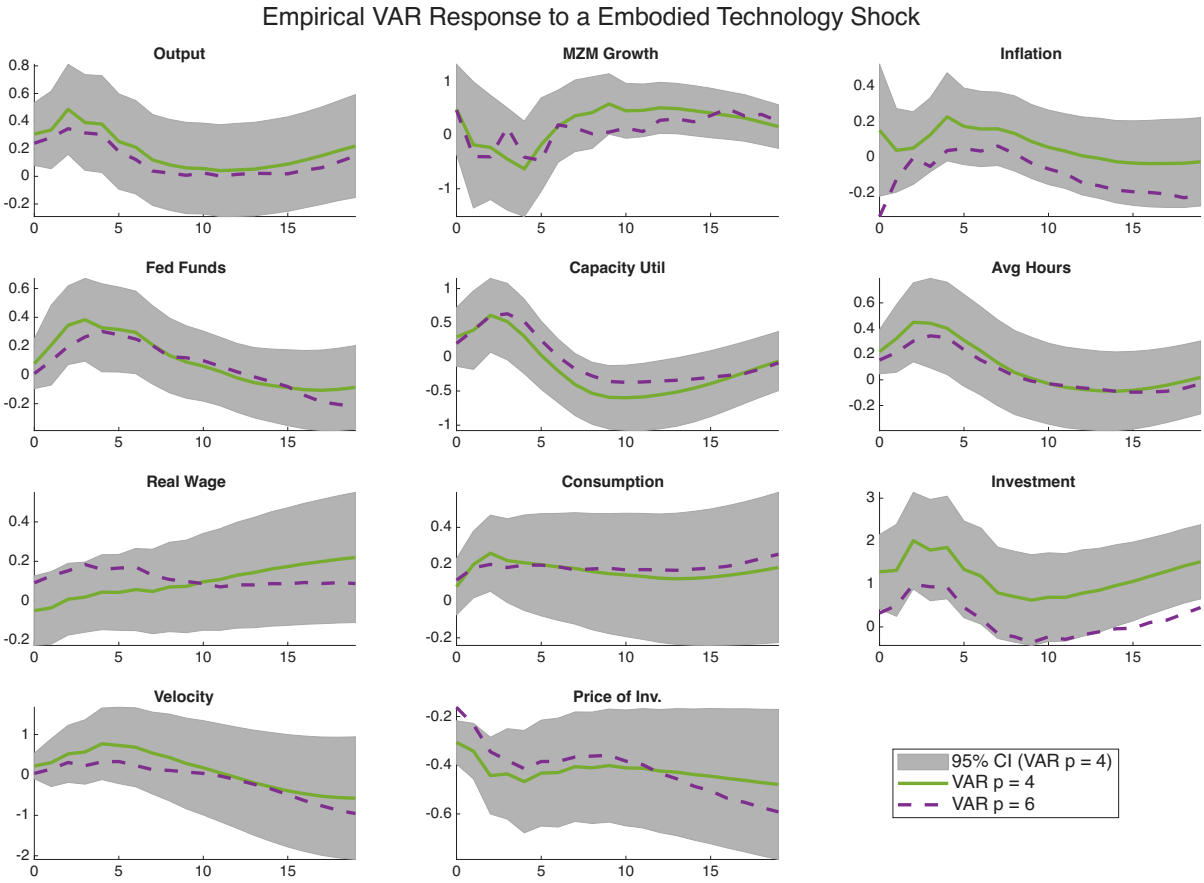


Figure 57: Embodied Technology Shock lag 6

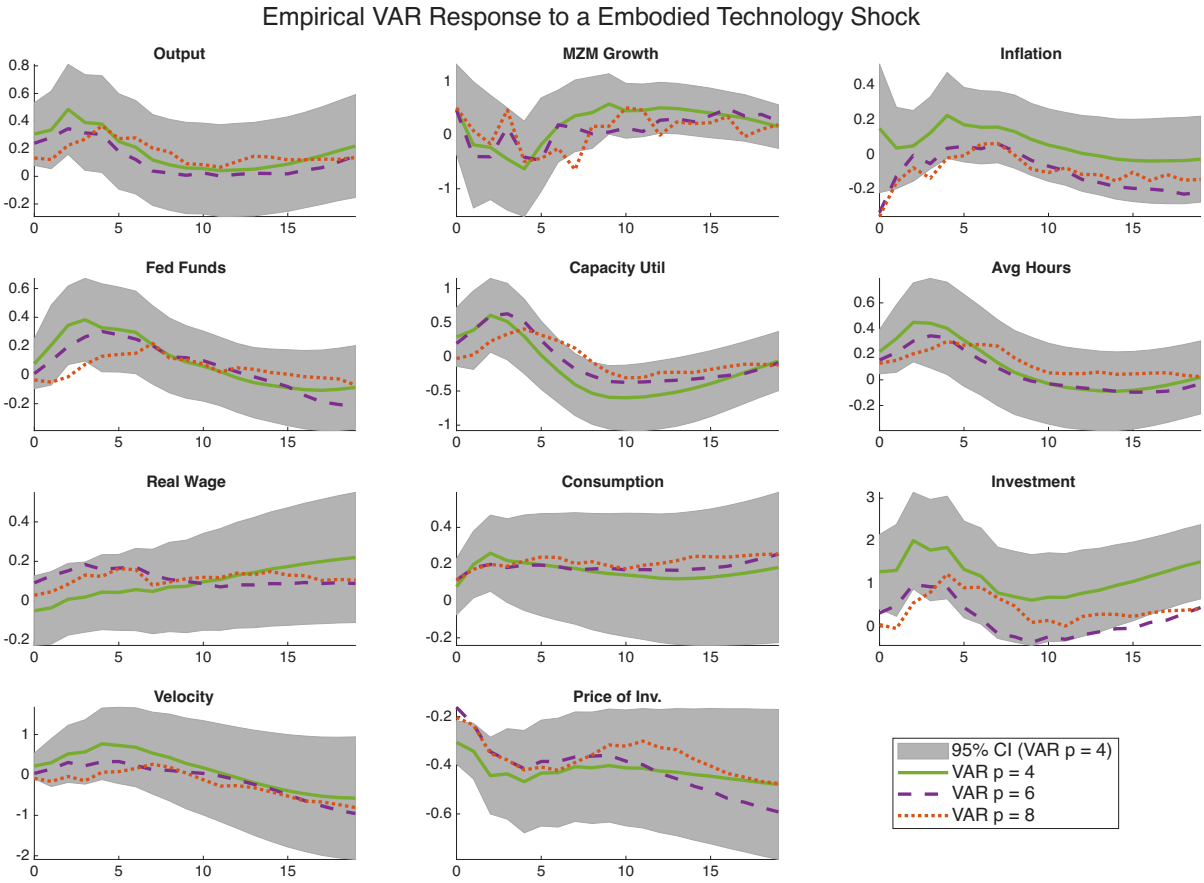


Figure 58: Embodied Technology Shock lag 8

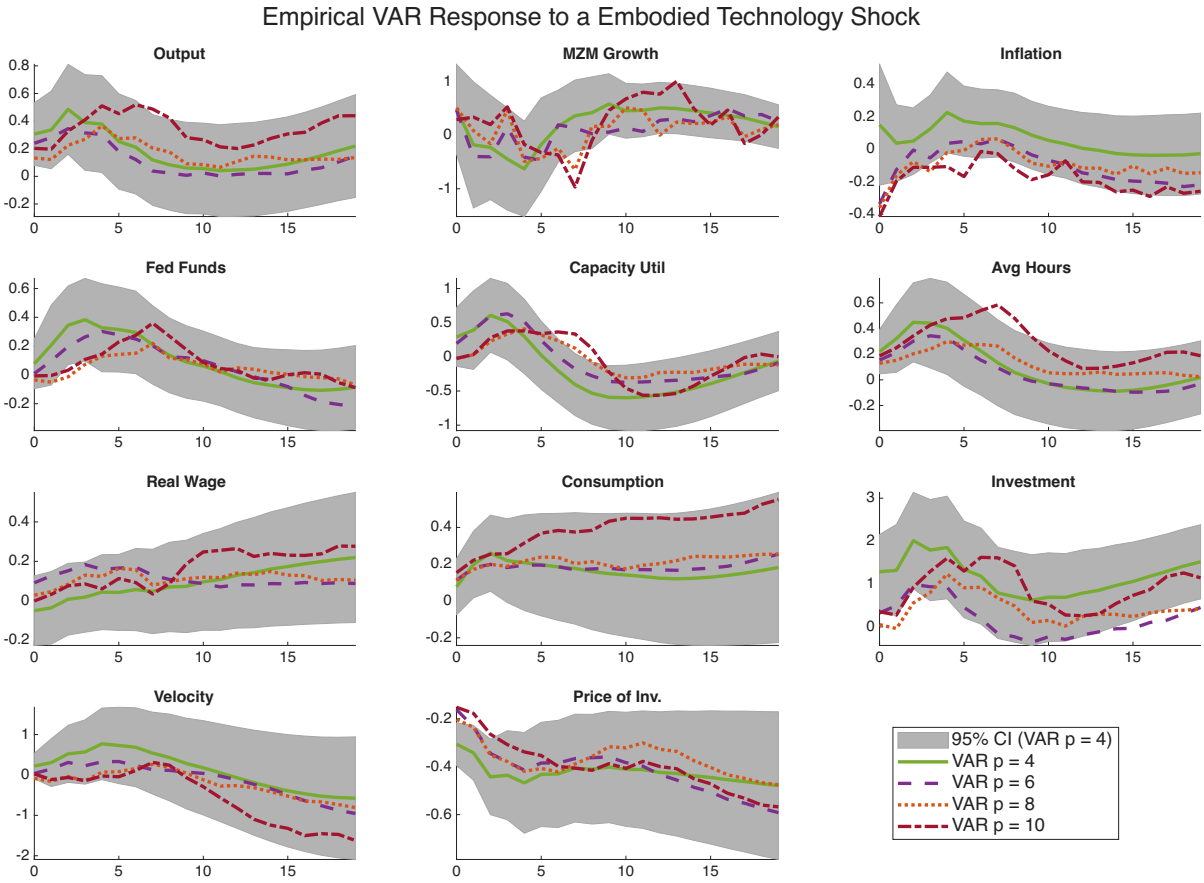


Figure 59: Embodied Technology Shock lag 10

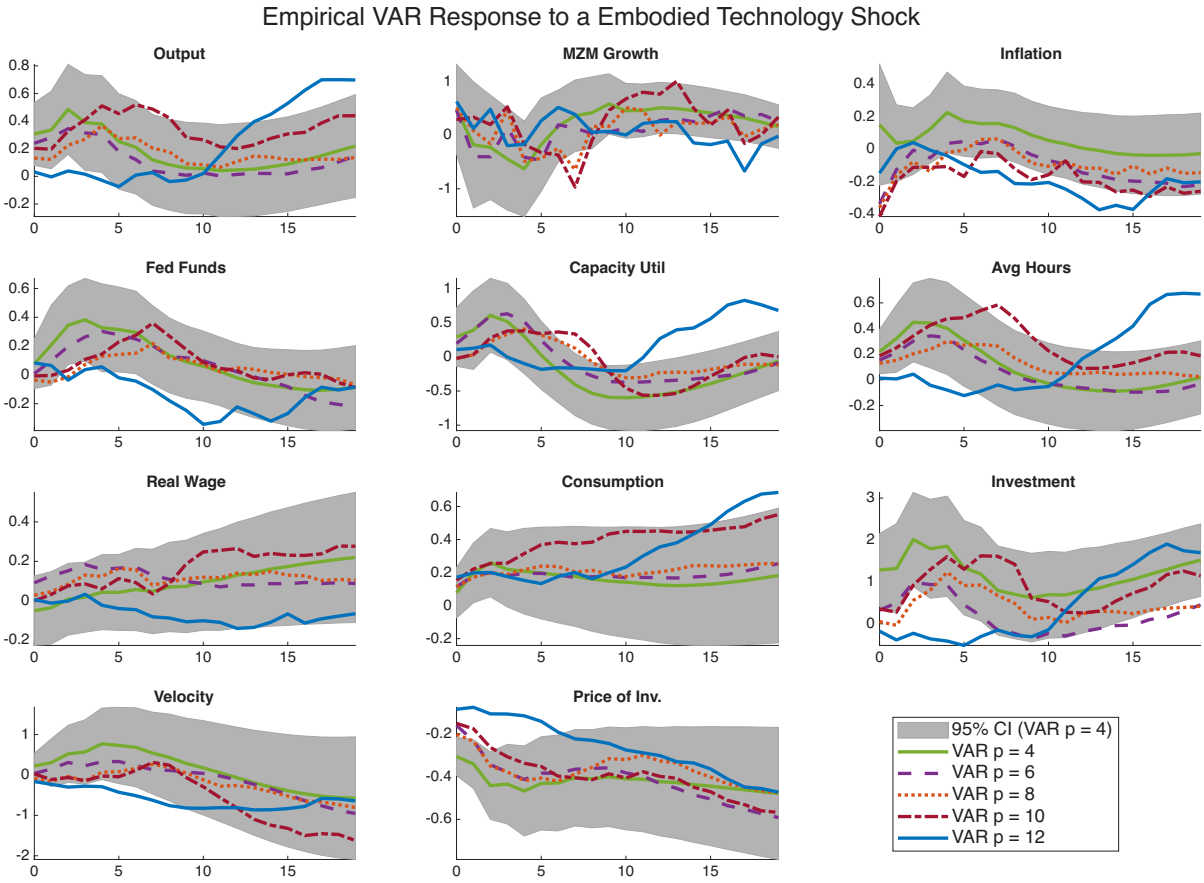


Figure 60: Embodied Technology Shock lag 12

C.14 ACEL 2005 VAR Across different lag structures Progressively Comparison: Neutral Technology Shock

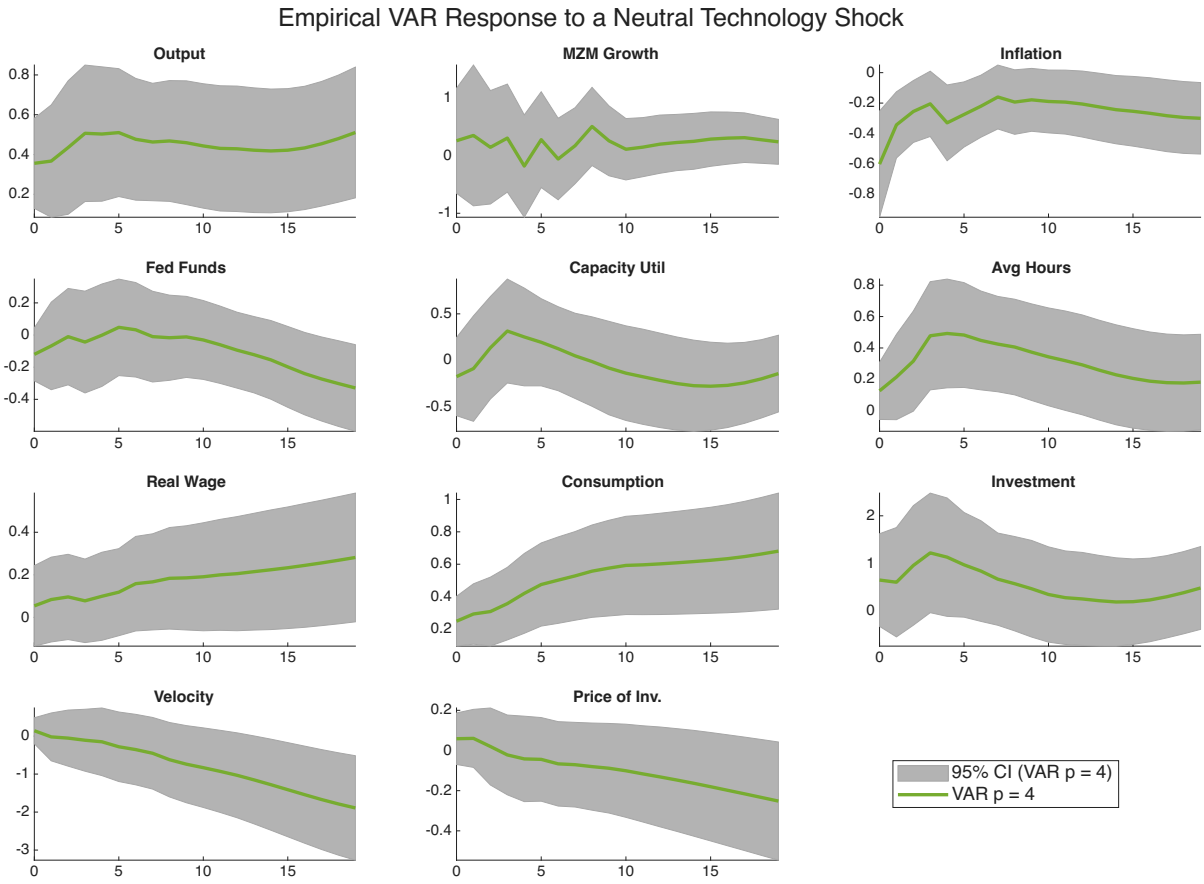


Figure 61: Neutral Technology Shock lag 4

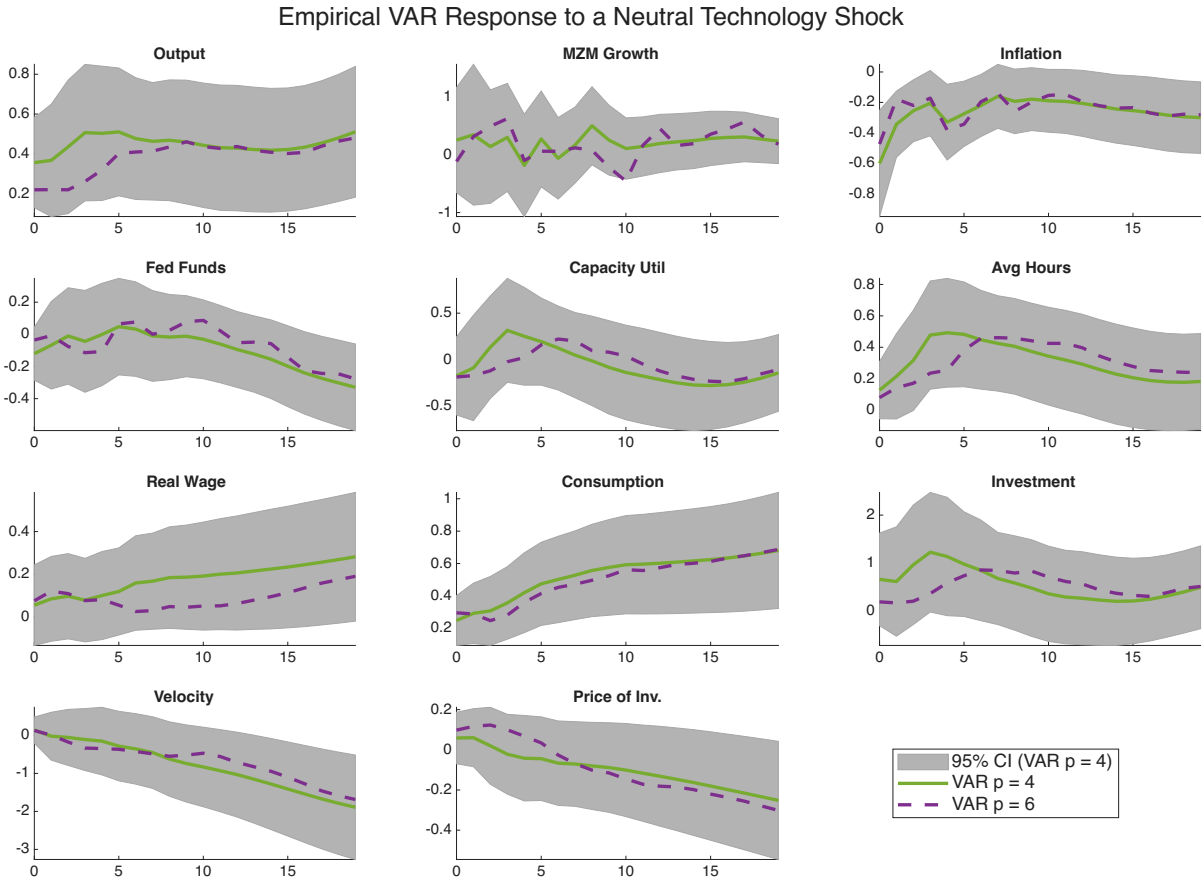


Figure 62: Neutral Technology Shock lag 6

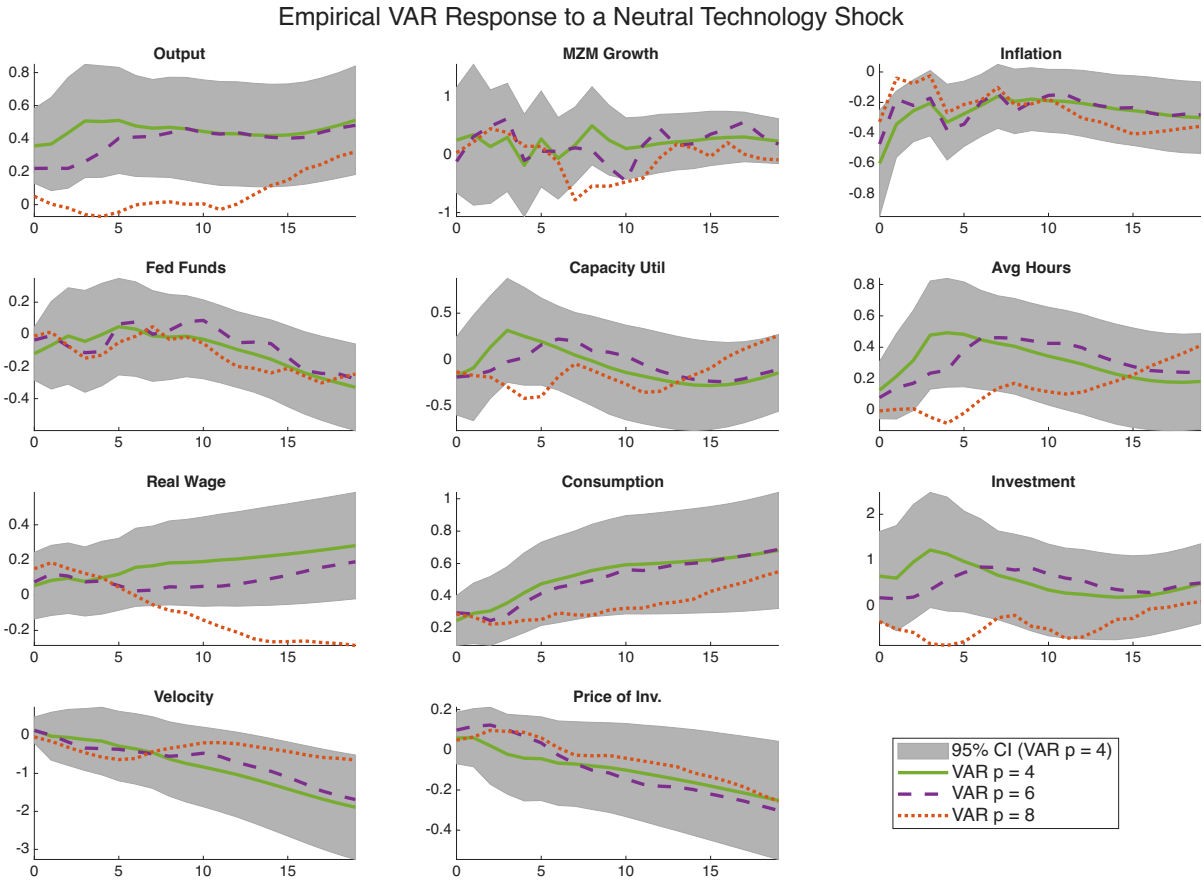


Figure 63: Neutral Technology Shock lag 8

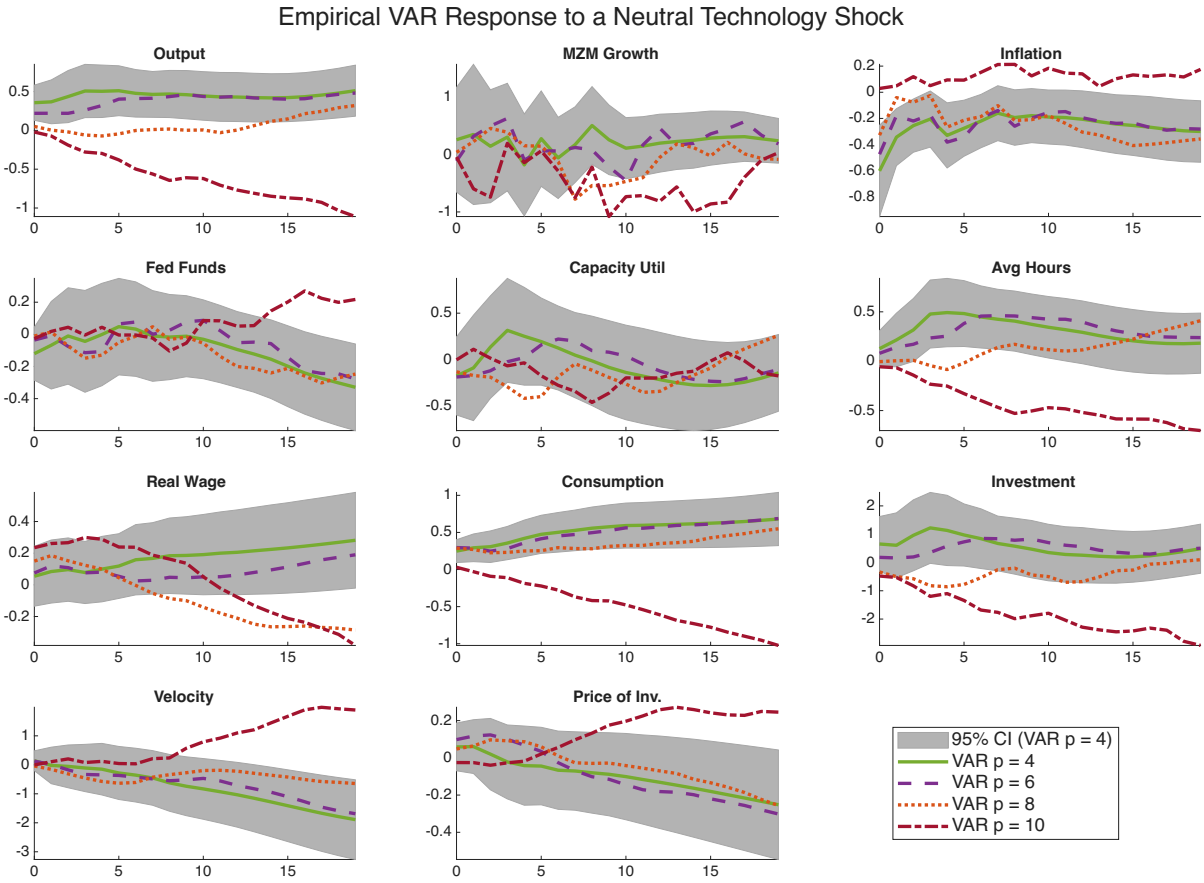


Figure 64: Neutral Technology Shock lag 10

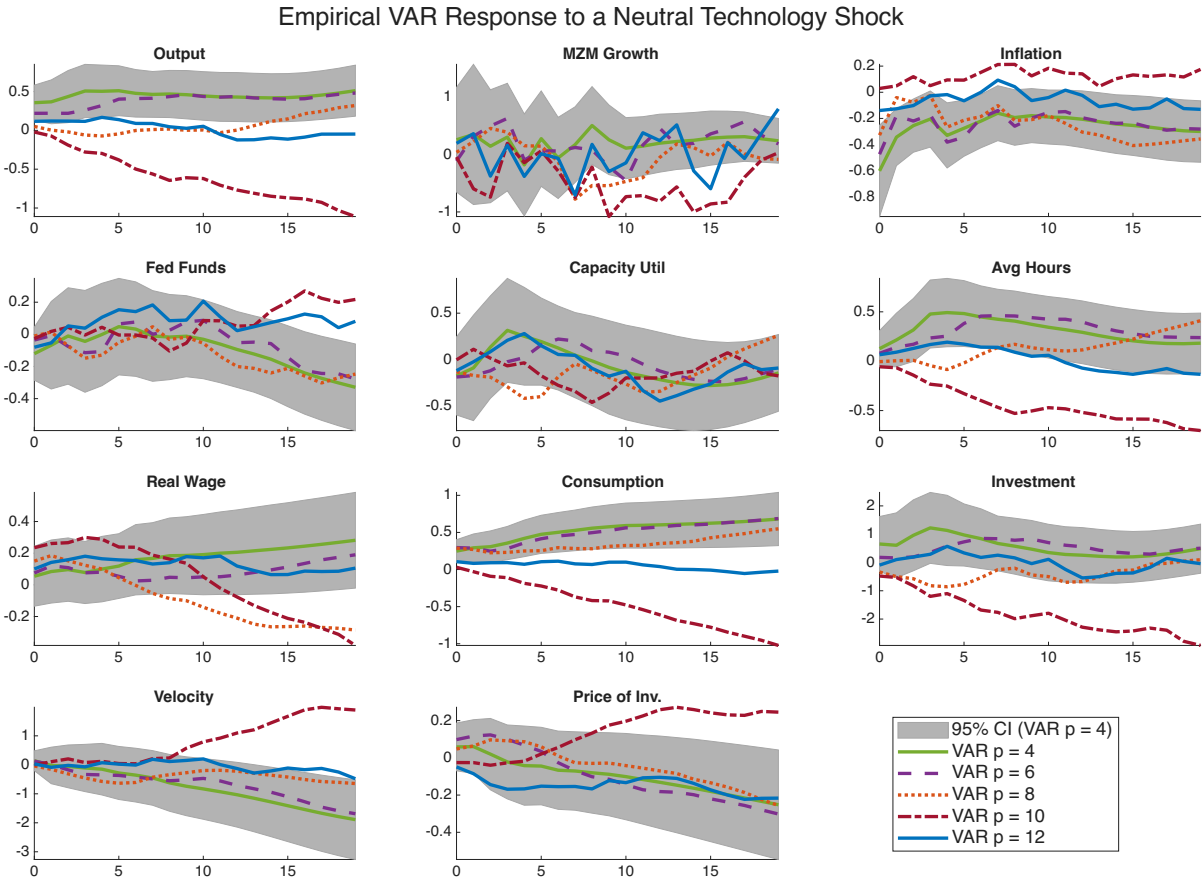


Figure 65: Neutral Technology Shock lag 12

C.15 ACEL 2005 DSGE Across different lag structures Progressively Comparison: Monetary Policy Shock

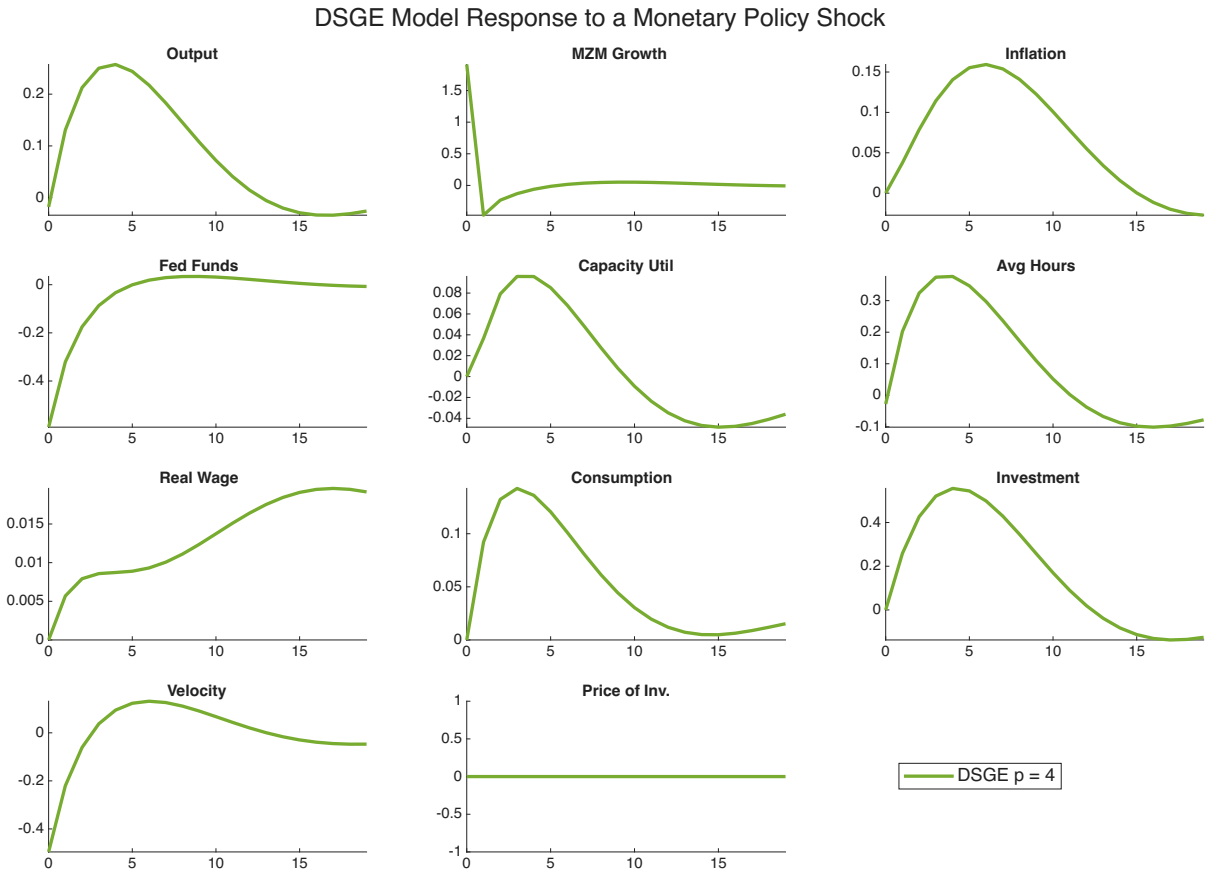


Figure 66: DSGE Monetary Policy Shock lag 4

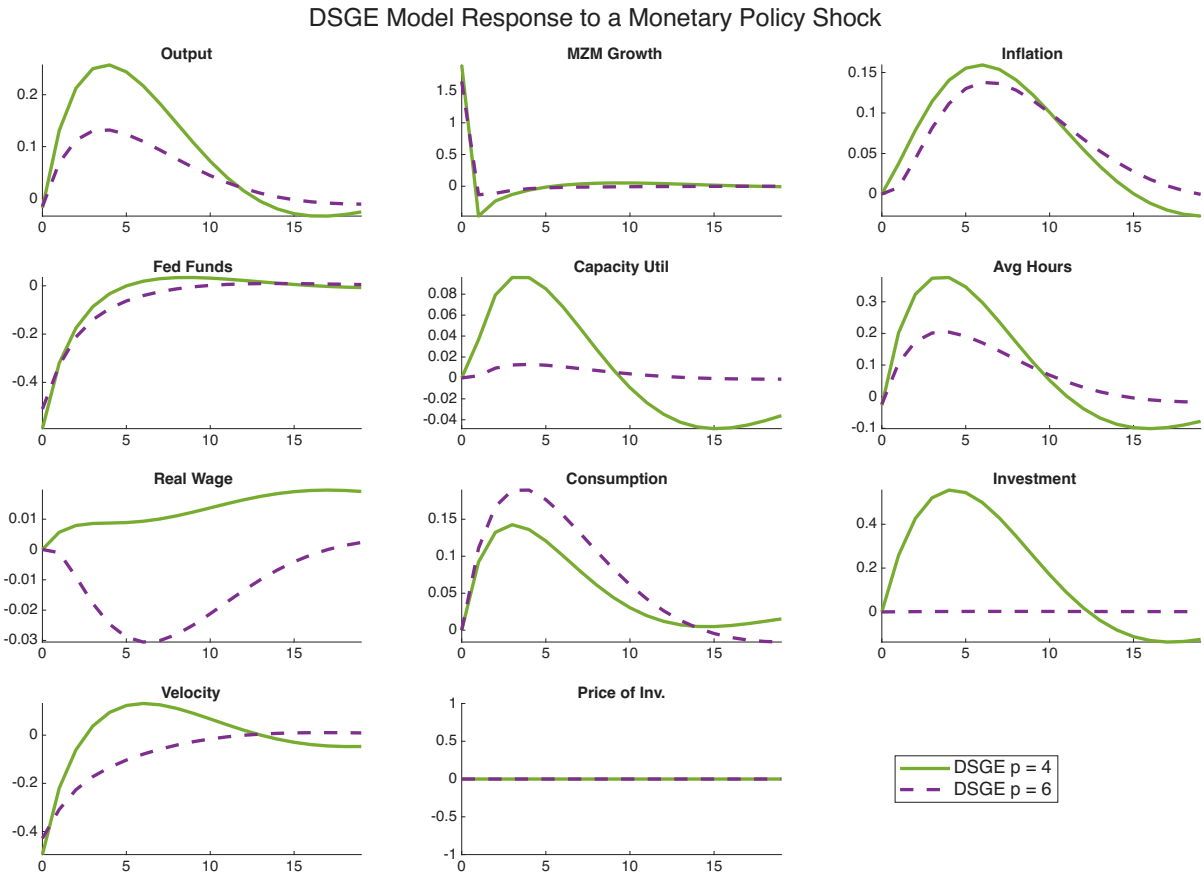


Figure 67: DSGE Monetary Policy Shock lag 6

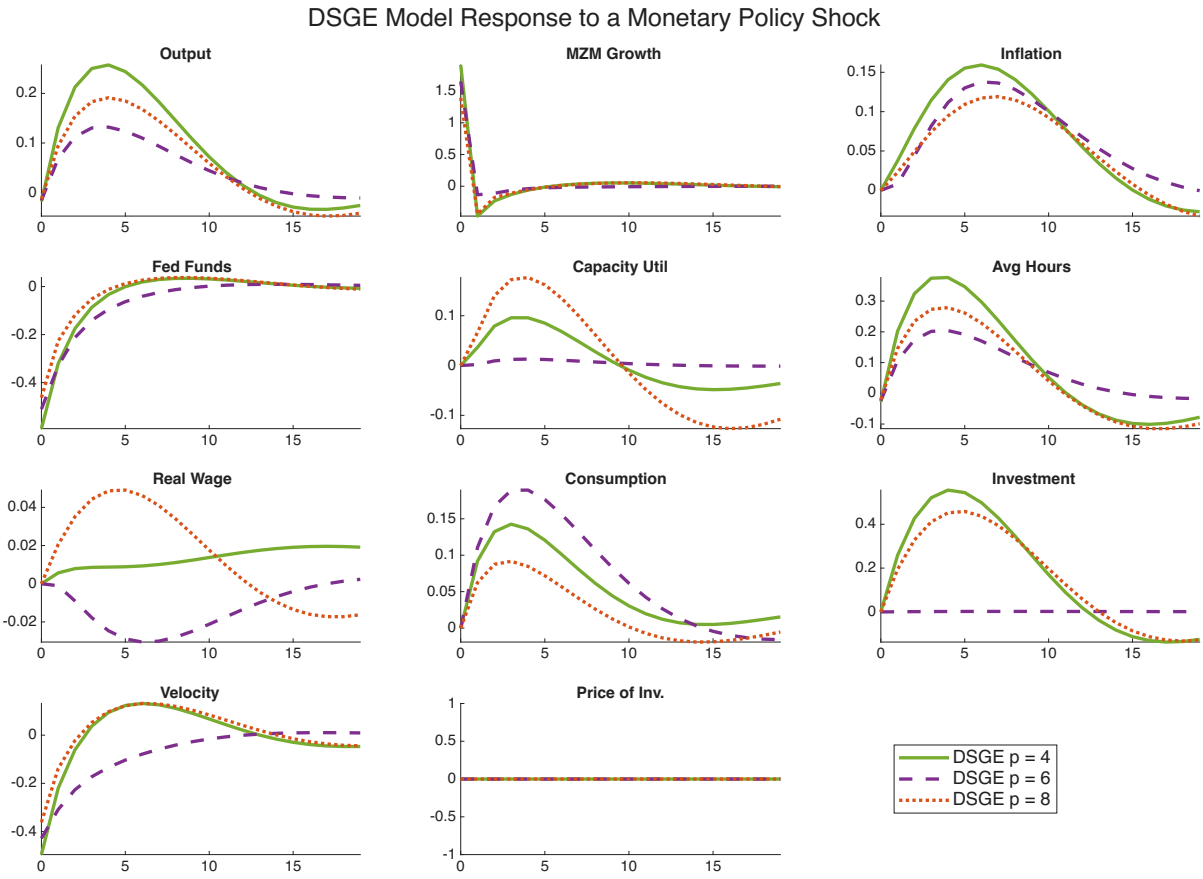


Figure 68: DSGE Monetary Policy Shock lag 8

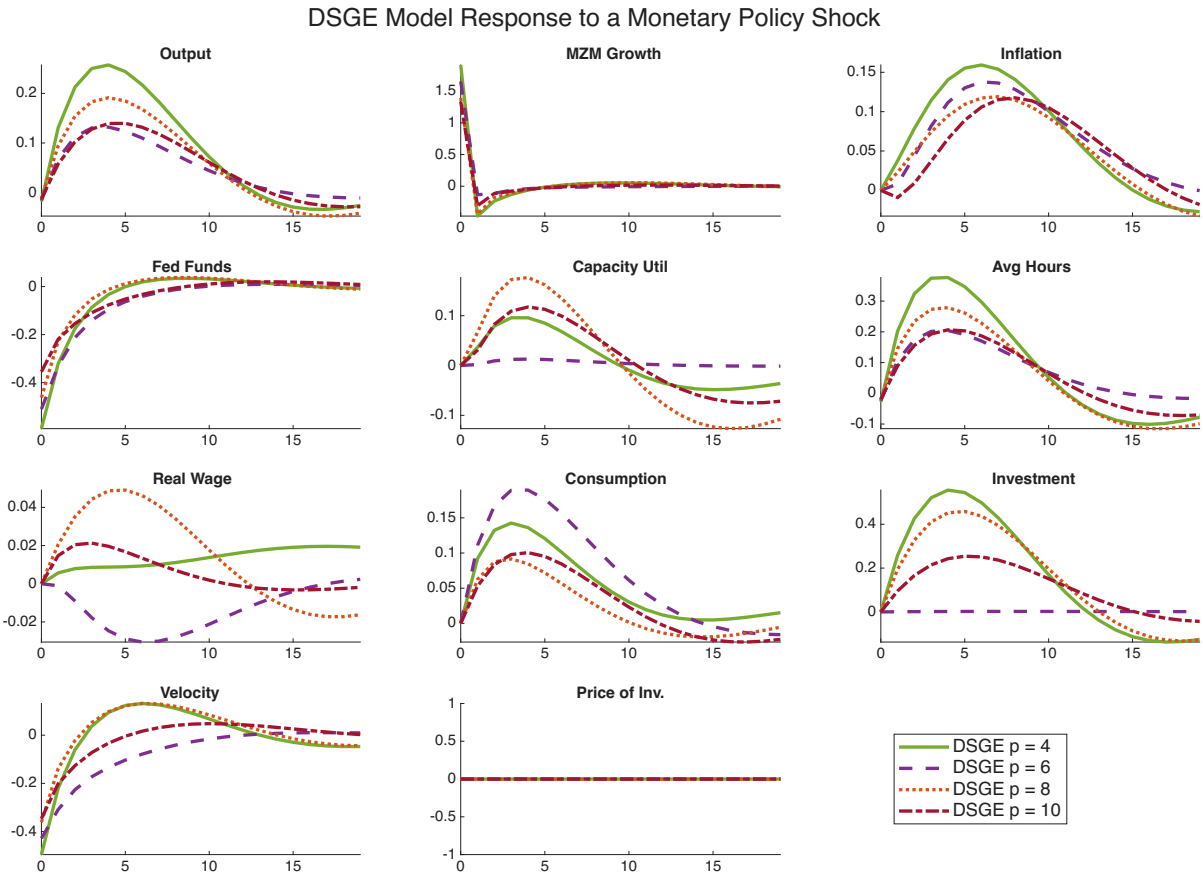


Figure 69: DSGE Monetary Policy Shock lag 10

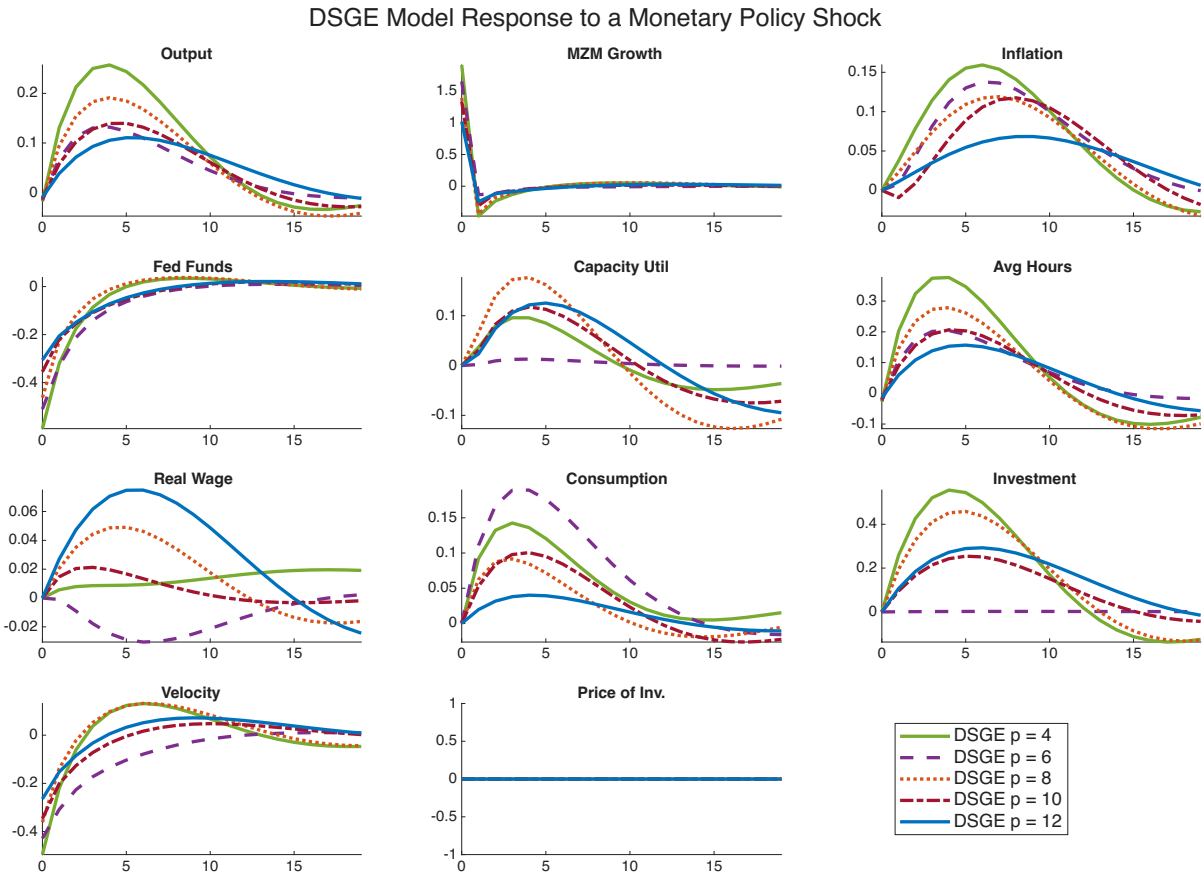


Figure 70: DSGE Monetary Policy Shock lag 12

C.16 ACEL 2005 DSGE Across different lag structures Progressively Comparison: Embodied Technology Shock

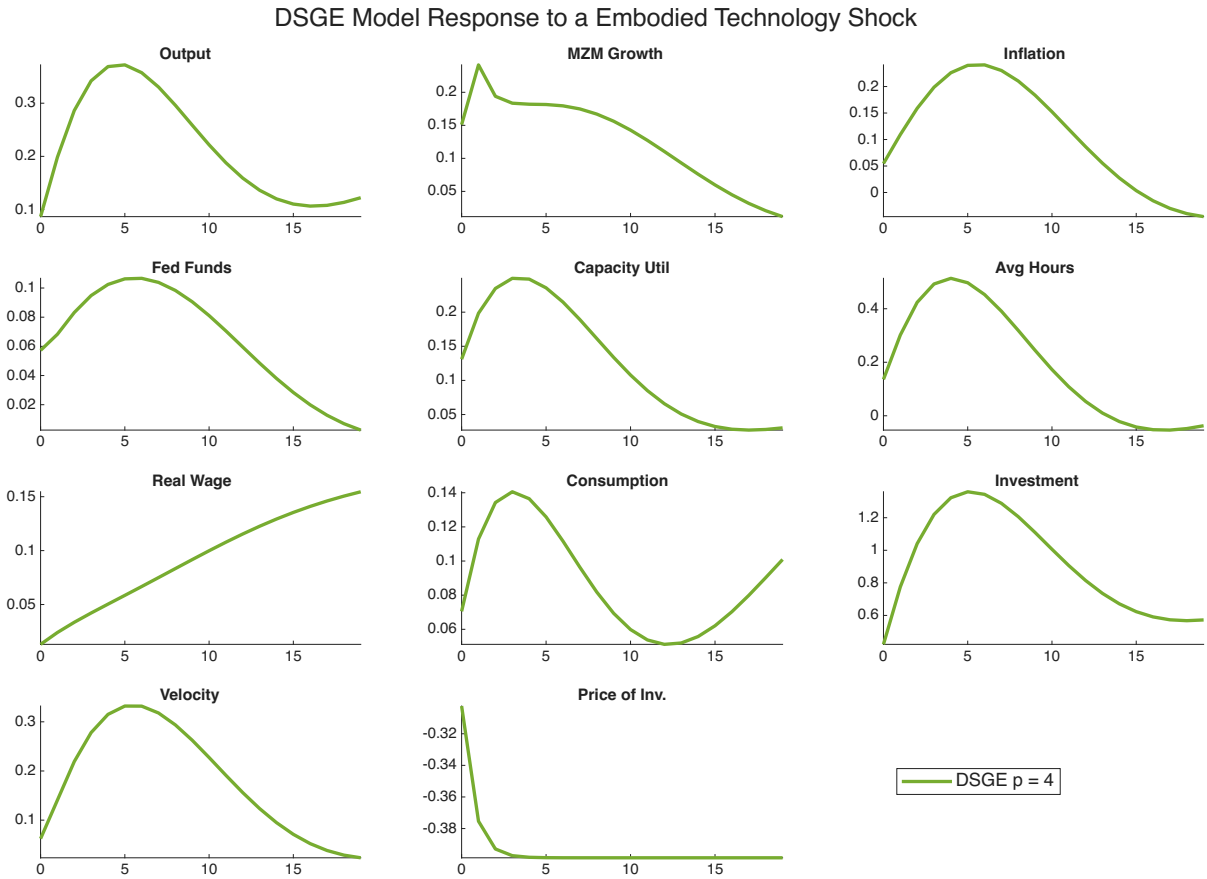


Figure 71: DSGE Embodied Technology Shock lag 4

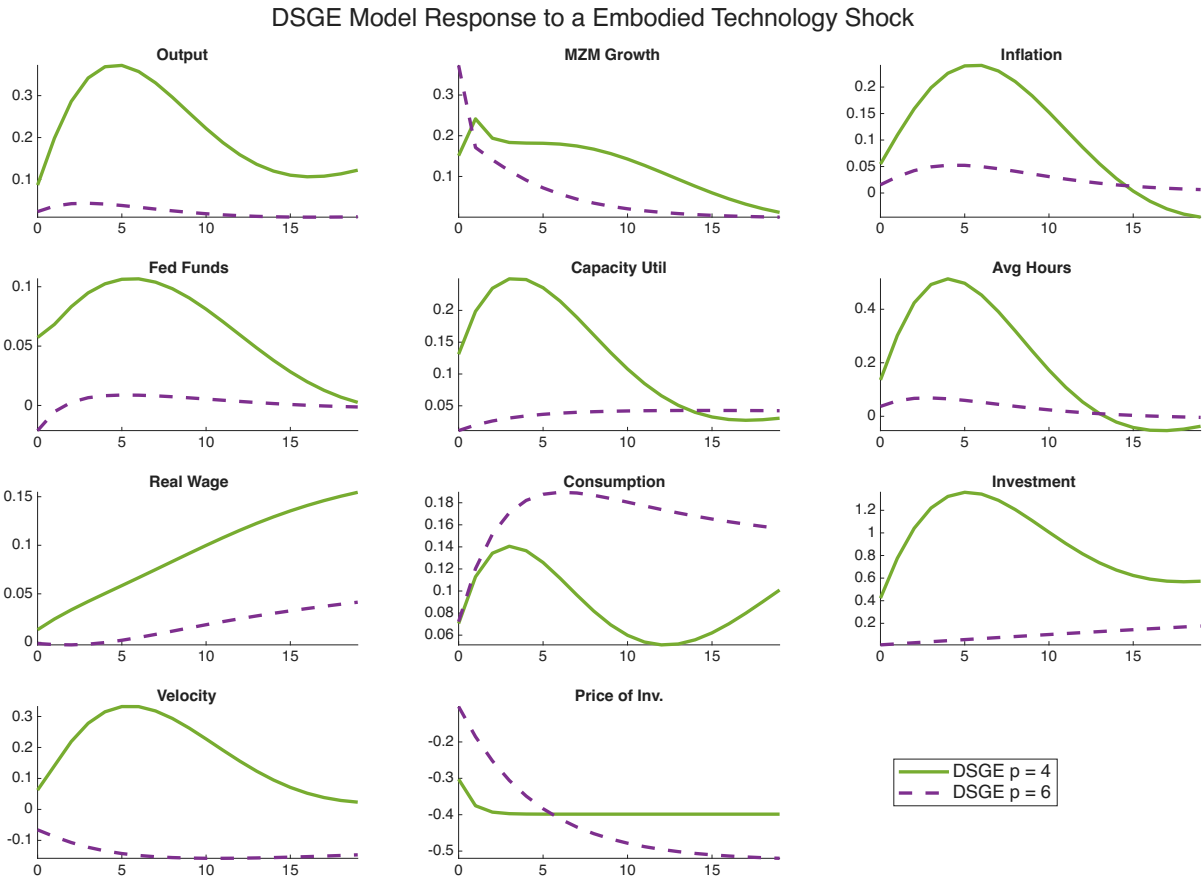


Figure 72: DSGE Embodied Technology Shock lag 6

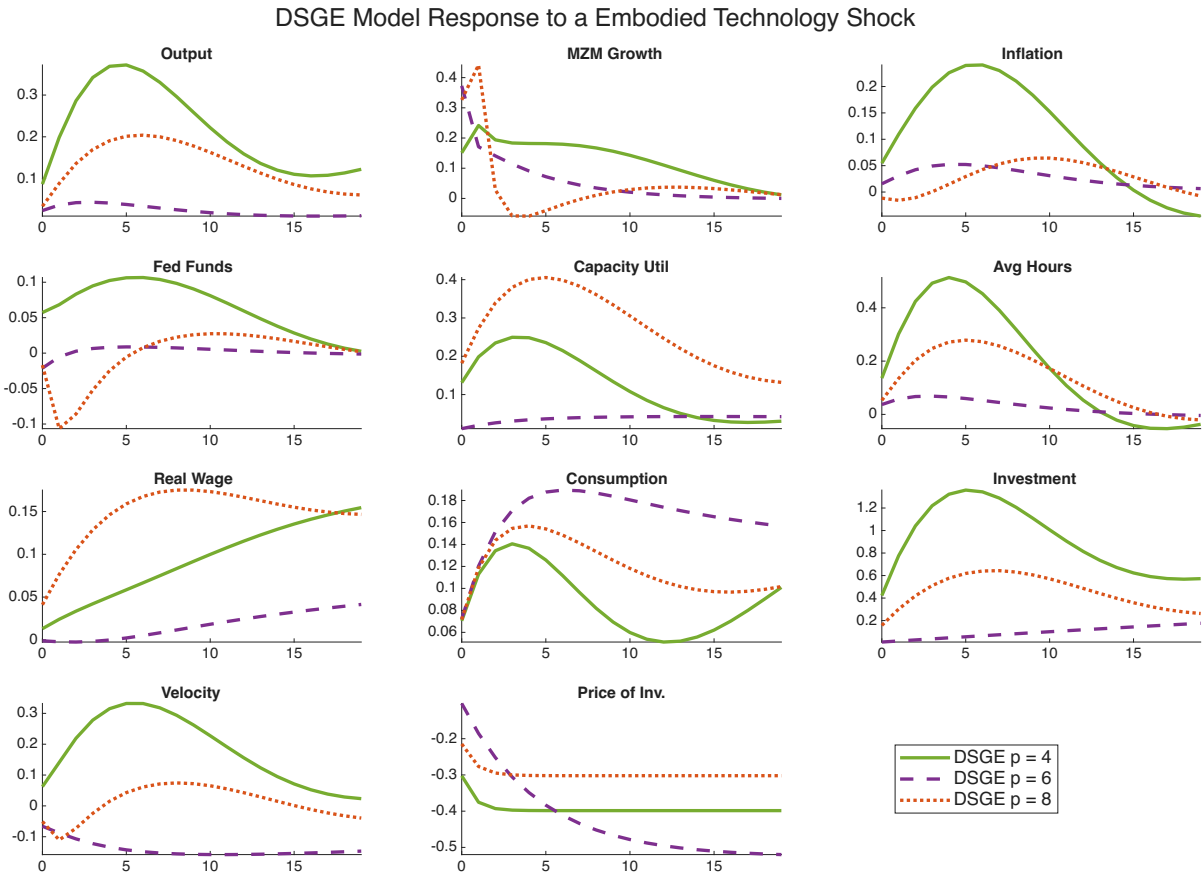


Figure 73: DSGE Embodied Technology Shock lag 8

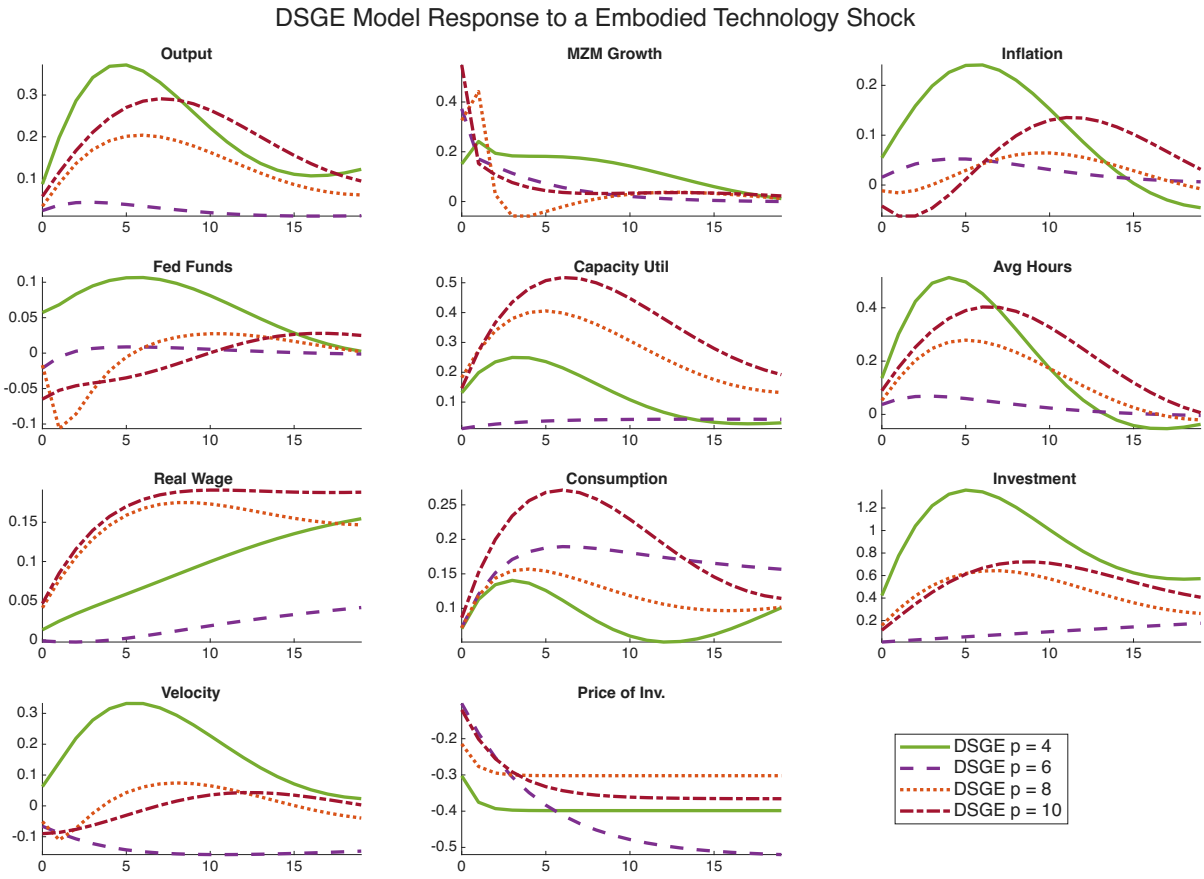


Figure 74: DSGE Embodied Technology Shock lag 10

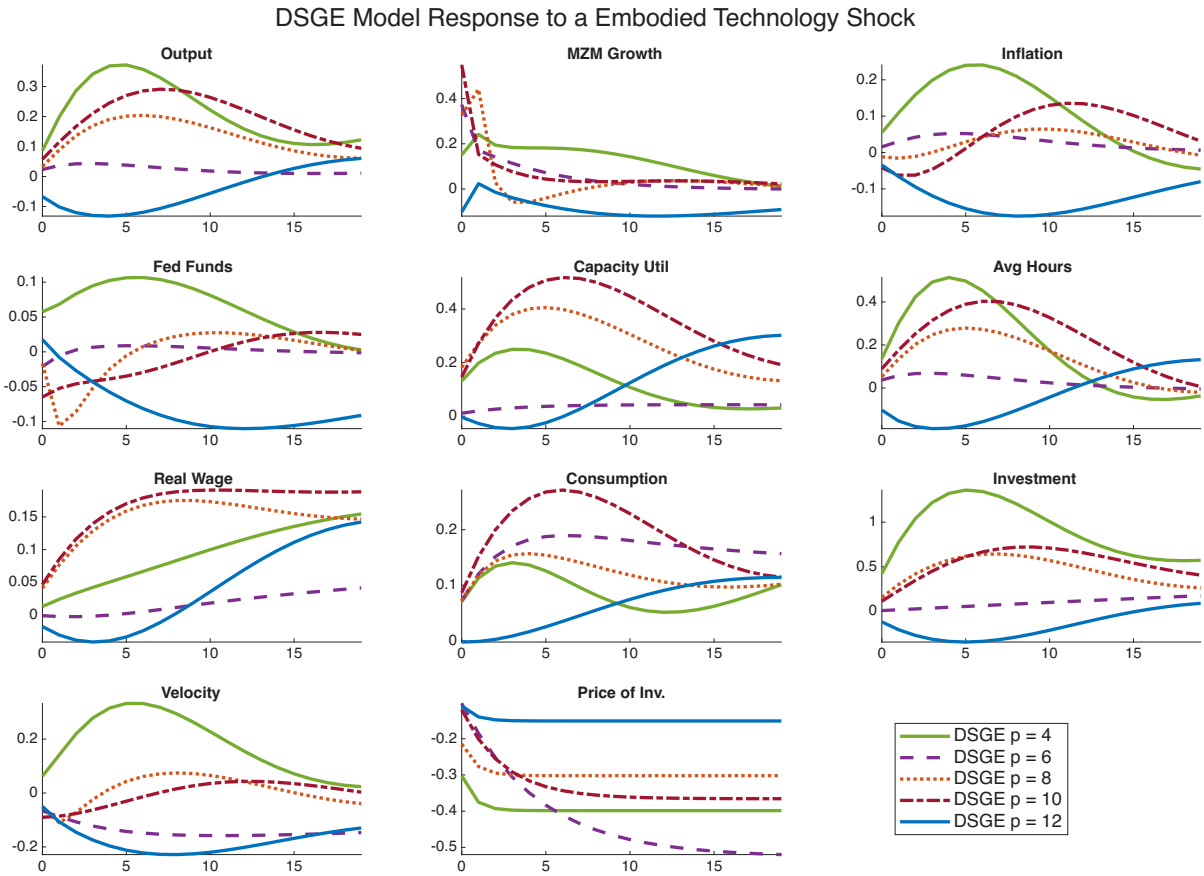


Figure 75: DSGE Embodied Technology Shock lag 12

C.17 ACEL 2005 DSGE Across different lag structures Progressively Comparison: Neutral Technology Shock

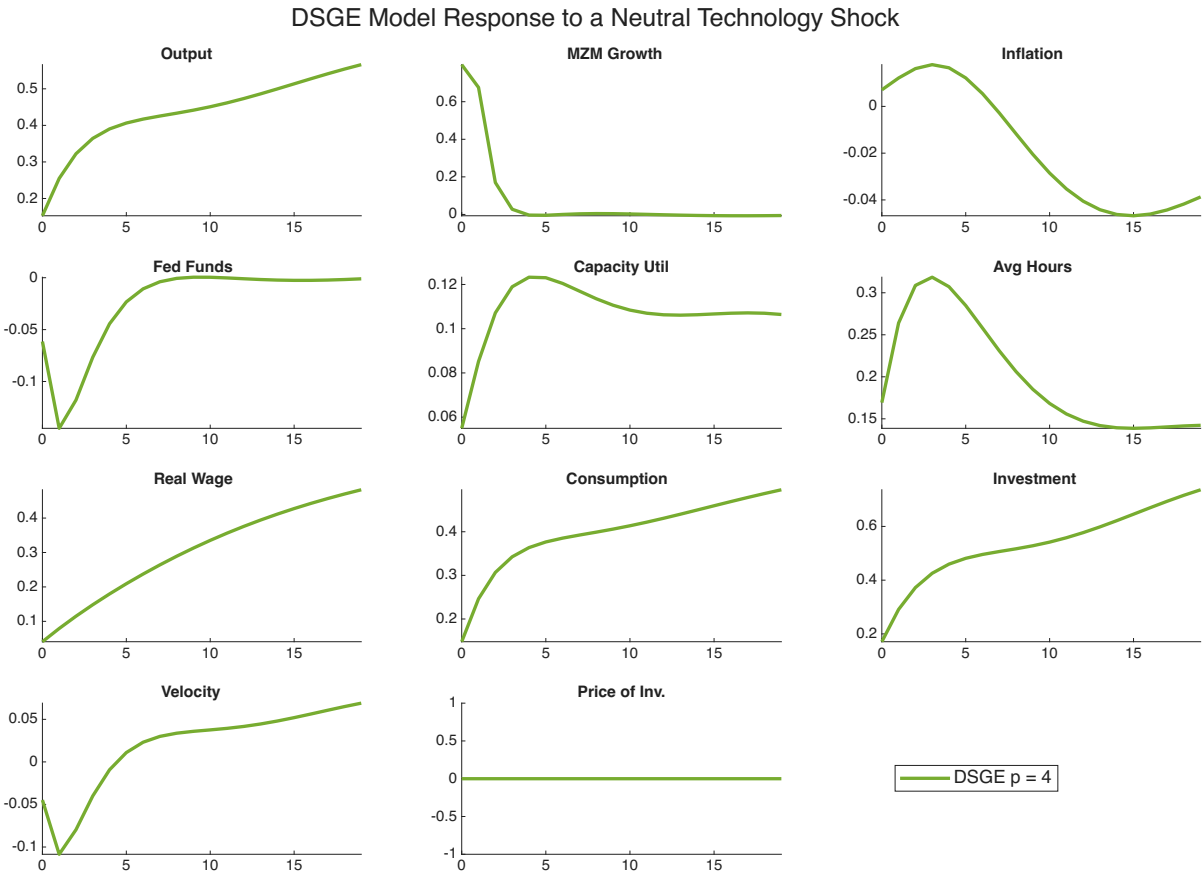


Figure 76: DSGE Neutral Technology Shock lag 4

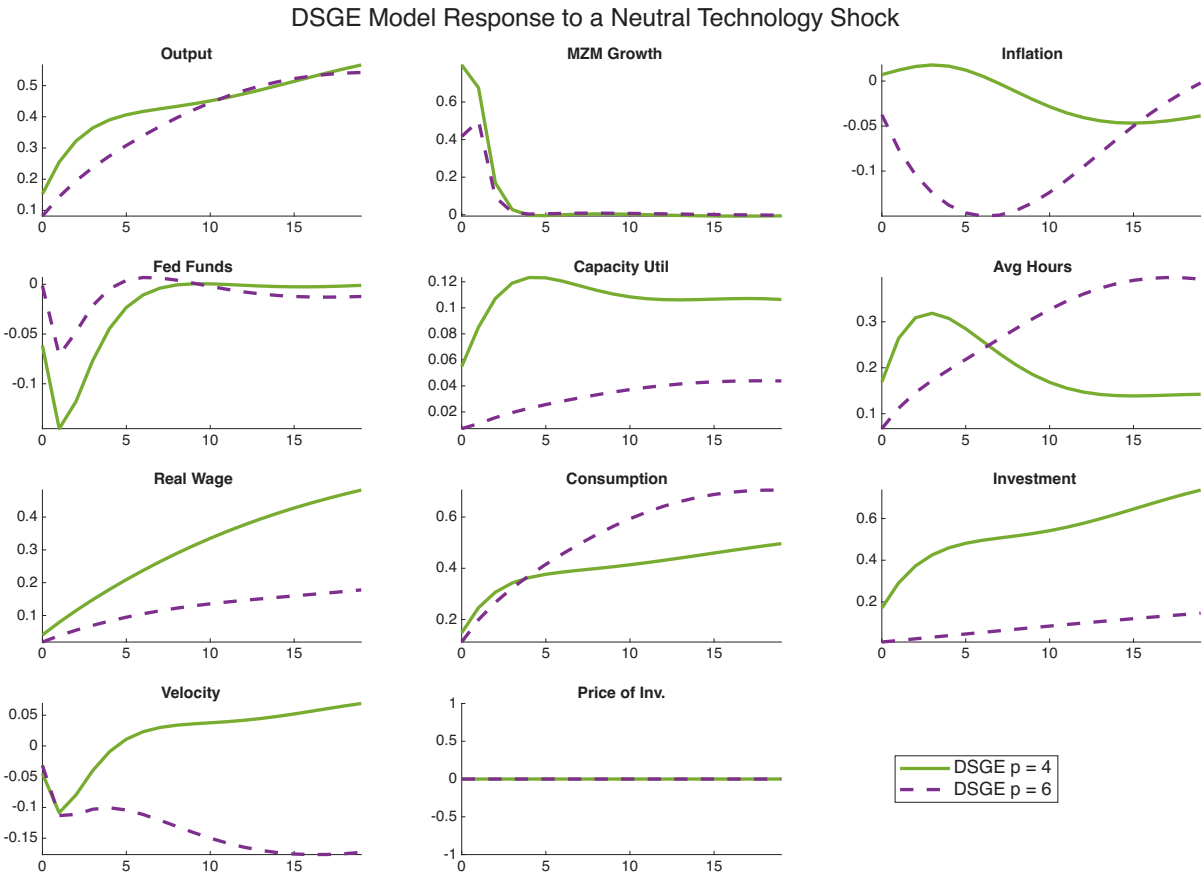


Figure 77: DSGE Neutral Technology Shock lag 6

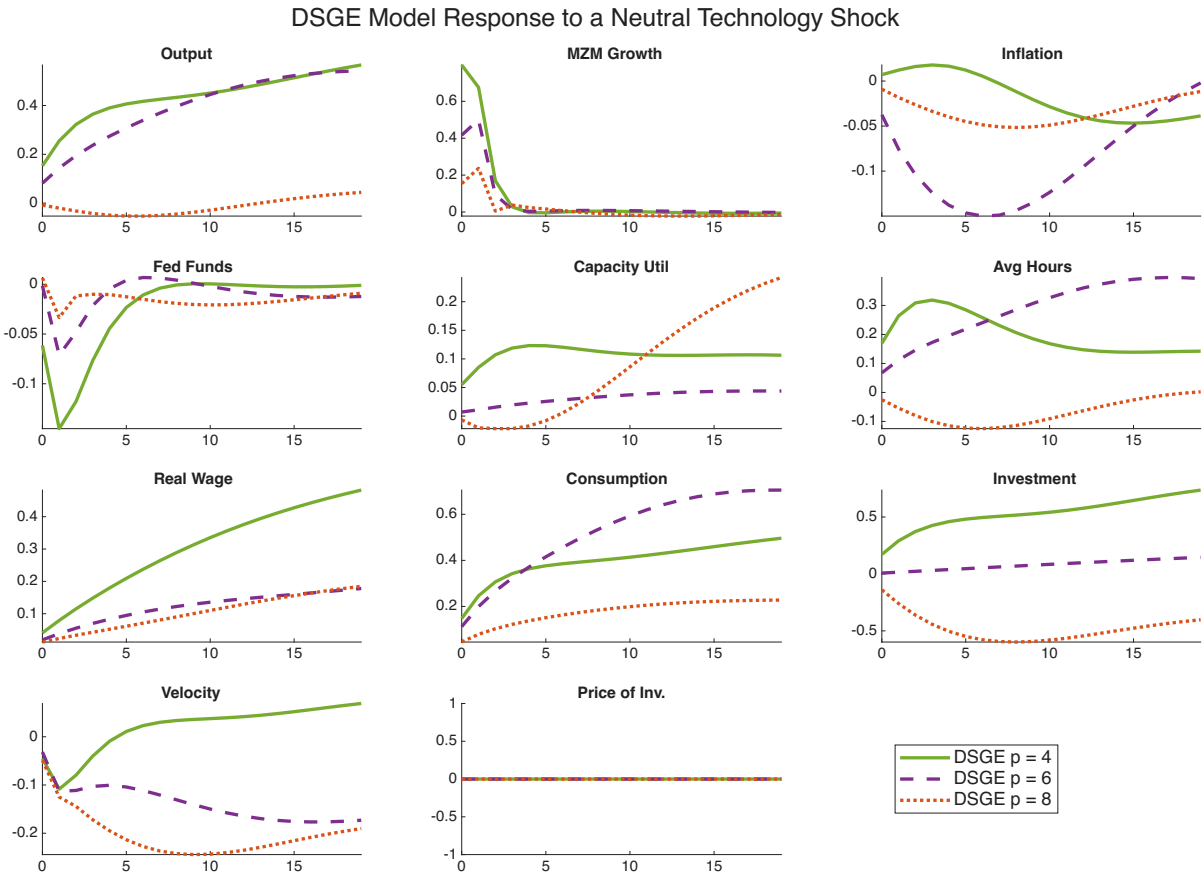


Figure 78: DSGE Neutral Technology Shock lag 8

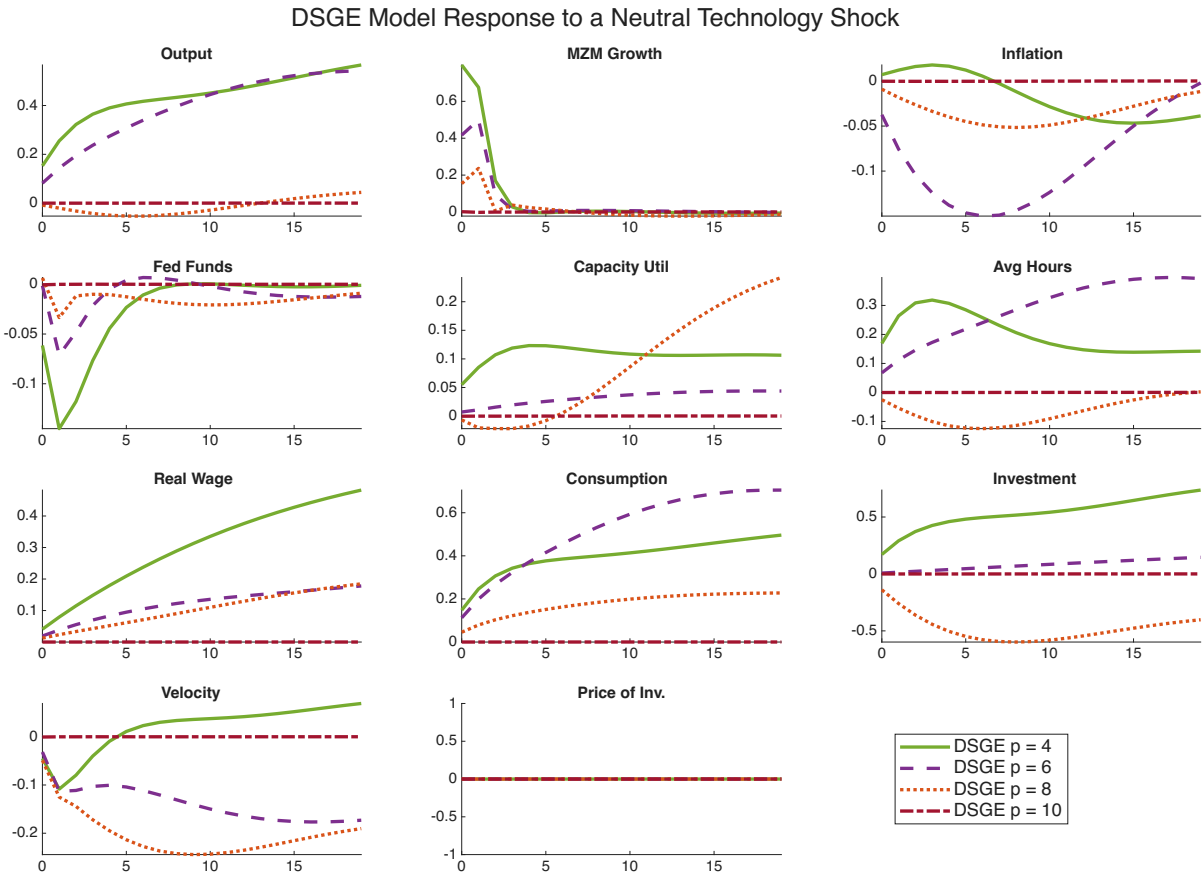


Figure 79: DSGE Neutral Technology Shock lag 10

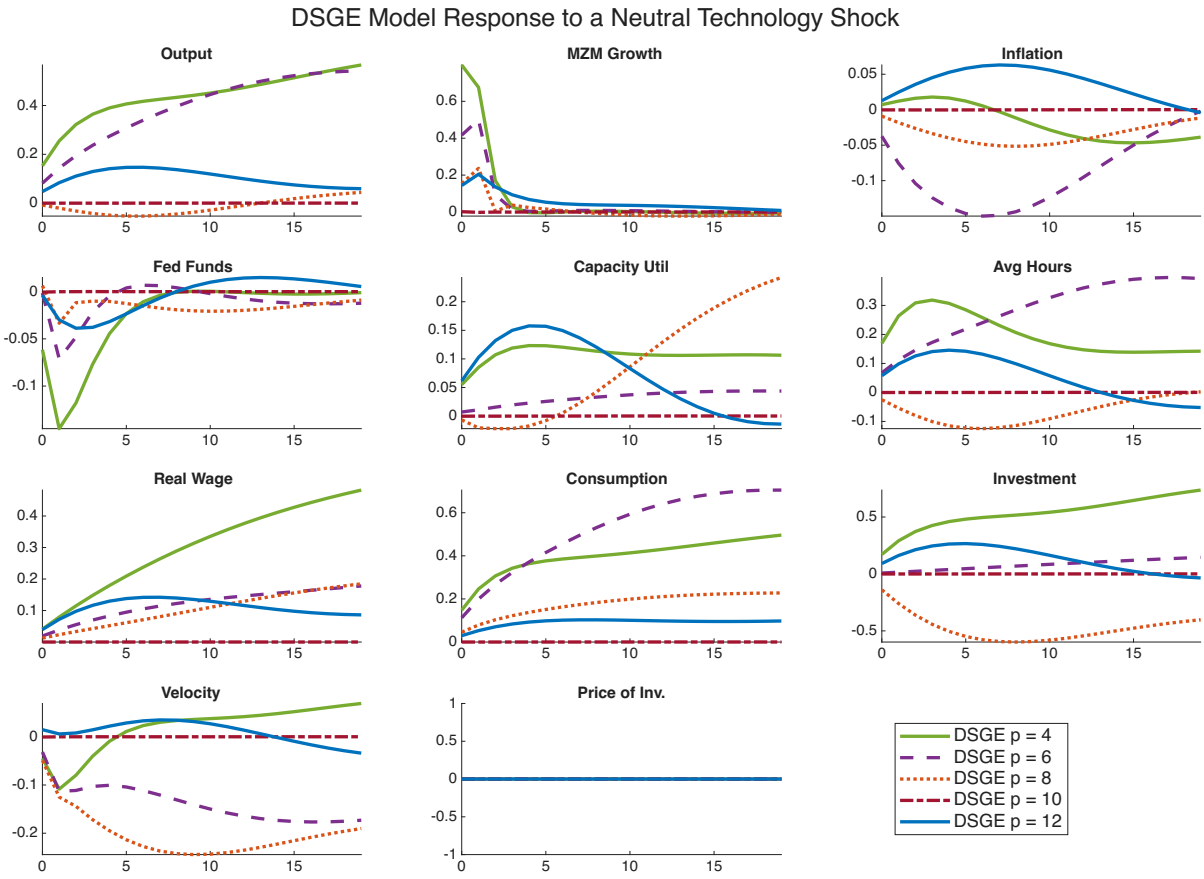


Figure 80: DSGE Neutral Technology Shock lag 12